Book II of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Document Status

This volume is the technical documentation on formal reconstruction of Book II of Euclid's *Elements*. It records the mathematical architecture of the reconstruction, the Lean formalization, the dependency structure of individual propositions, the reusable rectangle and scissors-congruence infrastructure, and the relation between the formal development and the classical sources.

The documentation accompanies the formal development. The Lean source is authoritative for theorem statements, hypotheses, dependencies, and proof status.

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Chapter 1

Proposition II.1

1.1 Introduction

Proposition II.1 opens Book II with a change of mathematical language. Book I has developed a synthetic theory of congruence, parallels, parallelograms, squares, and equal content. Book II begins to combine these ingredients in a systematic calculus of rectangles. A modern reader immediately recognizes in II.1 the pattern of a distributive law. That recognition is useful, but it must be stated carefully: Euclid does not introduce variables for numerical lengths, does not define multiplication of lengths, and does not prove an identity in a number field.

The primitive object in the proposition is a geometric figure: a rectangle "contained by" two straight lines. When one of the two lines is divided into parts, the large rectangle is divided into smaller rectangles having the same other side. The equality asserted by the proposition is therefore first of all a statement about decomposition of figures.

This is a natural point at which to speak of the beginning of the *geometrical algebra* characteristic of Book II. The phrase is an interpretive description of a stable pattern of geometric operations, not a claim that Euclid was secretly manipulating modern symbolic formulas. In the present formal reconstruction this distinction is especially sharp: the code contains rectangles, segment congruence, betweenness, parallelograms, right angles, and scissors relations, but no multiplication operation on segment lengths and no numerical area function.

The formal development separates three levels:

1. a concrete rectangle and its geometric cuts;
2. the Euclidean phrase "rectangle contained by" two given segments;
3. independence of the particular rectangle chosen to represent that phrase.

The final theorem is stronger than a bare equal-content statement. Its conclusion is the exact relation

```
HilbertScissorsEq
```

for the whole rectangle and the formal sum of its parts.

1.2 Euclid's "Rectangle Contained By"

1.2.1 The original geometric meaning

Book II begins with the terminology that a rectangular parallelogram is "contained by" the two straight lines which contain one of its right angles. Thus the expression does not initially denote a product of two numbers. It names a rectangle by two adjacent sides.

Suppose a rectangle has adjacent sides congruent to two given segments PQ and RS. Euclid's phrase

$$\text{the rectangle contained by } PQ, RS$$

should be read geometrically: choose a right-angled parallelogram whose two adjacent sides have those magnitudes. The rectangle itself remains a plane figure.

For exposition only, we will occasionally write

$$\mathrm{Rect}(PQ, RS).$$

This is documentary shorthand. It is not a Lean definition, it is not a number-valued function, and the comma is not multiplication.

1.2.2 Why this already looks algebraic

If BC is divided into consecutive parts BM, MN, and NC, II.1 says in this shorthand

$$\mathrm{Rect}(BC, PQ) \;=\; \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MN, PQ) + \mathrm{Rect}(NC, PQ).$$

A modern algebraist naturally sees the shape

$$(b_1 + b_2 + b_3)a = b_1a + b_2a + b_3a.$$

The formal proof, however, establishes the geometric statement on the first line, not the numerical identity on the second. The second formula is only a modern mnemonic for the combinatorial pattern of the dissection.

This is consistent with the modern synthetic reading of Euclid's area theory. Hartshorne treats Book II as a collection of equal-content statements about figures and describes the resulting manipulation as "geometrical algebra." The point is precisely that the manipulation precedes any assignment of numerical areas to the figures.

1.3 The Synthetic Rectangle Layer

1.3.1 Rectangles

The reconstruction introduces a small reusable rectangle API in `HilbertRectangle.lean`. A rectangle is a parallelogram with one specified right angle:

```
def IsRectangle
    (A B C D : Geo.Point) : Prop :=
  IsParallelogram Geo A B C D /\
  HilbertRightAngle Geo A B C
```

The parallelogram structure supplies the opposite-side relations, while the right-angle component distinguishes rectangles from arbitrary parallelograms. The remaining right angles can then be propagated synthetically.

1.3.2 The formal meaning of “contained by”

The Euclidean phrase is represented by the predicate

```
def IsRectangleContainedBy
    (A B C D P Q R S : Geo.Point) : Prop :=
  IsRectangle Geo A B C D /\
  Geo.Congruent A B P Q /\
  Geo.Congruent B C R S
```

Thus

```
IsRectangleContainedBy Geo A B C D P Q R S
```

means exactly that $ABCD$ is a concrete rectangle, $AB \cong PQ$, and $BC \cong RS$. Nothing in this definition assigns a number to PQ or RS. The two given segments specify the two adjacent geometric magnitudes of a representative rectangle.

The layer also proves existence of such a representative and, more importantly for the semantics of Book II, uniqueness of its scissors content. The theorem

```
rectangle_contained_by_unique
```

says that any two rectangles contained by the same ordered pair of segments satisfy the exact scissors relation. Consequently Euclid’s definite description “the rectangle contained by PQ, RS” becomes well-defined at the scissors level even though many different concrete quadrilaterals can represent it.

1.3.3 No segment multiplication and no numerical area

There are two tempting modernizations that the present reconstruction explicitly avoids.

First, we do not define a product of segment lengths $|PQ|\,|RS|$. The pair (PQ, RS) remains a pair of geometric segments, and the rectangle is related to that pair by congruence of adjacent sides.

Second, we do not map a rectangle to a real number called its area. The content of the proposition is expressed through relations between finite formal sums of triangles. This permits addition, dissection, transport, and comparison of figures at the synthetic level.

This is also conceptually distinct from Hilbert’s later arithmetic of segments, where arithmetic operations are introduced as an additional construction. No such field or segment-arithmetic layer is imported into II.1.

1.4 Exact Scissors Equality and Equal Content

The project uses two related but different content relations. They should not be conflated.

1.4.1 Exact scissors equality

`HilbertScissorsEq` records an exact finite cut-and-paste relation between formal scissors terms. In II.1 it is produced by explicit splits of the rectangle into smaller rectangles and by normalization of the triangulated parallelogram terms. Schematically,

$$P \simeq_{\mathrm{sc}} Q + R$$

means that the two sides can be matched by the exact scissors machinery already developed in the project.

The rectangle transport theorem is already available in this stronger form:

```
rectangle_transport_scissorsEq
```

for rectangles with congruent corresponding adjacent sides.

1.4.2 The more general equal-content relation

The broader project notion corresponding to equal content is

```
HilbertScissorsEquicomplementable
```

It allows suitable complementary terms to be adjoined before exact scissors equality is established. Thus it is designed to capture a more general synthetic equal-content relation.

The theorem

```
rectangle_transport
```

is obtained by weakening the exact transport theorem from `HilbertScissorsEq` to equicomplementability.

For II.1 no such weakening is necessary. The final theorem proves `HilbertScissorsEq` itself. Therefore II.1 certainly yields an equal-content consequence, but its formal statement retains the stronger fact that the displayed equality comes from an exact dissection.

1.5 The Classical and Formal Configurations

1.5.1 The binary split

The reusable core is a one-cut theorem. Let $B - M - C$ and let the corresponding point L divide the opposite side DE. The rectangle $BCED$ is cut into the rectangles $BMLD$ and $MCEL$:

$$BCED \simeq_{\mathrm{sc}} BMLD + MCEL.$$

In Lean the theorem is

```
rectangle_split
```

and it returns both the rectangle structure of the two pieces and the exact scissors equality.

The formal configuration contains one additional point X. It satisfies $D - X - C$ and $M - X - L$. This is not decorative data: the inherited exact split lemma from the I.47 scissors infrastructure uses the intersection of the cut line with a diagonal. The classical rectangle picture suppresses this witness; the formal theorem makes it explicit.

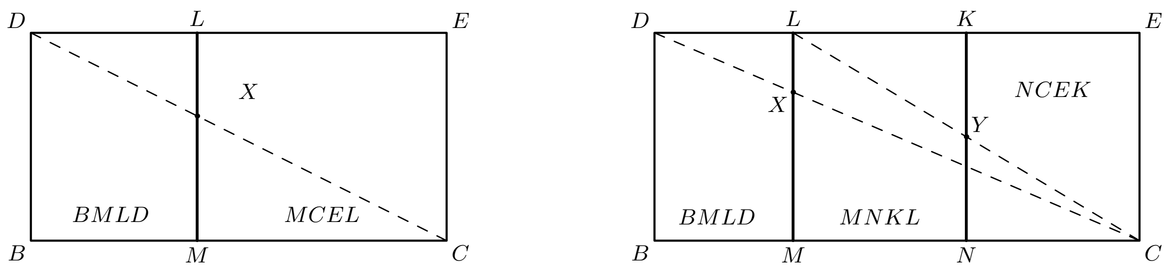
1.5.2 The three-part split

The three-part form applies the binary split twice. First cut at $M - L$; then cut the remaining rectangle at $N - K$. The result is

$$\begin{aligned} BCED \simeq_{\mathrm{sc}} &BMLD \\ &+ MNKL \\ &+ NCEK. \end{aligned}$$

The second split has its own diagonal-intersection witness Y, with $L - Y - C$ and $N - Y - K$.

Figure 1.1 shows both levels and includes the hidden witnesses X and Y required by the current Lean interface.



(a) binary split (b) three-part split

Figure 1.1: The exact rectangle splits underlying II.1. Panel (a) is the binary kernel: $B - M - C$ and the cut ML divide $BCED$ into $BMLD$ and $MCEL$; the dashed diagonal DC meets the cut at the formal witness X. Panel (b) applies the same construction twice: BC is divided into BM, MN, and NC, producing the three rectangles $BMLD$, $MNKL$, and $NCEK$. The dashed diagonals display the witnesses X and Y used by the two exact split calls.

The actual three-part theorem is

```
rectangle_split_three
```

Its proof is structurally transparent: obtain the first binary equality, split the remaining right-hand rectangle, add the unchanged left piece to that second equality, and compose by transitivity of `HilbertScissorsEq`.

1.5.3 Scope of the present formal statement

Euclid states II.1 for one line cut into an arbitrary finite number of parts. The current Lean file does not yet package an n-ary list-valued theorem. It proves the binary split and the explicit three-part form used in the standard II.1 configuration. The binary theorem is the reusable kernel from which a general finite version could later be obtained by iteration.

This is a deliberate statement-fidelity point. The documentation records Euclid's general formulation, but the final formal statement displayed below is the representative-independent three-part theorem actually present in `Proposition2_1.lean`.

1.6 From a Concrete Diagram to Euclid's Language

The geometric split alone produces concrete rectangles. Euclid II.1 is more abstract: it speaks of the rectangles *contained by* the whole segment and the several parts together with one fixed segment.

The theorem

```
rectangle_split_three_contained_by
```

bridges these levels. If $CE \cong PQ$, it packages the four concrete rectangles as

$$\begin{array}{rcl} BCED & : & \mathrm{Rect}(BC, PQ), \\ BMLD & : & \mathrm{Rect}(BM, PQ), \\ MNKL & : & \mathrm{Rect}(MN, PQ), \\ NCEK & : & \mathrm{Rect}(NC, PQ). \end{array}$$

The opposite-side congruences of the parallelograms are what transport the fixed side PQ through the two cuts. The conclusion still contains the same exact scissors equality between the concrete quadrilateral terms.

At this stage the geometry corresponding to Euclid's wording is visible, but the theorem still uses one particular drawn rectangle and its particular cut points. The final step removes this representational dependence.

1.7 Representative Independence

1.7.1 Why representatives matter formally

A phrase such as

$$\mathrm{Rect}(BC, PQ)$$

does not identify four vertices. Lean, however, must manipulate an actual term such as

```
hilbertParallelogramTerm Geo U0 U1 U2 U3
```

with named vertices. The final theorem therefore accepts arbitrary representatives for the whole rectangle and for each of the three part rectangles.

The concrete dissection first proves

$$BCED \simeq_{\text{sc}} BMLD + MNKL + NCEK.$$

Then four calls to

```
rectangle_contained_by_unique
```

transport the whole and the three pieces to arbitrary representatives U, V, W, and Z. Compatibility of `HilbertScissorsEq` with formal addition transports the nested sum. Transitivity completes the chain

$$\begin{array}{c}\text{arbitrary whole representative}\\ \Downarrow \\ \text{concrete whole rectangle}\\ \Downarrow \\ \text{concrete three-part dissection}\\ \Downarrow \\ \text{arbitrary representatives of the three parts.}\end{array}$$

This is the formal content of representative independence. It is not merely a convenience for Lean. It explains why the Euclidean expression "the rectangle contained by two segments" can function as a stable geometric magnitude without first quotienting segments into numerical lengths or assigning numerical areas to rectangles.

1.8 Commutativity by Rectangle Rotation

The second algebra-like feature is symmetry in the two containing segments. For a numerical product one would write $ab = ba$. The synthetic proof does not appeal to a commutative multiplication law. Instead it rotates the presentation of a rectangle.

If $ABCD$ is contained by PQ, RS, then the cyclically reoriented rectangle $BCDA$ has adjacent sides BC and CD. The first is already congruent to RS; the second is congruent to AB, hence to PQ. Thus the same geometric rectangle, with its vertices read from the next corner, is contained by RS, PQ.

The exact theorem available in the rectangle layer is

```
rectangle_contained_by_swap
```

It returns both the swapped `IsRectangleContainedBy` fact and a `HilbertScissorsEq` identifying the two quadrilateral presentations. The current rectangle API also packages the representative-independent consequence as

```
rectangle_contained_by_comm
```

so the commutativity needed by later Book II propositions is a theorem of rectangle geometry rather than an assumed law of segment multiplication.

Figure 1.2 shows the geometric mechanism. The operation is a cyclic change of the distinguished corner of the same rectangle; this exchanges the two adjacent side roles.

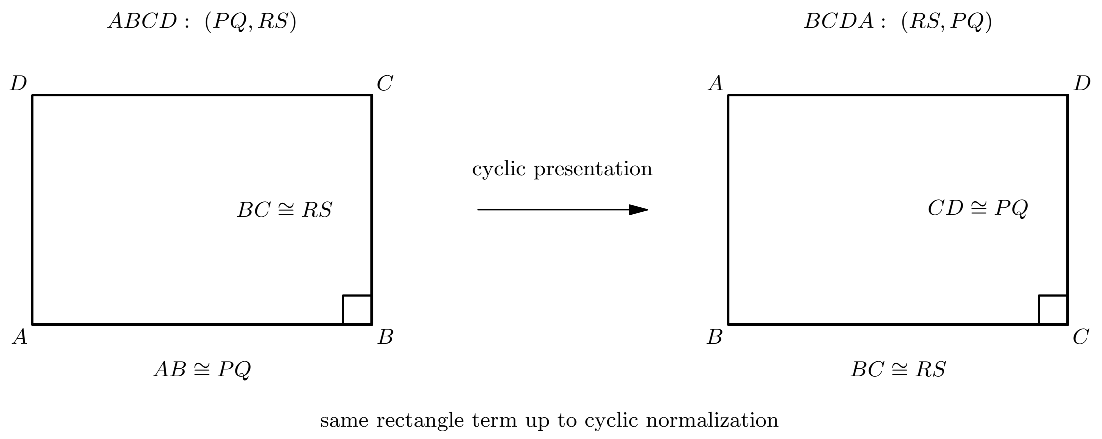


Figure 1.2: The geometric source of commutativity. On the left, $ABCD$ represents the rectangle contained by PQ, RS. Reading the same rectangle from the next vertex gives $BCDA$: now the first adjacent side represents RS and the second represents PQ. Exact scissors invariance under this cyclic presentation change yields the synthetic counterpart of exchanging the two factors.

This lemma is already operational in the next proposition. II.2 naturally writes rectangles in the order “whole side, part”, whereas the II.1 cut lemma produces them in the order “part, fixed side”. The implementation of II.2 uses the swap theorem to pass between these two presentations before applying the binary form of II.1.

1.9 Proof Architecture of the Reconstruction

The Lean proof is best read as five mathematical stages.

1.9.1 Step 1: exact binary dissection

```
rectangle_split_scissorsEq
rectangle_split_left
rectangle_split_right
rectangle_split
```

The inherited I.47 split theorem supplies the exact scissors decomposition. The two local rectangle helpers prove that each piece remains a rectangle by propagating right angles through the relevant parallelograms and along the same rays determined by betweenness.

1.9.2 Step 2: iterate the split

```
rectangle_split_three
```

The right-hand remainder of the first split is itself a rectangle, so the same binary theorem can be applied again. Addition and transitivity of exact scissors equality compose the two decompositions.

1.9.3 Step 3: attach the "contained by" data

```
rectangle_split_three_contained_by
```

Opposite-side congruence transports the fixed segment PQ through all three pieces. Each concrete piece is thereby promoted from `IsRectangle` to `IsRectangleContainedBy`.

1.9.4 Step 4: remove the choice of representatives

```
rectangle_contained_by_unique
euclid_proposition_2_1_three
```

The uniqueness theorem replaces the concrete whole and each concrete part by an arbitrary rectangle contained by the same pair of segments. This is the semantic step that makes the final statement faithful to Euclid's definite phrase "the rectangle contained by."

1.9.5 Step 5: expose the symmetric side order

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

Cyclic rotation of the rectangle exchanges the two containing segments. This is not needed to establish the displayed three-part split itself, but it is an important part of the Book II rectangle interface and is immediately useful in II.2 and later propositions.

1.10 Classical Proof and Formal Proof

1.10.1 Euclid's proof pattern

In the classical three-part configuration, take one undivided segment A and a segment BC divided at D and E. Erect a perpendicular at one endpoint of BC, lay off on it a segment equal to A, complete the outer rectangle, and through the division points draw parallels to the perpendicular side. The outer rectangle is visibly partitioned into three rectangles. Each smaller rectangle is contained by A and one of the three consecutive parts of BC. Common Notion 2 then combines the parts into the whole equality.

The classical DAG is therefore extremely small:

$$\begin{gathered}\text{divide one side}\\ \Downarrow\\ \text{draw the corresponding parallels}\\ \Downarrow\\ \text{obtain three subrectangles}\\ \Downarrow\\ \text{whole rectangle} = \text{sum of the three subrectangles.}\end{gathered}$$

1.10.2 The formal DAG

The current reconstruction expands two things that the classical diagram treats as immediate: exact scissors bookkeeping and independence of the chosen rectangle representatives.

$$\begin{gathered}\text{rectangle geometry} + \text{betweenness} + \text{cut parallelograms}\\ \Downarrow \\ \texttt{rectangle\_split} \\ \Downarrow \\ \texttt{rectangle\_split\_three} \\ \Downarrow \\ \texttt{rectangle\_split\_three\_contained\_by} \\ \Downarrow \\ \texttt{rectangle\_contained\_by\_unique} \\ \Downarrow \\ \texttt{euclid\_proposition\_2\_1\_three}.\end{gathered}$$

The proof is therefore not a mechanical transcription of Euclid. It preserves the geometric core, but makes the semantic and representational layers explicit.

1.11 Geometry, Content, and Representation

1.11.1 Geometry

The geometric data are substantial even though the proposition is visually simple. The current theorem receives the outer rectangle, strict betweenness relations on the divided side and its opposite, two diagonal-intersection witnesses, and four parallelogram facts for the two successive cuts. Right angles of the two resulting rectangles are not read from the picture; they are proved by right-angle transport and parallelogram geometry.

No coordinates are introduced. The cut points are controlled by `Between`, the side lengths by segment congruence, and the rectangular structure by parallels and synthetic right angles.

1.11.2 Content

Content enters in its strongest local form. The core statements are exact `HilbertScissorsEq` decompositions. The three-part theorem is composed from two exact binary splits by addition and transitivity. Representative transport is also exact scissors equality.

The more general equicomplementability relation is part of the surrounding Book I content theory, but II.1 does not need to weaken its conclusion to that level.

1.11.3 Representation

Representation is central to the formal meaning of the proposition. A rectangle contained by PQ, RS is not one canonical four-tuple of vertices. The predicate `IsRectangleContainedBy` describes a representative; `rectangle_contained_by_unique` proves that different representatives have the same exact scissors content.

There is a second representation issue inside the split itself. The inherited scissors theorem and the Book II statement use different cyclic orders for some parallelogram terms. The proof normalizes these by

```
parallelogram_term_rotateOne
```

before composing the scissors equalities. These cyclic changes are bookkeeping of a geometric representation, not new area facts.

1.12 Hidden Configuration Data

The classical drawing makes several facts appear automatic. Lean records them explicitly:

- $B - M - C$ for the first cut and $M - N - C$ for the second;
- $D - L - E$ and $L - K - E$ on the opposite side;
- $D - X - C$ together with $M - X - L$ for the first diagonal/cut intersection;
- $L - Y - C$ together with $N - Y - K$ for the second diagonal/cut intersection;
- explicit parallelogram witnesses for the left and right pieces of each split;
- congruence $CE \cong PQ$, which identifies the fixed side used in the "contained by" descriptions;
- four arbitrary rectangle representatives in the final theorem.

The points X and Y do not appear in Euclid's statement and need not be emphasized in an elementary picture of II.1. They are nevertheless real data of the current exact-dissection API, inherited from the triangulated proof of the rectangle split.

1.13 Axiomatic Status

The public II.1 theorems assume

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

so II.1 is Euclidean in the current project API. Group IV is available through this typeclass and is consumed transitively by the parallelogram and parallel infrastructure on which the rectangle layer and inherited split theorem depend. The proposition itself does not introduce a new parallel axiom.

No continuity typeclass, Archimedean axiom, numerical area function, or segment field is introduced in `Proposition2_1.lean`. No proposition-local `axiom` or `sorry` occurs in the audited source. The modular builds for `HilbertRectangle` and `Proposition2_1` are recorded as clean at the present project checkpoint.

The content status is also worth separating from the later converse arguments of Book I. II.1 proceeds from geometry to an exact dissection. It does not infer metric geometry from an equal-content hypothesis, so it does not need a proper-part or De Zolt-type strictness argument.

1.14 Relationship with Book I and the Following Propositions

The exact split theorem reuses the scissors infrastructure established during I.47. In particular, the underlying theorem historically named

```
i47_square_split
```

actually requires only parallelogram structure, and II.1 repackages it as a general rectangle split.

The rectangle layer also uses Book I results about parallelograms, opposite sides, congruent right angles, SAS/SSS transport, and the scissors representation of parallelograms. Thus Book II does not start a new foundational geometry; it starts a new way of organizing geometric magnitudes on top of the completed Book I infrastructure.

II.2 immediately imports II.1. Its formal proof illustrates why the symmetric rectangle API matters: Euclid's order of the two containing sides is opposite to the order produced by the binary cut theorem, so the representatives are swapped by rectangle rotation before II.1 is applied. II.3 continues the same rectangle calculus. In this sense II.1 is not only the first theorem of Book II but the API-level distributive kernel for the propositions that follow.

1.15 Source Audit

1.15.1 Euclid

Book II Definition 1 supplies the semantic key: the two lines "containing" a rectangle are the adjacent sides that contain a right angle. Proposition II.1 then divides one of two given lines into several parts and identifies the whole rectangle with the sum of the rectangles formed from the undivided line and those parts.

The proof is geometric throughout. The division points on one side are carried across the outer rectangle by parallels. The resulting subrectangles literally partition the original rectangle. Nothing in the proposition requires a numerical assignment to a segment or a plane figure.

1.15.2 Hartshorne

Hartshorne's modern synthetic discussion is a useful semantic control. He interprets Euclid's equalities of figures through equal content rather than numerical area, and notes that all the Book II propositions are valid at that level. He also uses "geometrical algebra" for the systematic manipulation of such geometric contents.

Our reconstruction is deliberately slightly sharper for II.1: the final result is not only equicomplementability but exact `HilbertScissorsEq`. This is possible because II.1 is itself a direct dissection theorem.

1.15.3 Hilbert

Hilbert is relevant here mainly as a warning against collapsing distinct layers. His later arithmetic of segments introduces arithmetic operations only after substantial synthetic development. The current Book II reconstruction does not import that arithmetic and does not use Hilbert's later numerical surface-area theory. It remains in the incidence/order/congruence plus scissors language already available after Book I.

1.15.4 The phrase "geometric algebra"

The phrase is therefore appropriate only with a qualification. II.1 exhibits an algebraic *pattern*: partition of one side corresponds to additive decomposition of a rectangle; exchanging the two containing sides corresponds to cyclic rotation; a square is the special case in which the two containing segments are the same. These operations anticipate the formal laws of a product-like calculus.

But the objects being manipulated by Euclid are still geometric figures and relations of equality between them. The documentation should therefore resist rewriting the theorem as if Euclid had stated

$$a(b + c + d) = ab + ac + ad$$

in symbolic algebra. That equation is a modern shadow of the configuration, not the proposition itself.

1.16 Dependency Summary

The actual formal dependency chain is

$$\begin{gathered}
\text{Hilbert incidence + order + congruence + Group IV} \\
\Downarrow \\
\text{right angles + parallels + parallelograms + opposite-side congruence} \\
\Downarrow \\
\text{Book I scissors terms and exact triangle/parallelogram splitting} \\
\Downarrow \\
\texttt{IsRectangle} \\
\Downarrow \\
\texttt{IsRectangleContainedBy} \\
\Downarrow \\
\texttt{rectangle_contained_by_unique} \\
\Downarrow \\
\texttt{rectangle_split} \\
\Downarrow \\
\texttt{rectangle_split_three} \\
\Downarrow \\
\texttt{rectangle_split_three_contained_by} \\
\Downarrow \\
\texttt{euclid_proposition_2_1_three}.
\end{gathered}$$

The symmetric side-order branch is

$$\begin{gathered}
\texttt{IsRectangleContainedBy}(PQ, RS) \\
\Downarrow \\
\text{cyclic rectangle rotation} \\
\Downarrow \\
\texttt{rectangle_contained_by_swap} \\
\Downarrow \\
\texttt{rectangle_contained_by_comm}.
\end{gathered}$$

New geometric axiom in II.1:	no
New proposition-local axiom:	no
New rectangle abstraction:	yes, in `HilbertRectangle`
New reusable split helper:	yes
New representative-independence theorem:	yes
Exact scissors dissection used:	yes
General equicomplementability required by II.1:	no
Numerical area semantics:	no
Segment multiplication / field theory:	no
Group IV available/used transitively:	yes
Continuity assumption in the file:	no
Proposition-local `sorry`/`admit`:	no
Modular build:	clean

1.17 Conclusion

Proposition II.1 is the first clean instance in Book II of a product-like law being realized entirely by geometry. A segment is divided; a rectangle is cut along corresponding parallels; exact scissors equality identifies the whole with the sum of the pieces. The rectangle layer then removes the accidental choice of a concrete drawing, so Euclid's phrase "the rectangle contained by" becomes a representative-independent synthetic magnitude.

The formalization also isolates the two structural laws that make the language look algebraic. Additivity comes from actual dissection. Commutativity comes from cyclic rotation of the rectangle. A square is the special case in which the two containing segments agree. None of these steps requires a product of segment lengths or a numerical area map.

Thus the "geometric algebra" of II.1 is best understood not as hidden modern symbolism but as a calculus of geometric magnitudes whose algebraic laws emerge from constructions, congruence, and finite decomposition.

The representative-independent three-part theorem at the end of the current formalization is:

```
theorem euclid_proposition_2_1_three
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    -- Concrete dissection diagram.
    (B C D E L M K N X Y P Q : Geo.Point)
    (hRect : IsRectangle Geo B C E D)
    (hCE_PQ : Geo.Congruent C E P Q)

    -- First cut.
    (hBMC : Geo.Between B M C)
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    -- Second cut.
    (hMNC : Geo.Between M N C)
```

```
    (hLKE : Geo.Between L K E)
    (hLYC : Geo.Between L Y C)
    (hNYK : Geo.Between N Y K)
    (hMiddlePar :
      IsParallelogram Geo K L M N)
    (hFinalPar :
      IsParallelogram Geo C E K N)

    -- Arbitrary representatives of the four rectangles.
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)
    (W0 W1 W2 W3 : Geo.Point)
    (Z0 Z1 Z2 Z3 : Geo.Point)

    (hWhole :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 B C P Q)

    (hPart1 :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B M P Q)

    (hPart2 :
      IsRectangleContainedBy Geo
        W0 W1 W2 W3 M N P Q)

    (hPart3 :
      IsRectangleContainedBy Geo
        Z0 Z1 Z2 Z3 N C P Q) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       (hilbertParallelogramTerm Geo W0 W1 W2 W3 +
        hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)) := by
```

Chapter 2

Proposition II.2

2.1 Introduction

Proposition II.2 is the first place where the rectangle calculus established in II.1 is reused rather than developed further. The classical proposition is visually elementary: a square on a straight line is cut by a parallel through an arbitrary point of that line, and the two resulting rectangles fill the square. The formal reconstruction preserves exactly that geometric content, but its proof architecture is deliberately different from Euclid's.

Euclid proves II.2 directly from a square construction and a parallel through the point of division. The current Lean theorem instead imports `Proposition2_1` and applies the representative-independent binary form of II.1. One additional operation is needed: Euclid names each part rectangle with the whole side first, whereas the II.1 cut theorem returns the divided part first. The proof therefore rotates each rectangle representative by one vertex, applies II.1, and then transports the result back to Euclid's original order.

This makes II.2 a particularly clean test of the Book II API. Two algebra-like laws are already available, but neither is primitive algebra:

- additivity is realized by an exact rectangle dissection;
- exchange of the two containing segments is realized by cyclic rotation of a rectangle.

No multiplication of segment lengths and no numerical area function occurs in the theorem.

2.2 Euclid's Statement and Classical Configuration

2.2.1 The statement

In Heath's wording, Proposition II.2 states:

> If a straight line be cut at random, the rectangle contained by the whole and both of the segments is equal to the square on the whole.

Here "both of the segments" means the two rectangles obtained by pairing the whole line with each of the two parts. If AB is cut at C, the geometric statement is therefore

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \;=\; \mathrm{Sq}(AB).$$

The notation is documentary shorthand only. It does not introduce a number-valued rectangle function.

2.2.2 The classical diagram

Euclid takes a straight line AB cut at C, constructs the square $ADEB$ on AB by I.46, and through C draws CF parallel to the two opposite sides AD and BE by I.31 [1]. The square is thereby partitioned into two rectangles. One is contained by the whole AB and the segment AC; the other is contained by the whole AB and the segment CB.

The current Lean configuration uses different letters:

classical role	Lean role
AB	BC
C	M
$ADEB$	$BCED$
CF	ML

Thus $B - M - C$, the outer figure $BCED$ is a square, and the cut ML divides it into the left and right rectangular pieces.

Figure 2.1 shows both the configured cut and the separate representation change used by the formal proof.

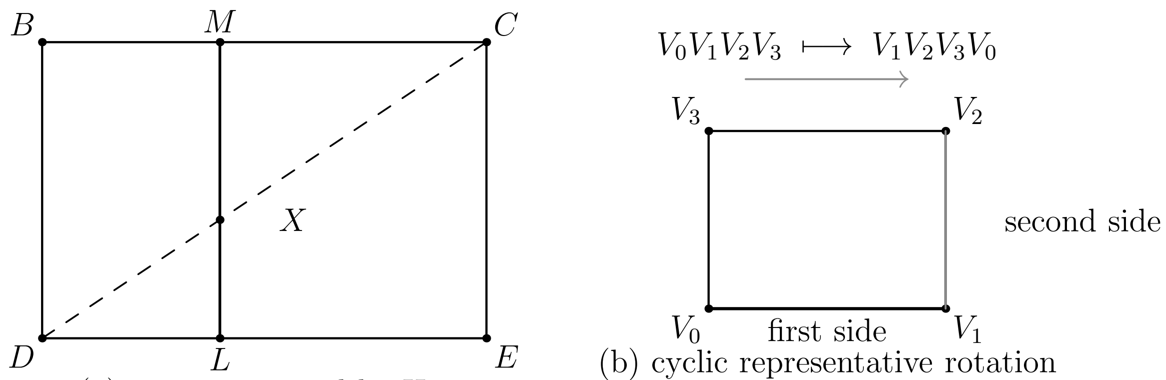


Figure 2.1: Two mechanisms used in the formal II.2 proof. Panel (a) is the instantiated square $BCED$ with $B - M - C$ and the cut ML; the dashed diagonal DC meets the cut at the hidden witness X, so the formal data include both $D - X - C$ and $M - X - L$. Panel (b) shows the cyclic presentation change $V_0V_1V_2V_3 \mapsto V_1V_2V_3V_0$, which exchanges the two adjacent side roles while preserving exact scissors content.

2.3 Classical Proof

The classical proof has only four mathematical steps.

First, construct the square on the whole segment. Second, draw through the point of section a line parallel to the appropriate pair of sides of the square. Third, observe that this line partitions the square into two rectangles. Finally, identify those rectangles by Book II Definition 1: their adjacent sides are the whole segment together with the two parts of the divided segment.

In the standard lettering this gives

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) = \mathrm{Sq}(AB).$$

Euclid does not invoke II.1 in this proof. The local classical dependency graph is instead

$$\begin{gathered}\text{I.46: construct the square on } AB\\ \Downarrow\\ \text{I.31: draw the parallel through } C\\ \Downarrow\\ \text{II.Def.1: identify the two rectangles}\\ \Downarrow\\ \text{the two pieces constitute the square.}\end{gathered}$$

This fact is important for documentation. The formal theorem proves the same geometric identity, but it is not a line-by-line formalization of Euclid's historical derivation.

2.4 The Geometric Identity

In the current Lean lettering the square side is BC and the cut point is M. The two Euclidean rectangles are therefore

$$\mathrm{Rect}(BC, BM), \qquad \mathrm{Rect}(BC, MC),$$

and the theorem asserts the exact scissors relation

$$\mathrm{Rect}(BC, BM) + \mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Sq}(BC).$$

A modern algebraic shadow is obtained by writing $x = y + z$:

$$x^2 = xy + xz.$$

This is useful as a mnemonic for the pattern, but it is not the object proved by Lean and it is not a claim that Euclid has defined multiplication on segment lengths. The formal terms remain rectangle representatives and their exact scissors relation.

2.5 Reusable Book II Infrastructure

II.2 imports

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction. Its proof consumes four pieces of the existing Book II infrastructure.

2.5.1 A square as a rectangle contained by the same side twice

From

```
hSquare : IsSquare Geo B C E D
```

the theorem

```
square_is_rectangle_contained_by_side
```

produces

```
IsRectangleContainedBy Geo B C E D B C B C
```

so the concrete square is recognized synthetically as a rectangle contained by BC, BC.

2.5.2 The binary form of II.1

The reusable distributive kernel is

```
euclid_proposition_2_1_two
```

whose geometric content is

$$\mathrm{Rect}(BC, PQ) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

when $B - M - C$ and the corresponding rectangular cut data are supplied.

For II.2 the fixed segment is specialized to $PQ = BC$. Thus the right-hand pieces naturally arrive in the order

$$\mathrm{Rect}(BM, BC), \qquad \mathrm{Rect}(MC, BC).$$

2.5.3 Exchanging the containing-side order

Euclid's statement uses the opposite order:

$$\mathrm{Rect}(BC, BM), \qquad \mathrm{Rect}(BC, MC).$$

The current proof does not treat this as "commutativity of multiplication." It calls

```
rectangle_contained_by_swap
```

on each arbitrary representative.

The theorem rotates a rectangle

$$A\,B\,C\,D \longmapsto B\,C\,D\,A,$$

returns the corresponding `IsRectangleContainedBy` fact with the two containing segments exchanged, and simultaneously returns a `HilbertScissorsEq` relating the two cyclic presentations.

The higher-level theorem `rectangle_contained_by_comm` belongs to the same rectangle API, but the public II.2 proof uses `rectangle_contained_by_swap` directly because its returned data are exactly the swapped representatives and exact scissors transports needed for the subsequent II.1 call.

2.6 Proof Architecture of the Reconstruction

The Lean proof has four stages.

2.6.1 Step 1: recognize the square

The square hypothesis is converted to the "contained by" language:

```
have hSquareContained :
    IsRectangleContainedBy Geo
      B C E D B C B C :=
  square_is_rectangle_contained_by_side
    Geo B C E D hSquare
```

The rectangle structure and the side congruence $CE \cong BC$ are then extracted from this fact.

2.6.2 Step 2: rotate the two arbitrary part representatives

The two hypotheses supplied in Euclid's order are

```
hPart1 :
  IsRectangleContainedBy Geo
    V0 V1 V2 V3 B C B M

hPart2 :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 B C M C
```

The current source then applies `rectangle_contained_by_swap` separately to `hPart1` and `hPart2`. Each call returns a swapped `IsRectangleContainedBy` fact and an exact scissors equality for the cyclically rotated representative. The resulting representatives are $V_1V_2V_3V_0$ and $W_1W_2W_3W_0$.

The mathematical result is that the rotated representatives now have the side orders required by the II.1 binary theorem.

2.6.3 Step 3: apply II.1 with fixed segment BC

The central call is

```
have hSplit :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B C E D)
      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
       hilbertParallelogramTerm Geo W1 W2 W3 W0) :=
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    B C
    hRect
```

```
    hCE_BC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    B C E D
    V1 V2 V3 V0
    W1 W2 W3 W0
    hSquareContained
    hPart1Swap
    hPart2Swap
```

This is the whole mathematical core of the reconstruction:

$$\mathrm{Sq}(BC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, BC) + \mathrm{Rect}(MC, BC).$$

2.6.4 Step 4: return to Euclid's ordering

The two exact rotation equalities are added:

```
have hPartsBack :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V1 V2 V3 V0 +
       hilbertParallelogramTerm Geo W1 W2 W3 W0)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate1)
    (HilbertScissorsEq.symm
      (Geo := Geo) hRotate2)
```

Transitivity composes `hSplit` with `hPartsBack`, and the final symmetry reverses the result into the order used by the public theorem: sum of the two Euclidean rectangles on the left, square on the right.

2.7 Lean Formalization

The current public theorem is the following instantiated diagram theorem:

```
theorem euclid_proposition_2_2
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (B C D E L M X : Geo.Point)

    (hSquare : IsSquare Geo B C E D)

    (hBMC : Geo.Between B M C)
```

```
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    (V0 V1 V2 V3 : Geo.Point)
    (W0 W1 W2 W3 : Geo.Point)

    (hPart1 :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B C B M)

    (hPart2 :
      IsRectangleContainedBy Geo
        W0 W1 W2 W3 B C M C) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo W0 W1 W2 W3)
      (hilbertParallelogramTerm Geo B C E D) := by
```

Three aspects of the signature are mathematically significant.

First, the theorem assumes an instantiated square and cut diagram. It is not an existence theorem constructing the square and the parallel cut from only a segment and a cut point. Euclid performs those constructions in the classical proof; the present public Lean theorem starts after they have been supplied.

Second, the two part rectangles are arbitrary representatives satisfying `IsRectangleContainedBy`. Their exact placement in the plane is not fixed by the statement.

Third, the conclusion is the exact relation `HilbertScissorsEq`. It is not merely a conclusion in the more general `HilbertScissorsEquicomplementable` relation. II.2 remains at the exact-dissection level because its proof is ultimately the binary exact split of II.1 plus exact representation transports.

2.8 Relationship with Proposition II.1

The contrast between the classical and formal dependency graphs is especially clear here.

Euclid's II.2 is proved directly from the square and the parallel cut. The current Lean file instead begins with

```
import CGJteamLab.Proposition2_1
```

and the public proof calls

```
euclid_proposition_2_1_two
```

once.

Thus II.2 is an explicit formal corollary of the binary II.1 API. No new dissection lemma is proved. The only additional mathematical work is conversion between the two possible orders of the containing segments.

This reuse also explains why the symmetric rectangle interface added at II.1 is not cosmetic. Without the side-order transport, the II.1 theorem would produce perfectly correct rectangles whose formal orientation does not match the language of II.2.

2.9 Relationship with Book I Infrastructure

The direct import is II.1, not a Book I proposition. Nevertheless the inherited chain underneath II.1 contains the Book I geometry on which the rectangle calculus rests.

The exact binary split is built over the scissors decomposition infrastructure developed around I.47. Rectangle structure uses Euclidean parallelogram theory and synthetic right angles; the square hypothesis uses the square infrastructure established in Book I. Consequently the formal proof of II.2 does not replay I.31 and I.46 as Euclid does. Those classical constructions have been absorbed into the established geometric API and into the hypotheses of the instantiated theorem.

2.10 Formalization Notes and Hidden Geometric Data

The visible square cut hides several obligations that are explicit parameters of the current theorem:

- $B - M - C$, fixing the orientation of the cut on the square side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the cut with the diagonal used by the inherited exact scissors split;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- two arbitrary representatives of the Euclidean part rectangles;
- two cyclic representative changes, each accompanied by an exact scissors equality.

The point X has no role in the elementary verbal statement. It belongs to the current exact-dissection interface inherited from the triangulated scissors proof. It is therefore documented as hidden formal geometry rather than presented as part of Euclid's original construction.

There is also an important negative observation. No `SameRay`, `SameSide`, or `OppositeSide` obligation appears in `Proposition2_2.lean` itself. The corresponding low-level geometry has already been discharged inside the reusable rectangle and II.1 theorems.

2.11 Geometry, Content, Representation, and Later Arithmetic

2.11.1 Geometry

The geometric layer consists of a square, betweenness on two opposite sides, the diagonal/cut witness, and two parallelogram pieces. Congruence of the square side with itself identifies the square as a rectangle contained by BC, BC.

2.11.2 Content / scissors

The conclusion and every transport used in the proof are exact `HilbertScissorsEq` relations. The proof uses symmetry, transitivity, and addition of exact scissors equalities. It does not weaken to the more general equicomplementability relation.

2.11.3 Representation

Representation is the distinctive issue in II.2. The theorem accepts arbitrary rectangles for

$$\mathrm{Rect}(BC, BM) \quad \text{and} \quad \mathrm{Rect}(BC, MC).$$

The binary II.1 theorem wants the factors in the reverse order. Cyclic rotation of each representative supplies the required conversion while preserving exact scissors content.

2.11.4 Later arithmetic / numerical area

Neither layer is used. In particular, the proof does not define a segment product, does not invoke Hilbert's later segment field, and does not assign real areas to the square or rectangles.

2.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore works in the same Euclidean project layer as II.1. No new proposition-specific geometric axiom or content axiom is declared in `Proposition2_2.lean`. Inspection of the current 162-line source shows one theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as "axiom-free." The theorem inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output was supplied with the source used for this documentation pass, so the transitive axiom list is not reproduced here.

The implementation checkpoint records that the II.2 Lean build had already been checked during theorem development. This documentation workspace does not contain the complete Lean project, so the Lean build is not re-run here; the LaTeX and Asymptote checks are reported separately.

2.13 Hartshorne's Semantic Control

Hartshorne places the end of Book I and all of Book II inside Euclid's theory of area, interpreted synthetically as equal content rather than as an unexplained numerical area function [5]. He also describes Book II as part of an "algebra of areas": rectangles can play a product-like role because Euclid has no operation that multiplies two line segments to produce another segment.

II.2 fits that interpretation exactly, provided the levels are kept separate. The square on the whole and the two rectangles are geometric figures; their equality is a content statement; the modern identity $x^2 = xy + xz$ is only an interpretive shadow.

The current Lean theorem is locally stronger than the weakest equal-content reading because its configured proof stays at exact scissors equality. That strength comes from the explicit binary dissection inherited from II.1, not from a general principle identifying every equal-content pair with an exact dissection.

2.14 Hilbert's Layering

Hilbert supplies a second semantic guardrail. His theory separates synthetic plane content from the later arithmetic of segments. Segment multiplication is introduced only in a later arithmetic construction, with a fixed unit segment and parallel geometry; it is not primitive rectangle notation.

The II.2 proof remains entirely on the earlier side of this boundary:

$$\begin{gathered}\text{square and rectangle geometry}\\ \Downarrow\\ \text{exact scissors dissection}\\ \Downarrow\\ \text{representative rotation and transport.}\end{gathered}$$

The apparent commutativity required by the formula $xy = yx$ is realized geometrically by `rectangle_contained_by_swap`, not imported from Hilbert's later segment arithmetic.

2.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{gathered}\text{Hilbert incidence + Euclidean plane layer}\\ \Downarrow\\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure}\\ \Downarrow\\ \texttt{HilbertRectangle}: \texttt{IsRectangleContainedBy}, \texttt{rectangle\_contained\_by\_swap}\\ \Downarrow\\ \text{Proposition II.1 binary API}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_1\_two}\\ \Downarrow\\ \text{two representative rotations + exact scissors addition/transitivity}\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_2}.\end{gathered}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{square cut into two rectangles} \longrightarrow \text{II.2.}$$

New geometric axiom in II.2:	no
New content axiom in II.2:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.2:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Direct side-order transport used:	yes, `rectangle_contained_by_swap`
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local `sorry`/`admit`:	no

2.16 Conclusion

Proposition II.2 is mathematically a square cut into two rectangles, but formally it does more than repeat the picture. It demonstrates that the II.1 rectangle calculus is already reusable: the binary dissection theorem supplies additivity, and cyclic rectangle rotation reconciles the two possible orders of the containing sides.

This is also the first sharp divergence between Euclid's local proof graph and the formal Book II graph. Euclid proves II.2 directly from I.46 and I.31. Lean uses II.1 as a library theorem and turns II.2 into a short transport-and-reuse argument. The theorem is therefore not a literal transcription of the classical proof, but it remains entirely synthetic and geometric.

The result can be remembered by the modern shadow

$$x^2 = xy + xz, \qquad x = y + z,$$

but the formal content is more precise: arbitrary representatives of the two rectangles contained by the whole side and the two parts are exactly scissors-equal, in sum, to the concrete square on the whole side. No numerical area and no multiplication of segment lengths is required.

Chapter 3

Proposition II.3

3.1 Introduction

Proposition II.3 is the second immediate reuse of the rectangle calculus established in Proposition II.1. Its classical form is again a statement about a straight line cut at an arbitrary point, but now one of the two resulting segments plays a distinguished role. The rectangle contained by the whole line and that segment is decomposed into a rectangle contained by the two parts and the square on the distinguished segment.

The current Lean reconstruction makes this structure unusually transparent. Its only direct import is the II.1 module:

```
import CGJteamLab.Proposition2_1
```

The proof first recognizes the supplied square on MC as a rectangle contained by MC, MC, and then applies the binary form of II.1 with the fixed segment specialized to MC. No new dissection theorem, side-order swap, or proposition-local helper theorem is needed.

This is an important contrast with II.2. There the Euclidean order of the two containing segments did not match the order produced by the binary II.1 theorem, so cyclic rectangle rotation was required. In II.3 the order already matches:

$$\mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, MC) + \mathrm{Rect}(MC, MC).$$

The square on MC is then simply a chosen representative of the final $\mathrm{Rect}(MC, MC)$ term.

Thus II.3 is not formalized through multiplication of segment lengths or through a numerical area function. Its apparently algebraic shape is realized by an exact geometric cut together with representative independence in the rectangle API.

3.2 Euclid's Statement and Classical Configuration

3.2.1 The statement

In Heath's wording, Proposition II.3 states:

> If a straight line be cut at random, the rectangle contained by the whole and one of

> the segments is equal to the rectangle contained by the segments and the square on the aforesaid segment.

If AB is cut at C and the distinguished part is BC, the geometric identity is

$$\mathrm{Rect}(AB, BC) = \mathrm{Rect}(AC, CB) + \mathrm{Sq}(BC).$$

As throughout the Book II documentation, Rect and Sq are expository shorthand for geometric figures and their content. They are not number-valued functions.

3.2.2 The classical diagram

Euclid constructs the square $CDEB$ on the distinguished segment CB by Proposition I.46. The side ED is produced, and through A a line is drawn parallel to the corresponding sides of the square by Proposition I.31 [1]. The resulting larger rectangle has adjacent side magnitudes AB and BC. The line through C separates it into a rectangle with side magnitudes AC, CB and the square on CB.

The current Lean lettering is related to the classical roles by

classical role	Lean role
AB	BC
C	M
AC	BM
CB	MC

so the formal cut relation is $B - M - C$ and the distinguished segment is MC. The configured outer rectangle is $BCED$, with $CE \cong MC$, while the cut through M meets the opposite side at L.

Figure 3.1 shows the geometry used by the reconstruction. The rectangle $BCED$ is drawn below the segment BC. Its right-hand piece has width MC and height congruent to MC, so it realizes the square-like $\mathrm{Rect}(MC, MC)$ part of the dissection. The supplied square $MCFG$ is drawn on the opposite side of MC to emphasize a formal point: the conclusion may use a square representative different from the concrete piece created by the cut.

3.3 Classical Proof

Let AB be cut at C, and choose BC as the segment named in the proposition. Construct the square $CDEB$ on CB by I.46. Produce one side of the square and through A draw a line parallel to the opposite sides by I.31. This forms a larger rectangle whose adjacent side magnitudes are AB and BC.

The line through C partitions that larger rectangle into two pieces. The first has adjacent side magnitudes AC and CB, and the second is precisely the square on CB. Hence

$$\mathrm{Rect}(AB, BC) = \mathrm{Rect}(AC, CB) + \mathrm{Sq}(BC).$$

This is Euclid's entire content argument: the whole rectangle is constituted by the two displayed subfigures.

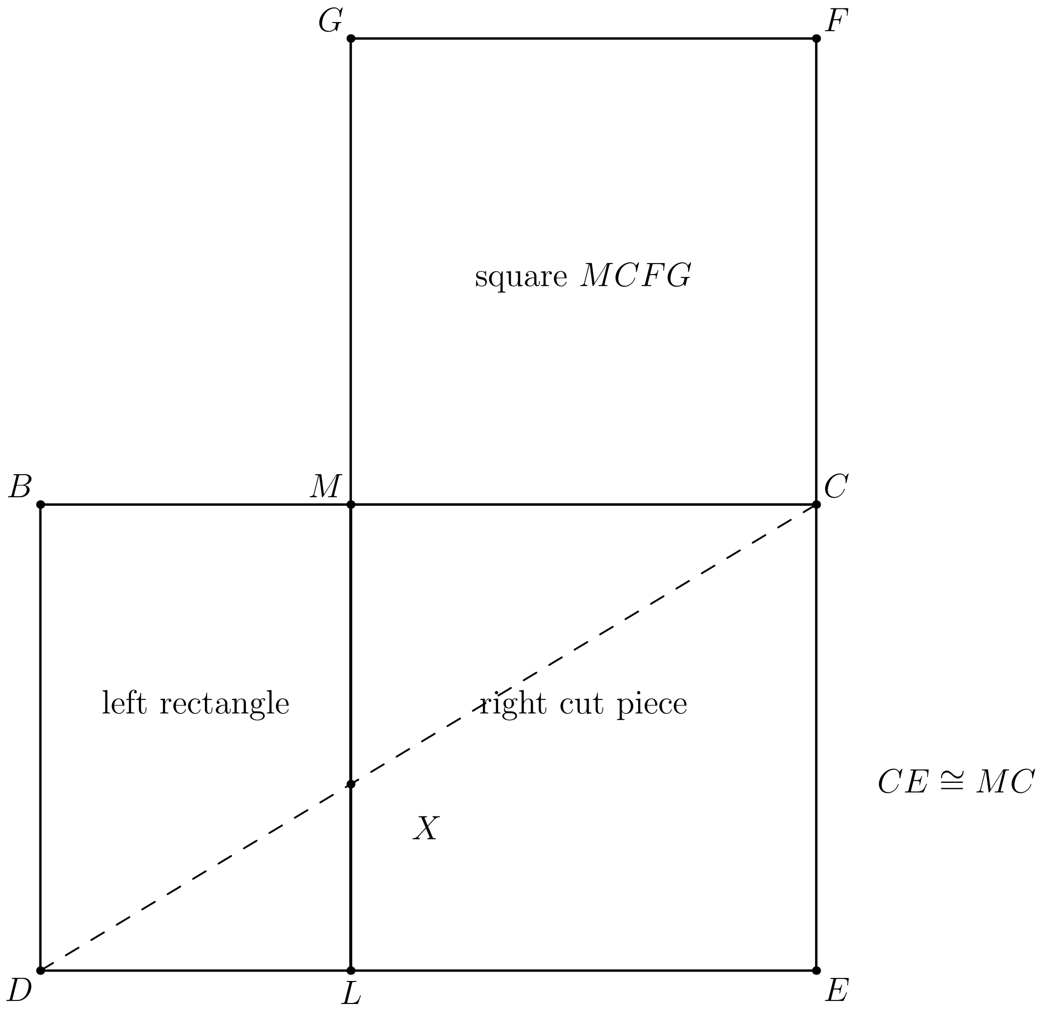


Figure 3.1: The geometric mechanism of II.3. The outer rectangle $BCED$ has height $CE \cong MC$ and is split by ML at the point $B - M - C$. The dashed diagonal DC meets the cut at the hidden witness X. The right-hand cut piece represents $\mathrm{Rect}(MC, MC)$; the separately drawn square $MCFG$ is the square representative used in the public theorem.

The local classical dependency graph is therefore

$$
\begin{gathered}
\text{I.46: construct the square on } CB \\
\Downarrow \\
\text{I.31: draw the required parallel through } A \\
\Downarrow \\
\text{Book II Definition 1: identify the rectangles} \\
\Downarrow \\
\text{whole rectangle} = \text{rectangle} + \text{square} \\
\Downarrow \\
\text{II.3.}
\end{gathered}
$$

Euclid does not cite Proposition II.1 in this proof. The current Lean proof, by contrast, is explicitly organized as a corollary of the reusable II.1 binary theorem. The classical and formal dependency graphs therefore prove the same geometric identity through different library organizations.

3.4 The Geometric Identity

In the current Lean lettering, $B - M - C$. The theorem fixes the segment MC and applies it to the whole BC and the two parts BM, MC. Its exact geometric content is

$$\mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, MC) + \mathrm{Sq}(MC).$$

A modern algebraic shadow is obtained by writing

$$x = y + z,$$

where x corresponds to BC, y to BM, and z to MC. Then the pattern is

$$xz = yz + z^2,$$

or equivalently

$$(y + z)z = yz + z^2.$$

This formula is useful only as a mnemonic. The Lean proof never forms the products xz or yz as segment-arithmetic objects. The actual objects are rectangle representatives and a square, related by exact scissors equality.

3.5 Reusable Book II Infrastructure

The current file begins with

```
import CGJteamLab.Proposition2_1
```

and introduces no new rectangle abstraction.

3.5.1 The binary form of II.1

The reusable theorem is

```
euclid_proposition_2_1_two
```

with geometric content

$$\mathrm{Rect}(BC, PQ) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

whenever $B - M - C$ and the required rectangular cut data are supplied.

For II.3 the fixed segment is specialized directly to

$$PQ = MC.$$

Therefore II.1 returns

$$\mathrm{Rect}(BC, MC) \simeq_{\mathrm{sc}} \mathrm{Rect}(BM, MC) + \mathrm{Rect}(MC, MC),$$

which already has exactly the containing-side order required by the public II.3 theorem.

3.5.2 A square as $\mathrm{Rect}(MC, MC)$

The square hypothesis

```
hSquare : IsSquare Geo M C F G
```

is converted by

```
square_is_rectangle_contained_by_side
```

into

```
IsRectangleContainedBy Geo M C F G M C M C
```

so the supplied square is a concrete representative of the final $\mathrm{Rect}(MC, MC)$ term.

3.5.3 What II.3 does not use

Unlike II.2, the current proof does not call

```
rectangle_contained_by_swap
rectangle_contained_by_comm
```

The absence is mathematically informative. The binary II.1 theorem already places BM and MC first and the fixed segment MC second, exactly as II.3 needs. No geometric analogue of exchanging factors is required.

3.6 Proof Architecture of the Reconstruction

The public proof has only two mathematical stages.

3.6.1 Step 1: recognize the square

The first local fact is

```
have hSquareContained :
    IsRectangleContainedBy Geo
      M C F G M C M C :=
  square_is_rectangle_contained_by_side
    Geo M C F G hSquare
```

This moves from the square-specific geometric predicate to the general "rectangle contained by" language used by II.1.

3.6.2 Step 2: instantiate II.1 with the fixed segment MC

The remainder of the proof is one direct call:

```
exact
  euclid_proposition_2_1_two
    Geo
    B C D E L M X
    M C
    hRect
    hCE_MC
    hBMC
    hDLE
    hDXC
    hMXL
    hLeftPar
    hRightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    M C F G
    hWhole
    hProduct
    hSquareContained
```

No scissors transitivity, addition, symmetry, or representative rotation is performed locally in II.3. Those mechanisms have already been packaged inside the representative-independent binary theorem of II.1.

This makes the formal DAG exceptionally short:

$$\begin{gathered}\text{square } MCFG \\ \Downarrow \\ \texttt{square\_is\_rectangle\_contained\_by\_side} \\ \Downarrow \\ \mathrm{Rect}(MC, MC) \text{ representative} \\ \Downarrow \\ \texttt{euclid\_proposition\_2\_1\_two} \text{ with } PQ = MC \\ \Downarrow \\ \texttt{euclid\_proposition\_2\_3}.\end{gathered}$$

3.7 Lean Formalization

The current public theorem is:

```
theorem euclid_proposition_2_3
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (B C D E L M X : Geo.Point)
    (hRect : IsRectangle Geo B C E D)
    (hCE_MC : Geo.Congruent C E M C)

    (hBMC : Geo.Between B M C)
```

```
    (hDLE : Geo.Between D L E)
    (hDXC : Geo.Between D X C)
    (hMXL : Geo.Between M X L)
    (hLeftPar :
      IsParallelogram Geo L D B M)
    (hRightPar :
      IsParallelogram Geo C E L M)

    (F G : Geo.Point)
    (hSquare : IsSquare Geo M C F G)

    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)

    (hWhole :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 B C M C)

    (hProduct :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 B M M C) :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo V0 V1 V2 V3 +
       hilbertParallelogramTerm Geo M C F G) := by
```

Several features of this signature are mathematically significant.

First, the theorem assumes a configured outer rectangle and its cut. It does not construct that diagram from the bare data $B - M - C$. Euclid performs the square and parallel constructions in his proof; the formal theorem receives the corresponding geometry as explicit hypotheses.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of

$$\mathrm{Rect}(BC, MC) \quad \text{and} \quad \mathrm{Rect}(BM, MC).$$

The conclusion is therefore not restricted to the literal pieces drawn in the configured cut.

Third, the square $MCFG$ is concrete, but the II.1 representative-independence machinery permits it to stand for the $\mathrm{Rect}(MC, MC)$ piece even when it is not the internal right-hand rectangle of the supplied cut diagram.

Fourth, the conclusion is exactly

```
HilbertScissorsEq
```

rather than the more general relation

```
HilbertScissorsEquicomplementable
```

and it is not equality of real numbers.

3.8 Relationship with Earlier Book II Propositions

3.8.1 Proposition II.1

II.3 is a direct formal corollary of II.1. The import is

```
import CGJteamLab.Proposition2_1
```

and the only substantial theorem called in the final proof is

```
euclid_proposition_2_1_two
```

The specialization $PQ = MC$ converts the general binary distributive pattern into II.3 immediately.

The relationship can be written schematically as

$$\mathrm{Rect}(BC, PQ) = \mathrm{Rect}(BM, PQ) + \mathrm{Rect}(MC, PQ)$$

followed by

$$PQ := MC.$$

The final term then becomes $\mathrm{Rect}(MC, MC)$, represented by the square on MC.

3.8.2 Proposition II.2

The current Lean file does *not* import Proposition II.2, and its public theorem is not used in the proof. II.2 and II.3 are therefore sibling corollaries of the II.1 binary API rather than a formal chain II.1 $\to$ II.2 $\to$ II.3.

The contrast is nevertheless useful. II.2 required cyclic representative rotation because its chosen side order was opposite to the order returned by II.1. II.3 chooses the fixed segment MC in precisely the position expected by II.1, so even that transport disappears. This is evidence that the II.1 API is not merely sufficient but well aligned with the geometry of the next Book II identities.

3.9 Relationship with Book I Infrastructure

Euclid's local proof explicitly calls I.46 to construct the square and I.31 to draw the parallel. The current Lean theorem does not replay those constructions. Instead it assumes the resulting rectangle, cut, parallelograms, side congruence, and square as data.

Below II.1, however, the formal dependency chain still rests on the Book I infrastructure for rectangles, right angles, squares, parallelograms, and exact scissors decomposition. The binary split ultimately uses the exact rectangle cut machinery developed over the Book I scissors layer. Thus the difference is one of modular organization, not a change from synthetic to numerical geometry.

3.10 Formalization Notes and Hidden Geometric Data

The elementary picture hides several obligations that appear explicitly in the current theorem signature:

- $B - M - C$, fixing the cut point and its orientation on the whole side;
- $D - L - E$, locating the opposite endpoint of the cut;
- $D - X - C$ and $M - X - L$, recording the intersection X of the diagonal and the cut line used by the inherited exact scissors theorem;
- `IsParallelogram Geo L D B M` for the left piece;
- `IsParallelogram Geo C E L M` for the right piece;
- $CE \cong MC$, making the outer rectangle have the required fixed second side magnitude;
- a supplied square $MCFG$;
- arbitrary representatives of the whole rectangle and the BM-by-MC product rectangle.

The point X is invisible in Euclid's verbal statement. It belongs to the triangulated exact-dissection interface inherited from II.1. It should therefore be read as hidden formal geometry rather than as an additional step in Euclid's classical construction.

There are also useful negative observations. No `SameRay`, `SameSide`, or `OppositeSide` hypothesis occurs in `Proposition2_3.lean` itself, and no side-order swap is performed. Those lower-level obligations have already been absorbed by the reusable rectangle and II.1 layers.

3.11 Geometry, Content, Representation, and Later Arithmetic

3.11.1 Geometry

The geometric layer consists of the rectangle $BCED$, the cut point M, the opposite cut point L, the diagonal witness X, the two parallelogram pieces, the congruence $CE \cong MC$, and the square $MCFG$.

3.11.2 Content / scissors

The conclusion is the exact relation `HilbertScissorsEq`. The public proof does not weaken the result to general equicomplementability. Exactness is inherited from the binary rectangle split and representative transports already proved in II.1.

3.11.3 Representation

Representation is present but simpler than in II.2. The theorem accepts an arbitrary rectangle representative for $\mathrm{Rect}(BC, MC)$ and an arbitrary representative for $\mathrm{Rect}(BM, MC)$, together with a concrete square on MC. The binary II.1 theorem is already independent of the particular concrete rectangles appearing in the cut, so no additional representation transport is required locally.

3.11.4 Later arithmetic / numerical area

Neither layer is used. No segment product is defined, no fixed unit segment is introduced, and no real-valued area function appears. The expression $xz = yz + z^2$ is only the modern algebraic shadow of a geometric scissors identity.

3.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore lives in the same Euclidean project layer as II.1 and II.2. Inspection of the current 96-line source shows one public theorem declaration and no proposition-local `axiom`, `sorry`, or `admit`.

This should not be described as "axiom-free." The result inherits the axioms and typeclass assumptions of the established Hilbert/Euclidean/scissors library transitively. No current `#print axioms` output for II.3 was supplied with the documentation sources, so a transitive axiom list is not asserted here.

The implementation checkpoint records that the II.3 Lean theorem had already been build-checked during formalization. The present documentation pass uses the current source as authoritative but does not contain the full Lean project needed to repeat that build locally.

3.13 Hartshorne's Semantic Control

Hartshorne's general analysis of Euclid's area theory states that the results of Book II are valid when Euclid's equality of figures is read as equality of content [5]. His condensed list of selected Book II propositions does not separately single out II.3, so the source should not be cited as providing a special proof of this proposition.

The general semantic point is sufficient. II.3 compares a rectangle with the union of another rectangle and a square. In the present reconstruction this comparison is sharpened, for this configured case, to exact scissors equality. The modern identity

$$xz = yz + z^2$$

therefore belongs to Hartshorne's "algebra of areas" only as an interpretive shadow of operations on geometric figures.

II.3 also clarifies the representative issue. The term called "the square on MC" need not be the literal internal square-shaped piece of the cut; the formal theorem accepts the separately supplied square $MCFG$ because the rectangle calculus already proves independence of representatives at the scissors level.

3.14 Hilbert's Layering

Hilbert provides the complementary warning against reading later segment arithmetic into the proof. His synthetic theory of plane content distinguishes finite equidecomposition from the more general content relations, while his arithmetic of segments is introduced later and requires a fixed unit segment and additional parallel constructions [2].

II.3 remains entirely below that arithmetic boundary. Its formal chain is

$$\begin{gathered}\text{rectangle and square geometry}\\ \Downarrow\\ \text{exact binary dissection from II.1}\\ \Downarrow\\ \text{representative-independent exact scissors equality.}\end{gathered}$$

In fact II.3 is even more direct than II.2: no geometric commutativity theorem is needed because the ordered containing segments already agree with the II.1 binary interface.

3.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{gathered}\text{Hilbert incidence + Euclidean plane layer}\\ \Downarrow\\ \text{Book I right-angle, square, parallelogram, and scissors infrastructure}\\ \Downarrow\\ \texttt{HilbertRectangle}\\ \texttt{IsRectangleContainedBy}\\ \texttt{square\_is\_rectangle\_contained\_by\_side}\\ \Downarrow\\ \text{Proposition II.1 binary API}\\ \texttt{euclid\_proposition\_2\_1\_two}\\ \text{specialized by } PQ = MC\\ \Downarrow\\ \texttt{euclid\_proposition\_2\_3.}\end{gathered}$$

The classical local chain is different:

$$\text{I.46} \longrightarrow \text{I.31} \longrightarrow \text{rectangle cut into rectangle + square} \longrightarrow \text{II.3.}$$

New geometric axiom in II.3:	no
New content axiom in II.3:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.3:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses an earlier Book II proposition:	yes, II.1
Imports II.2:	no
Direct side-order transport used:	no
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem build:	previously checked; not re-run here
Full relevant Lean build:	previously checked; not re-run here
Proposition-local `sorry/admit`:	no

3.16 Conclusion

Proposition II.3 is one of the cleanest tests of the Book II rectangle API. Classically, Euclid constructs a square on one segment and enlarges it by a parallel construction until the whole rectangle contained by the original line and that segment is obtained. Formally, the same identity is already contained in the binary theorem of II.1.

The decisive specialization is simply
$$PQ = MC.$$

It turns the second part rectangle of II.1 into $\mathrm{Rect}(MC, MC)$, and the supplied square on MC provides exactly that representative. Nothing else is needed.

This also explains the precise relation between II.2 and II.3. Both are formal corollaries of II.1, but II.2 needs geometric side-order transport while II.3 does not. The result is therefore a very short Lean proof without being an algebraic proof in the modern numerical sense: the content remains rectangles, squares, congruent sides, and exact scissors equality throughout.

Chapter 4

Proposition II.4

4.1 Introduction

Proposition II.4 is the first proposition of Book II in the present reconstruction whose formal proof is naturally expressed as a composition of *two different earlier Book II propositions.* The classical theorem says that when a segment is cut into two parts, the square on the whole is equal to the two squares on the parts together with twice the rectangle contained by the parts.

The modern algebraic shadow is the familiar identity

$$(y+z)^2 = y^2 + z^2 + 2yz.$$

The formal proof, however, does not define a product of segment lengths and does not evaluate a numerical area. It remains entirely inside the rectangle and exact-scissors layer established at the beginning of Book II.

The current Lean source begins with

```
import CGJteamLab.Proposition2_2
import CGJteamLab.Proposition2_3
```

and the proof uses those imports exactly as the architecture suggests:

1. Proposition II.2 splits the square on the whole segment into two rectangles pairing the whole with the two parts;

2. Proposition II.3 expands each of those rectangles into one square and one cross rectangle;

3. representation transport identifies the two cross rectangles as two representatives of the same "rectangle contained by" the two parts;

4. addition and transitivity in `HilbertScissorsEq` combine the results.

Thus the substantial new fact at II.4 is not a new geometric construction or a new arithmetic law. It is the observation that the Book II rectangle API is already compositional enough to express the square-of-a-sum pattern using previously proved exact scissors theorems.

4.2 Euclid's Statement and Classical Configuration

4.2.1 The statement

In Heath's wording, Proposition II.4 states:

> If a straight line be cut at random, the square on the whole is equal to the squares on the segments and twice the rectangle contained by the segments.

Let AB be cut at C. In the geometric shorthand used throughout this volume, the statement is

$$\mathrm{Sq}(AB) = \mathrm{Sq}(AC) + \mathrm{Sq}(CB) + 2\,\mathrm{Rect}(AC, CB).$$

Here Sq and Rect are expository names for geometric figures and their content. The symbol 2 means that two equal rectangle contents occur in the decomposition; it is not scalar multiplication inside the Lean segment language.

4.2.2 The classical diagram

Euclid describes the square $ADEB$ on AB by I.46 and joins the diagonal BD. Through the cut point C he draws CF parallel to the vertical sides of the square, and through the intersection G of CF with BD he draws HK parallel to AB and DE by I.31 [1]. The square is thereby divided into four subfigures named AG, CK, HF, and GE.

The figure is the classical Euclidean configuration. It is deliberately not presented as the literal Lean input diagram. The formal theorem receives three configured rectangle-cut diagrams inherited from II.2 and II.3, rather than reconstructing this single classical picture point by point.

4.3 Classical Proof

Euclid's proof first identifies the lower-right parallelogram $CGKB$ as a square on CB. Since $CF \parallel AD$, the diagonal BD gives the relevant angle equality by I.29. The square has $BA = AD$, so I.5 gives equal base angles in triangle ADB. Combining these angle equalities and using I.6 yields

$$BC = CG.$$

Opposite sides in the surrounding parallelograms are equal by I.34, so the four sides of $CGKB$ are equal. Right-angle information from the parallels and the square then shows that $CGKB$ is a square on CB [1].

The same reasoning identifies HF as the square on AC. The two remaining subfigures AG and GE are complements about the diagonal of the outer square, and Proposition I.43 gives

$$AG = GE.$$

Moreover AG has adjacent side magnitudes AC and CG, while $CG = CB$. Hence AG is the rectangle contained by AC, CB, and $AG + GE$ is twice that rectangle. Since the four subfigures constitute the whole square $ADEB$, Euclid obtains the required equality [1].

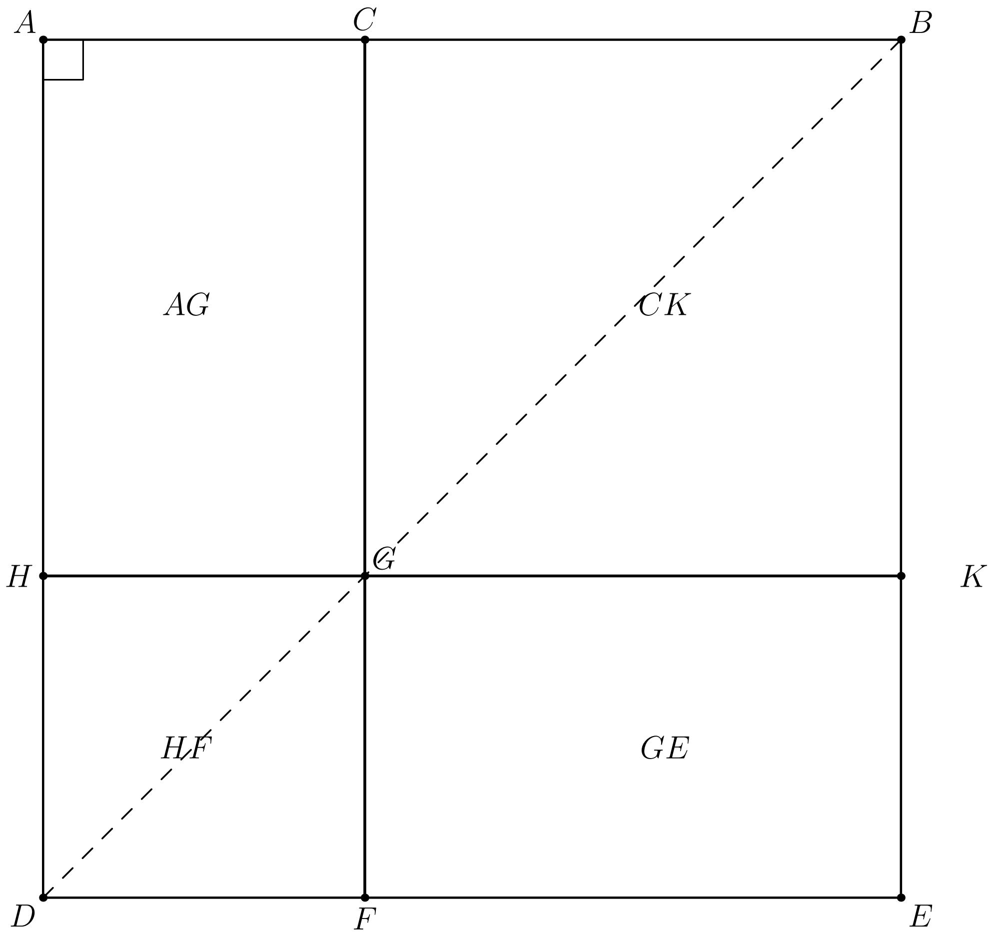


Figure 4.1: The classical configuration of Euclid II.4. The square $ADEB$ is cut at C by CF, the diagonal BD meets it at G, and the parallel HK through G produces four subfigures. Euclid proves that HF and CK are the squares on AC and CB, while the complements AG and GE are equal; AG is the rectangle contained by AC, CB.

The local classical dependency graph is therefore

$$\begin{gathered}
\text{I.46: square on } AB \quad + \quad \text{I.31: required parallels} \\
\Downarrow \\
\text{I.29 + I.5 + I.6: } BC = CG \\
\Downarrow \\
\text{I.34: recognize the two smaller squares} \\
\Downarrow \\
\text{I.43: complements } AG = GE \\
\Downarrow \\
\mathrm{Sq}(AB) = \mathrm{Sq}(AC) + \mathrm{Sq}(CB) + 2\,\mathrm{Rect}(AC, CB) \\
\Downarrow \\
\text{II.4.}
\end{gathered}$$

This classical DAG is important because it is *not* the DAG of the Lean proof. In particular, the formal reconstruction does not locally call I.43 and does not prove the two cross rectangles equal by diagonal complements. That equality has already been absorbed into the representative-independent rectangle calculus developed after Book I.

4.4 The Geometric Identity in the Current Formalization

The public Lean theorem starts with a strict cut

$$A - C - B.$$

It receives a square on AB, two square representatives on the two parts, and two arbitrary representatives of the cross rectangle contained by AC and CB.

At the level of exact scissors terms the final conclusion has the form

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CA)) + (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB)).$$

The use of CA rather than AC in the first square is only an endpoint orientation choice. Segments are unoriented for congruence purposes, so this is the same geometric side magnitude.

The two occurrences of the cross rectangle are deliberately represented by independent quadrilaterals. The theorem therefore says more than "the same literal rectangle occurs twice in one drawing." It says that any two representatives satisfying

```
IsRectangleContainedBy Geo ... A C C B
```

may occupy the two cross-term positions, because the rectangle layer has already proved representative independence at the exact scissors level.

A modern mnemonic is

$$x = y + z \qquad \Longrightarrow \qquad x^2 = y^2 + z^2 + 2yz.$$

This is useful for recognizing the combinatorial pattern only. No object corresponding to yz is constructed by `HilbertSegmentMul` in the II.4 proof.

4.5 Reusable Book II Infrastructure

4.5.1 Proposition II.2: split the square on the whole

The first reusable theorem gives

$$\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \simeq_{\mathrm{sc}} \mathrm{Sq}(AB).$$

In the current source it is called exactly as

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2

--------------------------------------------------------------------
-- II.2:
--
--   Rect(AB,AC) + Rect(AB,CB) = Square(AB).
--------------------------------------------------------------------

have hII2 :
    HilbertScissorsEq Geo
```

```
      (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
       hilbertParallelogramTerm Geo V0 V1 V2 V3)
      (hilbertParallelogramTerm Geo A B E0 D0) :=
  euclid_proposition_2_2
    Geo
    A B D0 E0 L0 C X0
    hSquareAB
    hACB
    hD0L0E0
    hD0X0B
    hCX0L0
    hII2LeftPar
    hII2RightPar
    U0 U1 U2 U3
    V0 V1 V2 V3
    hAB_AC
    hAB_CB
```

The final II.4 proof uses the symmetry of this relation, so the square becomes the starting point and the two whole-by-part rectangles become the two branches to be expanded.

4.5.2 Proposition II.3: expand a whole-by-part rectangle

Proposition II.3 has the generic geometric pattern

$$\mathrm{Rect}(XY, ZY) \simeq_{\mathrm{sc}} \mathrm{Rect}(XZ, ZY) + \mathrm{Sq}(ZY)$$

for a strict division point Z on XY, with the configured rectangle cut supplied explicitly.

II.4 calls this theorem twice. The right branch uses the original orientation $A - C - B$ directly:

$$\mathrm{Rect}(AB, CB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB).$$

The left branch reads the same line in the reverse direction $B - C - A$:

$$\mathrm{Rect}(BA, CA) \simeq_{\mathrm{sc}} \mathrm{Rect}(BC, CA) + \mathrm{Sq}(CA).$$

After endpoint reversal and rectangle rotation, its cross term is another representative of $\mathrm{Rect}(AC, CB)$.

4.5.3 Rectangle rotation and endpoint reversal

The only side-order normalization that is mathematically visible in II.4 is

```
rectangle_contained_by_swap
```

applied to the second cross-rectangle representative. It cyclically rotates the concrete rectangle and exchanges the roles of its containing sides while returning an exact scissors certificate for the change of presentation.

Endpoint reversal is handled by the already established congruence theorem

```
CongruentSwapSecond
```

so that AB may be read as BA, AC as CA, and CB as BC when the reversed II.3 invocation requires that orientation. These steps are representation transport, not new metric statements.

4.5.4 Exact scissors addition

Once the two II.3 branches are available, the constructor

```
HilbertScissorsEq.add
```

combines them componentwise. Transitivity then attaches the resulting four-term expression to the square through the symmetric II.2 equality.

No subtraction, cancellation, positivity, content order, or numerical area operation occurs.

4.6 Proof Architecture of the Reconstruction

4.6.1 Step 1: reverse the cut orientation for the left branch

From $A - C - B$ the proof derives $B - C - A$:

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the only explicit betweenness reversal performed locally.

4.6.2 Step 2: apply II.2 to the square on the whole

The theorem `euclid_proposition_2_2` produces the exact decomposition of the square into the two rectangles $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$. Nothing about the smaller squares is used yet.

4.6.3 Step 3: prepare the reversed left II.3 call

The first II.2 rectangle is formally contained by AB, AC, while the reversed II.3 call expects BA, CA. The same concrete rectangle is repackaged using endpoint reversal. The second final cross-rectangle representative is then rotated from the side order AC, CB to CB, AC and read, after endpoint reversal, as BC, CA.

The current source makes these transports explicit:

```
  have hAB_AC_rev :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 B A C A :=
    And.intro
      hAB_AC.1
      (And.intro
        (CongruentSwapSecond
          Geo U0 U1 A B hAB_AC.2.1)
```

```
      (CongruentSwapSecond
        Geo U1 U2 A C hAB_AC.2.2))

  ---------------------------------------------------------------
  -- Rotate the second cross rectangle:
  --
  --   Rect(AC,CB) -> Rect(CB,AC).
  --
  -- Endpoint reversal then reads the rotated representative as
  -- Rect(BC,CA), exactly what the reversed II.3 needs.
  ---------------------------------------------------------------

  have hSwapResult :=
    rectangle_contained_by_swap
      Geo
      T20 T21 T22 T23
      A C C B
      hCross2

  have hCross2Swap :=
    hSwapResult.1

  have hRotateCross2 :=
    hSwapResult.2

  have hCross2ForLeft :
      IsRectangleContainedBy Geo
        T21 T22 T23 T20 B C C A :=
    And.intro
      hCross2Swap.1
      (And.intro
        (CongruentSwapSecond
          Geo T21 T22 C B hCross2Swap.2.1)
        (CongruentSwapSecond
          Geo T22 T23 A C hCross2Swap.2.2))
```

4.6.4 Step 4: expand the left rectangle by II.3

The reversed invocation gives

$$\mathrm{Rect}(BA, CA) \simeq_{\mathrm{sc}} \mathrm{Rect}(BC, CA) + \mathrm{Sq}(CA).$$

The rotated cross term is then transported back to the arbitrary final AC, CB representative. The exact source block is:

```
  have hLeftRaw :
      HilbertScissorsEq Geo
        (hilbertParallelogramTerm Geo U0 U1 U2 U3)
        (hilbertParallelogramTerm Geo T21 T22 T23 T20 +
         hilbertParallelogramTerm Geo C A F G) :=
    euclid_proposition_2_3
      Geo
      B A D1 E1 L1 C X1
      hRectLeft
      hAE1_CA
```

```
    hBCA
    hD1L1E1
    hD1X1A
    hCX1L1
    hLeftLeftPar
    hLeftRightPar
    F G
    hSquareCA
    U0 U1 U2 U3
    T21 T22 T23 T20
    hAB_AC_rev
    hCross2ForLeft

------------------------------------------------------------------
-- Return the left cross term to the original arbitrary
-- representative T20-T21-T22-T23.
------------------------------------------------------------------

have hRotateCross2Back :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo T21 T22 T23 T20)
      (hilbertParallelogramTerm Geo T20 T21 T22 T23) :=
  HilbertScissorsEq.symm
    (Geo := Geo)
    hRotateCross2

have hLeftTarget :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3)
      (hilbertParallelogramTerm Geo T20 T21 T22 T23 +
       hilbertParallelogramTerm Geo C A F G) :=
  HilbertScissorsEq.trans
    (Geo := Geo)
    hLeftRaw
    (HilbertScissorsEq.add
      (Geo := Geo)
      hRotateCross2Back
      (HilbertScissorsEq.refl
        (Geo := Geo)
        (hilbertParallelogramTerm Geo C A F G)))
```

4.6.5 Step 5: expand the right rectangle by II.3

The right branch needs no reversal of the whole segment. It directly produces

$$\mathrm{Rect}(AB, CB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB).$$

4.6.6 Step 6: add the branches and attach them to the square

The two II.3 relations are added. Finally the result is composed with the symmetric II.2 relation. The closing source is:

```
have hRight :
    HilbertScissorsEq Geo
```

```
      (hilbertParallelogramTerm Geo V0 V1 V2 V3)
      (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
       hilbertParallelogramTerm Geo C B H K) :=
  euclid_proposition_2_3
    Geo
    A B D2 E2 L2 C X2
    hRectRight
    hBE2_CB
    hACB
    hD2L2E2
    hD2X2B
    hCX2L2
    hRightLeftPar
    hRightRightPar
    H K
    hSquareCB
    V0 V1 V2 V3
    T10 T11 T12 T13
    hAB_CB
    hCross1

-------------------------------------------------------------
-- Expand both II.2 summands simultaneously.
-------------------------------------------------------------

have hExpanded :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo U0 U1 U2 U3 +
       hilbertParallelogramTerm Geo V0 V1 V2 V3)
      ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
        hilbertParallelogramTerm Geo C A F G) +
       (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
        hilbertParallelogramTerm Geo C B H K)) :=
  HilbertScissorsEq.add
    (Geo := Geo)
    hLeftTarget
    hRight

-------------------------------------------------------------
-- Square(AB)
--    = Rect(AB,AC) + Rect(AB,CB)
--    = the four terms of II.4.
-------------------------------------------------------------

exact
  HilbertScissorsEq.trans
    (Geo := Geo)
    (HilbertScissorsEq.symm
      (Geo := Geo)
      hII2)
    hExpanded
```

The actual Lean DAG is therefore

$$
\begin{tikzcd}
& \texttt{euclid\_proposition\_2\_2} \arrow[d, Rightarrow] & \\
& \mathrm{Sq}(AB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB) \arrow[dl] \arrow[dr] & \\
\text{reversed II.3} \arrow[dr] & & \text{direct II.3} \arrow[dl] \\
& \texttt{HilbertScissorsEq.add} \arrow[d, Rightarrow] & \\
& \text{two squares + two cross rectangles} \arrow[d, Rightarrow] & \\
& \texttt{euclid\_proposition\_2\_4}. &
\end{tikzcd}
$$

This is substantially shorter conceptually than Euclid's local geometric DAG, but it is not an algebraic shortcut. The earlier propositions already package the relevant geometric dissections as reusable exact-scissors theorems.

4.7 Lean Formalization

The current public theorem signature is copied from the compiling `Proposition2_4.lean` source:

```
theorem euclid_proposition_2_4
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C : Geo.Point)

    ------------------------------------------------------------------
    -- II.2 diagram: square on AB cut at C.
    ------------------------------------------------------------------

    (D0 E0 L0 X0 : Geo.Point)
    (hSquareAB : IsSquare Geo A B E0 D0)

    (hACB : Geo.Between A C B)
    (hD0L0E0 : Geo.Between D0 L0 E0)
    (hD0X0B : Geo.Between D0 X0 B)
    (hCX0L0 : Geo.Between C X0 L0)

    (hII2LeftPar :
      IsParallelogram Geo L0 D0 A C)
    (hII2RightPar :
      IsParallelogram Geo B E0 L0 C)

    -- Arbitrary representatives of Rect(AB,AC) and Rect(AB,CB).
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)

    (hAB_AC :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 A B A C)

    (hAB_CB :
```

```
    IsRectangleContainedBy Geo
      V0 V1 V2 V3 A B C B)

  ------------------------------------------------------------------
  -- II.3 diagram for the left term.
  --
  -- We read the divided whole segment in the reverse direction:
  --
  --   B-C-A.
  --
  -- Thus II.3 gives
  --
  --   Rect(BA,CA) = Rect(BC,CA) + Square(CA).
  ------------------------------------------------------------------

  (D1 E1 L1 X1 : Geo.Point)

  (hRectLeft :
    IsRectangle Geo B A E1 D1)

  (hAE1_CA :
    Geo.Congruent A E1 C A)

  (hD1L1E1 : Geo.Between D1 L1 E1)
  (hD1X1A : Geo.Between D1 X1 A)
  (hCX1L1 : Geo.Between C X1 L1)

  (hLeftLeftPar :
    IsParallelogram Geo L1 D1 B C)
  (hLeftRightPar :
    IsParallelogram Geo A E1 L1 C)

  -- Square on CA, i.e. on the same unoriented segment as AC.
  (F G : Geo.Point)
  (hSquareCA : IsSquare Geo C A F G)

  ------------------------------------------------------------------
  -- II.3 diagram for the right term:
  --
  --   Rect(AB,CB) = Rect(AC,CB) + Square(CB).
  ------------------------------------------------------------------

  (D2 E2 L2 X2 : Geo.Point)

  (hRectRight :
    IsRectangle Geo A B E2 D2)

  (hBE2_CB :
    Geo.Congruent B E2 C B)

  (hD2L2E2 : Geo.Between D2 L2 E2)
  (hD2X2B : Geo.Between D2 X2 B)
  (hCX2L2 : Geo.Between C X2 L2)

  (hRightLeftPar :
    IsParallelogram Geo L2 D2 A C)
  (hRightRightPar :
    IsParallelogram Geo B E2 L2 C)
```

```
-- Square on CB.
(H K : Geo.Point)
(hSquareCB : IsSquare Geo C B H K)

------------------------------------------------------------------
-- Two arbitrary representatives of the repeated cross rectangle
-- Rect(AC,CB).
------------------------------------------------------------------

(T10 T11 T12 T13 : Geo.Point)
(T20 T21 T22 T23 : Geo.Point)

(hCross1 :
  IsRectangleContainedBy Geo
    T10 T11 T12 T13 A C C B)

(hCross2 :
  IsRectangleContainedBy Geo
    T20 T21 T22 T23 A C C B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo A B E0 D0)
  ((hilbertParallelogramTerm Geo T20 T21 T22 T23 +
    hilbertParallelogramTerm Geo C A F G) +
   (hilbertParallelogramTerm Geo T10 T11 T12 T13 +
    hilbertParallelogramTerm Geo C B H K)) := by
```

Several aspects of the signature deserve explicit interpretation.

First, one strict division relation $A - C - B$ is shared by the theorem, but the public statement receives *three* configured cut diagrams:

- the II.2 square-cut diagram, with points D_0, E_0, L_0, X_0;
- the left II.3 rectangle-cut diagram, with points D_1, E_1, L_1, X_1;
- the right II.3 rectangle-cut diagram, with points D_2, E_2, L_2, X_2.

The theorem does not construct these configurations from A, C, B alone. Rather, it proves the content identity once the corresponding geometric witnesses have been supplied.

Second, $U_0U_1U_2U_3$ and $V_0V_1V_2V_3$ are arbitrary representatives of the two rectangles produced at the II.2 level. They need not be literal subrectangles of the square ABE_0D_0.

Third, the cross terms are represented independently by $T_{10}T_{11}T_{12}T_{13}$ and $T_{20}T_{21}T_{22}T_{23}$. Both satisfy the same `IsRectangleContainedBy` predicate for AC, CB. This makes the formal "twice the rectangle" statement representative-independent rather than tied to one duplicated quadrilateral.

Fourth, the conclusion is exactly

```
HilbertScissorsEq Geo ...
```

and not the weaker

```
HilbertScissorsEquicomplementable Geo ...
```

relation. Exactness is inherited from II.2, II.3, rectangle rotation, formal addition, and transitivity.

4.8 Relationship with Earlier Book II Propositions

4.8.1 Proposition II.1

II.4 does not import `Proposition2_1` directly. Nevertheless II.1 is the underlying distributive core because both II.2 and II.3 are themselves built from its binary rectangle split. The dependency is therefore transitive:

$$\text{II.1} \longrightarrow \text{II.2 and II.3} \longrightarrow \text{II.4}.$$

This is the first point where the investment in II.1 becomes visibly compositional at the proposition level. II.2 and II.3 were sibling specializations of the binary II.1 API; II.4 uses both siblings together.

4.8.2 Proposition II.2

II.2 supplies exactly the outer decomposition needed by II.4:

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, CB).$$

The current II.4 source imports II.2 explicitly and calls its public theorem once.

4.8.3 Proposition II.3

II.3 supplies the expansion of each whole-by-part rectangle. The current proof calls the same public theorem twice, once after reversing the line orientation and once directly. Thus II.4 is a genuine composition theorem over the public Book II API rather than a restatement of one earlier helper.

4.9 Relationship with Book I Infrastructure

Euclid's own local proof depends explicitly on I.46, I.31, I.29, I.5, I.6, I.34, and I.43. The formal II.4 theorem does not replay any of those arguments locally. Their geometric content has entered the current project through the established square, parallel, parallelogram, rectangle, and scissors layers and through the already proved II.2 and II.3 theorems.

The contrast with I.43 is especially instructive. Euclid needs I.43 to prove that the two complements AG and GE in one concrete square are equal. Lean instead receives two arbitrary cross-rectangle representatives and normalizes one of them with `rectangle_contained_by_swap`. Representative independence is already built into the rectangle API, so no local complement theorem is required.

The formal proof is therefore not a transcription of Euclid's diagram. It is a proof over a stronger reusable library whose theorems were themselves proved synthetically from the established Hilbert/Book I geometry.

4.10 Formalization Notes and Hidden Geometric Data

The polished identity hides a substantial amount of explicit positional data. The current signature contains:

- $A - C - B$, fixing the strict order of the original segment;
- a complete square-cut configuration for the II.2 call;
- two complete rectangle-cut configurations for the two II.3 calls;
- three diagonal/cut intersection witnesses X_0, X_1, X_2;
- six parallelogram hypotheses certifying the two pieces in each cut;
- one square on the whole and one square on each part;
- two arbitrary II.2 rectangle representatives;
- two arbitrary representatives of the repeated cross rectangle.

The proof then adds two representational obligations not visible in Euclid's statement:

- reverse $A - C - B$ to $B - C - A$ for the left II.3 call;
- rotate one cross-rectangle representative and reverse segment endpoints so that the ordered side data match the reversed II.3 interface.

There is also useful negative information. No `SameRay`, `SameSide`, or `OppositeSide` datum occurs in the public II.4 proof. No new right-angle propagation theorem, no new parallel theorem, and no new scissors constructor is introduced. Those obligations are encapsulated inside the imported rectangle and Book II layers.

4.11 Geometry, Content, Representation, and Later Arithmetic

4.11.1 Geometry

The geometry consists of one segment cut $A - C - B$, one square on the whole, two squares on the parts, and three configured rectangle dissections. The classical source packages these ideas in one square diagram; the Lean theorem packages them as three reusable instances of already formalized cut theorems.

4.11.2 Content / scissors

The conclusion is exact `HilbertScissorsEq`. The proof only uses reflexivity, symmetry, transitivity, compatibility with addition, and exact scissors certificates supplied by II.2, II.3, and rectangle rotation. General equicomplementability is not required.

4.11.3 Representation

Representation is the genuinely nontrivial local layer. Endpoint order on the containing segments must be normalized for the reversed II.3 branch, and one cross rectangle must be cyclically rotated before it can be used in that branch. The final theorem nevertheless permits two completely independent representatives of $\mathrm{Rect}(AC, CB)$.

This is the precise formal content of "twice the rectangle" in the current architecture: two scissors terms representing the same geometric rectangle magnitude occur as separate summands.

4.11.4 Later arithmetic / numerical area

Neither later layer is used. The proof imports no segment multiplication module, chooses no unit segment, proves no distributive law for a product of segment classes, and evaluates no real-valued area function.

The formula

$$(y+z)^2 = y^2 + z^2 + 2yz$$

should therefore remain explicitly labelled as the modern algebraic shadow of the synthetic construction.

4.12 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and therefore belongs to the same Euclidean project layer as II.1–II.3.

The current `Proposition2_4.lean` file introduces no proposition-local `axiom`, `sorry`, or `admit`. The theorem was compiled successfully in the project. The supplied transitive axiom audit is:

```
#print axioms Geometry.euclid_proposition_2_4

'Geometry.euclid_proposition_2_4' depends on axioms:
[propext, Classical.choice, bookZero_nullSegment1, Quot.sound]
```

Accordingly II.4 should not be described as "axiom-free." The precise statement is that it introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are exactly the four declarations listed above. In particular the list contains no `sorryAx`.

The compile result reported for the current theorem is clean. A separate full `lake build` result was not part of the documentation input used here, so the status table below does not claim that stronger check.

4.13 Hartshorne's Semantic Control

Hartshorne's Section 22 supplies the appropriate general semantic control for Book II: Euclid's equalities of plane figures can be interpreted through equal content without first assigning numerical areas [5]. He also explains why the algebraic-looking manipulations of rectangles and squares may be regarded as a kind of geometrical algebra.

II.4 is a particularly vivid example of that interpretation. Its modern shadow is a quadratic identity, but Euclid's objects are still a large square, two smaller squares, and two rectangles. The current formalization sharpens the configured result from general equal content to exact scissors equality, because every step is built from exact dissection and exact representation transport.

Hartshorne's "Brief Euclid" explicitly lists II.4 in the form that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. Thus II.4 has a more direct Hartshorne-side textual counterpart than II.2 or II.3. The surrounding Section 22 discussion still supplies the essential semantic interpretation: Euclid's equality is read as equal content, not as prior equality of numerical areas.

The stronger exact-scissors certificate remains a fact about the present Lean architecture. It should not be attributed to Hartshorne's general content relation.

4.14 Hilbert's Layering

Hilbert provides the complementary architectural warning. His plane-content theory and his later arithmetic of segments are distinct layers. Segment multiplication is introduced only later, using a fixed unit segment and parallel constructions, after which commutativity and distributivity become actual arithmetic theorems [2].

II.4 may look like the first place where one should invoke those later laws, because the shadow

$$(y+z)^2 = y^2 + z^2 + 2yz$$

contains both distributivity and commutativity. The compiling source shows that this move is unnecessary. The proof stays below segment arithmetic:

$$\begin{array}{c}\text{rectangle/square geometry}\\ \Downarrow \\ \text{exact II.2 and II.3 scissors identities}\\ \Downarrow \\ \text{geometric representation transport}\\ \Downarrow \\ \text{exact II.4 scissors identity.}\end{array}$$

This is a useful architectural checkpoint. Even a full quadratic-looking Book II identity can be derived from the geometric rectangle calculus without crossing into Hilbert's later segment field.

4.15 Dependency Summary

The actual Lean dependency chain is

$$\begin{array}{c}
\text{Hilbert incidence + Euclidean plane layer} \\
\Downarrow \\
\text{Book I square, parallelogram, right-angle, and scissors infrastructure} \\
\Downarrow \\
\texttt{HilbertRectangle} + \text{Proposition II.1 binary API} \\
\Downarrow \\
\texttt{euclid_proposition_2_2} \quad + \quad \texttt{euclid_proposition_2_3} \\
\Downarrow \\
\text{endpoint reversal} + \texttt{rectangle_contained_by_swap} \\
\Downarrow \\
\texttt{HilbertScissorsEq.add} + \texttt{trans} \\
\Downarrow \\
\texttt{euclid_proposition_2_4}.
\end{array}$$

The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.29} + \text{I.5} + \text{I.6} + \text{I.34} + \text{I.43} \longrightarrow \text{II.4}.$$

New geometric axiom in II.4:	no
New content axiom in II.4:	no
New proposition-local helper theorem:	no
New reusable helper theorem:	no
New `HilbertRectangle` theorem introduced by II.4:	no
New Interface / Book Zero / Scissors theorem:	no
Reuses earlier Book II propositions:	yes, II.2 and II.3
Direct import of II.1:	no; inherited through II.2 and II.3
Direct side-order transport used:	yes, `rectangle_contained_by_swap`
Endpoint-reversal transport used:	yes
Exact `HilbertScissorsEq`:	yes
General equicomplementability required:	no
Numerical area:	no
Segment multiplication:	no
Public Lean theorem compile:	yes, clean
Full relevant `lake build`:	not separately reported here
Proposition-local `sorry/admit`:	no
`sorryAx` in reported axiom list:	no

4.16 Conclusion

Proposition II.4 is the first strong compositional test of the Book II API. Euclid proves the theorem inside one carefully constructed square: two subfigures are shown to be squares, and I.43 identifies the two remaining complements. The current Lean proof instead works one level higher. It takes II.2 as the outer square decomposition and applies II.3 to each of the two resulting rectangles.

The only local difficulty is representational. One II.3 call uses the reversed order $B - C - A$, and one cross rectangle must be rotated so that its ordered containing sides match that call. Once these transports are made, `HilbertScissorsEq.add` and transitivity close the theorem.

Consequently II.4 requires no new rectangle theorem, no new scissors theorem, no new proposition-local helper, and no segment multiplication. Its apparent quadratic algebra is already present in the synthetic rectangle calculus as a composition of exact geometric decompositions. The reported axiom audit adds no new proposition-specific debt and contains no `sorryAx`.

Chapter 5

Proposition II.5

5.1 Introduction

Proposition II.5 is the first Book II theorem in the present reconstruction to combine a *midpoint condition* with the rectangle/scissors calculus. Let C bisect AB, and let D be a second cut point. In the ordering chosen by the Lean theorem,

$$A - D - C - B,$$

so that C is the midpoint of AB and D lies in the left half. The theorem asserts, in geometric content language,

$$\mathrm{Rect}(AD, DB) + \mathrm{Sq}(CD) = \mathrm{Sq}(CB).$$

The familiar modern algebraic shadow is obtained by writing the half-length as a and the displacement of the unequal cut from the midpoint as x:

$$(a - x)(a + x) + x^2 = a^2.$$

This formula is only an interpretive mnemonic. The Lean proof does not define multiplication of segment lengths and does not evaluate numerical area. Its objects are concrete rectangle and square representatives, while equality is expressed by exact scissors equivalence.

The current source begins with

```
import CGJteamLab.Proposition2_4
```

but this import fact should not be confused with the proof DAG. The proof does not invoke `euclid_proposition_2_4`. It directly reuses the binary rectangle theorem from II.1 and Proposition II.3, together with the reusable rectangle representation API. Thus II.5 provides an especially useful example of the difference between module dependency and mathematical proof dependency.

5.2 Euclid's Statement and Classical Configuration

5.2.1 The statement

In the notation used here, Euclid's proposition may be stated as follows. A straight line AB is divided equally at C and unequally at D. Then the rectangle contained by AD and DB, together with the square on CD, is equal in content to the square on the half CB [1].

The phrase "rectangle contained by AD, DB" names a plane figure whose adjacent sides have those magnitudes. It is not a product already formed in a segment field. Likewise, the addition in the displayed identity describes the content of two figures taken together.

5.2.2 The classical diagram

The principal figure below follows the standard Euclidean configuration with the unequal cut D chosen in the right half, so its baseline order is $A - C - D - B$. This is the mirror version of the Lean convention $A - D - C - B$; the content identity is unchanged. Euclid erects the square $CEFB$ on the half CB, joins BE, draws the parallel through D, then the horizontal through its intersection H with the diagonal, and finally the parallel through A [1].

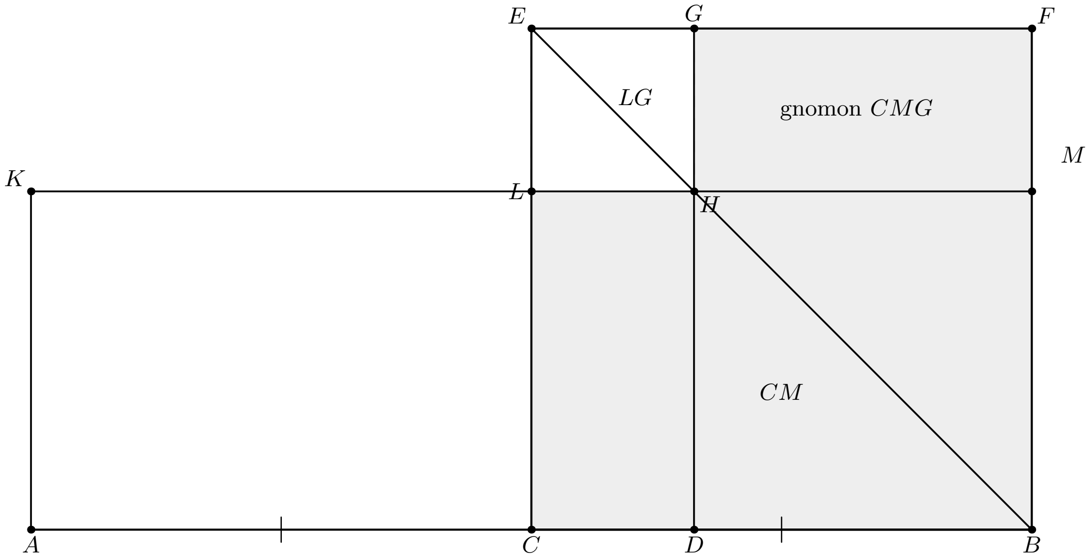


Figure 5.1: The classical mechanism of Euclid II.5. The square $CEFB$ is built on the half CB. The point D is the unequal cut, H lies on diagonal BE, and LG is the small square on CD. The remaining part of $CEFB$ is the gnomon used in Euclid's content argument. The formal Lean proof does not introduce a primitive gnomon object.

5.3 Classical Proof

The classical proof uses the gnomon structure of Figure 5.1. The diagonal BE and the parallels create complementary parallelograms. By I.43 the complements CH and HF are equal. Adding the same rectangle DM to the relevant figures yields an equality in which the rectangle CM can be replaced by AL: these are equal parallelograms on equal bases AC and CB and in the same parallels, using I.36 and the midpoint condition $AC = CB$ [1].

The construction also gives $DH = DB$ by the square-of-a-sum geometry already available at II.4 (in Euclid's classical dependency chain). Consequently the figure AH is the rectangle contained by the unequal parts AD and DB. Euclid identifies that rectangle with the gnomon obtained by removing the small corner square LG from the square $CEFB$. Finally LG is the square on the segment CD, and adjoining it to the gnomon recovers the entire square on CB.

Thus the classical local DAG is schematically

$$\begin{gathered}
\text{I.43: equal complements} \quad + \quad \text{I.36: equal parallelograms on equal bases} \\
\text{together with the II.4 corollary} \\
\Downarrow \\
\text{rectangle on } AD, DB \ = \ \text{gnomon in the square on } CB \\
\Downarrow \\
\text{add the square on } CD \\
\Downarrow \\
\mathrm{Rect}(AD, DB) + \mathrm{Sq}(CD) = \mathrm{Sq}(CB).
\end{gathered}$$

This is *not* the Lean DAG. In particular, the current formal proof does not reconstruct the classical gnomon, does not call I.43 locally, and does not call II.4 despite importing the II.4 module.

5.4 The Geometric Identity in the Current Formalization

The public Lean theorem chooses the strict order

$$A - D - C - B$$

through two hypotheses:

```
(hMidC : HilbertIsMidpoint Geo C A B)
(hADC : Geo.Between A D C)
```

The midpoint hypothesis supplies $A - C - B$ and $AC \cong CB$. From $A - D - C$ and $A - C - B$, a private order helper derives $D - C - B$ and $A - D - B$. Thus the full four-point order is a theorem of the hypotheses, not an additional assumption.

At the exact-scissors level the target is

$$\mathrm{Rect}(DB, AD) + \mathrm{Sq}(DC) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

The reversed order DB, AD rather than AD, DB is immaterial at the geometric rectangle level because side-order exchange is itself proved by rectangle rotation/transport. It is not treated as primitive commutativity of a segment product.

A notable semantic point is that no gnomon type occurs in the source. The classical gnomon is represented extensionally by formal sums of rectangle and square scissors terms. The proof then rearranges and composes those sums by `HilbertScissorsEq.add`, symmetry, transitivity, and the existing rectangle transport lemmas.

5.5 Reusable Book II Infrastructure

5.5.1 The binary rectangle split from II.1

The theorem `euclid_proposition_2_1_two` is used twice. First it splits DB at C, with AD held as the other containing side:

$$\mathrm{Rect}(DB, AD) \simeq_{\mathrm{sc}} \mathrm{Rect}(DC, AD) + \mathrm{Rect}(CB, AD).$$

Later the same binary theorem, now applied to AC split at D, assembles

$$\mathrm{Rect}(CB, AD) + \mathrm{Rect}(CB, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, AC),$$

after geometric side-order transports.

5.5.2 Proposition II.3

Proposition II.3 supplies the first quadratic-looking decomposition:

$$\mathrm{Rect}(AC, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC).$$

The current source calls `euclid_proposition_2_3` directly. The midpoint congruence $AC \cong CB$ is then used to transport the left-hand rectangle to a representative contained by CB, DC.

5.5.3 Rectangle representation and transport

The proof relies on the established Book II rectangle API rather than numerical products. In particular it uses

- `rectangle_transport_scissorsEq` to change containing segments along segment congruences;
- `rectangle_contained_by_comm` and `rectangle_contained_by_swap` to exchange the two adjacent side roles geometrically;
- `rectangle_contained_by_unique` to compare arbitrary representatives of the same contained-by data;
- `square_is_rectangle_contained_by_side` to view a square as a rectangle contained by one side twice.

5.6 Proof Architecture of the Reconstruction

The production file preserves the private helper names created during the incremental proof search. The names contain `testXX`, but all these lemmas are `private`; they are implementation history, not public API. Their mathematical roles are stable and form the following chain.

5.6.1 Step 1: recover the full order and midpoint congruence

`proposition2_5_test01_order` derives

$$D - C - B, \qquad A - D - B, \qquad AC \cong CB, \qquad CA \cong CB.$$

This is the only place where the four-point order is consolidated.

5.6.2 Step 2: apply II.3 on the left half

`proposition2_5_test02_left_decomposition` proves

$$\mathrm{Rect}(AC, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC).$$

5.6.3 Step 3: transport the midpoint side

`proposition2_5_test03_midpoint_rectangle_transport` uses $AC \cong CB$ to prove

$$\mathrm{Rect}(AC, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, DC).$$

By symmetry and transitivity, `proposition2_5_test04_first_assembled_identity` obtains

$$\mathrm{Rect}(CB, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC).$$

5.6.4 Step 4: split the unequal part DB

`proposition2_5_test05_db_decomposition` applies the binary II.1 theorem to $D - C - B$:

$$\mathrm{Rect}(DB, AD) \simeq_{\mathrm{sc}} \mathrm{Rect}(DC, AD) + \mathrm{Rect}(CB, AD).$$

The next two helpers use geometric commutativity to normalize the first term:

$$\mathrm{Rect}(DB, AD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AD, DC) + \mathrm{Rect}(CB, AD).$$

5.6.5 Step 5: the scissors bridge

The private lemma `proposition2_5_test08_scissors_bridge` is an abstract content calculation. Given

$$P \simeq X + Y, \qquad Z \simeq X + S,$$

it proves

$$P + S \simeq Y + Z.$$

Crucially, this is *not* a cancellation principle. It is obtained by adding exact scissors equalities, reassociating and commuting the formal multiset sum, and using transitivity. Applying it to the two preceding identities gives

$$\boxed{\mathrm{Rect}(DB, AD) + \mathrm{Sq}(DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, AD) + \mathrm{Rect}(CB, DC).}$$

5.6.6 Step 6: assemble the right side

`proposition2_5_test09_assemble_right_side` rotates the two right-hand rectangle representatives, applies the binary II.1 split to $A - D - C$, and rotates the result back. It proves

$$\mathrm{Rect}(CB, AD) + \mathrm{Rect}(CB, DC) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, AC).$$

5.6.7 Step 7: recognize the half-square

Finally `proposition2_5_test10_half_rectangle_is_square` uses $AC \cong CB$ and representative uniqueness to identify

$$\mathrm{Rect}(CB, AC) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

This is where the midpoint hypothesis enters a second time: first it controlled order, and now it turns the final rectangle into the square on the half.

5.7 Lean Formalization

5.7.1 The public theorem signature

The following is the current compiling public signature. It is intentionally large because the theorem receives the concrete rectangle-cut configurations and representatives required by the earlier Book II APIs rather than hiding them behind choice constructions.

```
theorem euclid_proposition_2_5
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hADC : Geo.Between A D C)

    ----------------------------------------------------------
    -- Data used by test 04 inside test 08.
    ----------------------------------------------------------

    (E4 H4 L4 X4 : Geo.Point)
    (hRect4 : IsRectangle Geo A C H4 E4)
    (hCH_DC : Geo.Congruent C H4 D C)
    (hELE4 : Geo.Between E4 L4 H4)
    (hEXC4 : Geo.Between E4 X4 C)
    (hDXL4 : Geo.Between D X4 L4)
    (hLeftPar4 : IsParallelogram Geo L4 E4 A D)
    (hRightPar4 : IsParallelogram Geo C H4 L4 D)

    -- Square on DC.
    (F G : Geo.Point)
    (hSquareDC : IsSquare Geo D C F G)

    -- Rect(AC,DC).
    (R0 R1 R2 R3 : Geo.Point)
    (hAC_DC :
      IsRectangleContainedBy Geo
        R0 R1 R2 R3 A C D C)

    -- Common Rect(AD,DC).
    (T0 T1 T2 T3 : Geo.Point)
    (hAD_DC :
      IsRectangleContainedBy Geo
        T0 T1 T2 T3 A D D C)

    -- Rect(CB,DC).
    (Z0 Z1 Z2 Z3 : Geo.Point)
    (hCB_DC :
      IsRectangleContainedBy Geo
        Z0 Z1 Z2 Z3 C B D C)

    ----------------------------------------------------------
    -- Data used by test 07 inside test 08.
    ----------------------------------------------------------

    (E7 H7 L7 X7 : Geo.Point)
```

```
(hRect7 : IsRectangle Geo D B H7 E7)
(hBH_AD : Geo.Congruent B H7 A D)
(hELE7 : Geo.Between E7 L7 H7)
(hEXB7 : Geo.Between E7 X7 B)
(hCXL7 : Geo.Between C X7 L7)
(hLeftPar7 : IsParallelogram Geo L7 E7 D C)
(hRightPar7 : IsParallelogram Geo B H7 L7 C)

-- Rect(DB,AD), the unequal-parts rectangle.
(P0 P1 P2 P3 : Geo.Point)
(hDB_AD :
  IsRectangleContainedBy Geo
    P0 P1 P2 P3 D B A D)

-- Rect(DC,AD), before commutativity normalization.
(Q0 Q1 Q2 Q3 : Geo.Point)
(hDC_AD :
  IsRectangleContainedBy Geo
    Q0 Q1 Q2 Q3 D C A D)

-- Rect(CB,AD).
(Y0 Y1 Y2 Y3 : Geo.Point)
(hCB_AD :
  IsRectangleContainedBy Geo
    Y0 Y1 Y2 Y3 C B A D)

---------------------------------------------------------
-- Data for test 09:
-- assemble Rect(CB,AD) + Rect(CB,DC) into Rect(CB,AC).
---------------------------------------------------------

(E9 H9 L9 X9 : Geo.Point)
(hRect9 : IsRectangle Geo A C H9 E9)
(hCH_CB9 : Geo.Congruent C H9 C B)
(hELE9 : Geo.Between E9 L9 H9)
(hEXC9 : Geo.Between E9 X9 C)
(hDXL9 : Geo.Between D X9 L9)
(hLeftPar9 : IsParallelogram Geo L9 E9 A D)
(hRightPar9 : IsParallelogram Geo C H9 L9 D)

-- Rect(CB,AC).
(W0 W1 W2 W3 : Geo.Point)
(hCB_AC :
  IsRectangleContainedBy Geo
    W0 W1 W2 W3 C B A C)

---------------------------------------------------------
-- Data for test 10: square on CB.
---------------------------------------------------------

(S0 S1 S2 S3 : Geo.Point)
(hSquareCB : IsSquare Geo S0 S1 S2 S3)
(hS01_CB : Geo.Congruent S0 S1 C B) :

HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo P0 P1 P2 P3 +
   hilbertParallelogramTerm Geo D C F G)
  (hilbertParallelogramTerm Geo S0 S1 S2 S3) := by
```

Several layers of data are visible in the signature.

1. A, B, C, D and the midpoint/betweenness hypotheses encode the one-dimensional configuration.
2. The groups labelled in the source as data for tests 04, 07, and 09 are concrete rectangle-cut witnesses needed by II.3 or II.1.
3. F, G give a square on DC.
4. $P_0P_1P_2P_3$ represents the rectangle contained by DB, AD.
5. $W_0W_1W_2W_3$ represents the intermediate rectangle contained by CB, AC.
6. $S_0S_1S_2S_3$ is an arbitrary square representative whose side is congruent to CB.

Thus the theorem proves representative-independent content equality for supplied geometric witnesses; it does not claim that all of those witnesses are constructed internally from four points alone.

5.7.2 The final three-stage assembly

After all private helpers have been established, the public theorem is short in conceptual structure. The current source builds exactly three equations:

1. `h08`: unequal-parts rectangle plus the DC square equals two rectangles with common side CB;
2. `h09`: those two rectangles assemble into the rectangle $\text{Rect}(CB, AC)$;
3. `h10`: midpoint congruence turns that rectangle into the square on CB.

The final transitivity block is:

```
have h08 :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo P0 P1 P2 P3 +
       hilbertParallelogramTerm Geo D C F G)
      (hilbertParallelogramTerm Geo Y0 Y1 Y2 Y3 +
       hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3) :=
  proposition2_5_test08_core_identity
    Geo
    A B C D
    hMidC hADC
    E4 H4 L4 X4
    hRect4 hCH_DC hELE4 hEXC4 hDXL4
    hLeftPar4 hRightPar4
    F G hSquareDC
    R0 R1 R2 R3
    hAC_DC
    T0 T1 T2 T3
    hAD_DC
    Z0 Z1 Z2 Z3
    hCB_DC
    E7 H7 L7 X7
```

```
    hRect7 hBH_AD hELE7 hEXB7 hCXL7
    hLeftPar7 hRightPar7
    P0 P1 P2 P3
    hDB_AD
    Q0 Q1 Q2 Q3
    hDC_AD
    Y0 Y1 Y2 Y3
    hCB_AD

--------------------------------------------------------------
-- test09:
-- Rect(CB,AD) + Rect(CB,DC) = Rect(CB,AC).
--------------------------------------------------------------

have h09 :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo Y0 Y1 Y2 Y3 +
       hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)
      (hilbertParallelogramTerm Geo W0 W1 W2 W3) :=
  proposition2_5_test09_assemble_right_side
    Geo
    A B C D
    hADC
    E9 H9 L9 X9
    hRect9 hCH_CB9 hELE9 hEXC9 hDXL9
    hLeftPar9 hRightPar9
    Y0 Y1 Y2 Y3
    hCB_AD
    Z0 Z1 Z2 Z3
    hCB_DC
    W0 W1 W2 W3
    hCB_AC

--------------------------------------------------------------
-- test10:
-- Rect(CB,AC) = Square(CB).
--------------------------------------------------------------

have h10 :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo W0 W1 W2 W3)
      (hilbertParallelogramTerm Geo S0 S1 S2 S3) :=
  proposition2_5_test10_half_rectangle_is_square
    Geo
    A B C D
    hMidC hADC
    W0 W1 W2 W3
    hCB_AC
    S0 S1 S2 S3
    hSquareCB
    hS01_CB

--------------------------------------------------------------
-- Full II.5.
--------------------------------------------------------------

exact
  HilbertScissorsEq.trans
```

```
      (Geo := Geo)
      h08
      (HilbertScissorsEq.trans
        (Geo := Geo)
        h09
```

The formal result is therefore exact `HilbertScissorsEq`, not the weaker relation `HilbertScissorsEquicomplementable`, and not equality of real numbers.

5.8 Relationship with Earlier Book II Propositions

5.8.1 Proposition II.1

II.1 is the basic additive engine of the proof. Its binary theorem is called directly in the decomposition of DB and again in the assembly of AC. Thus II.5 genuinely reuses the rectangle split calculus rather than reproving the corresponding dissections from Book I geometry.

5.8.2 Proposition II.3

II.3 is also called directly. It supplies the identity

$$\mathrm{Rect}(AC, DC) = \mathrm{Rect}(AD, DC) + \mathrm{Sq}(DC),$$

at the exact scissors level. This is the step that inserts the square on the difference between the two section points.

5.8.3 Proposition II.4

The source file imports `CGJteamLab.Proposition2_4`, so II.4 is a direct *module* dependency. Nevertheless the theorem `euclid_proposition_2_4` is not invoked in the proof term. This sharply contrasts with Euclid's classical proof, where a corollary of II.4 participates in the identification of the relevant rectangle and small square.

This distinction should be recorded explicitly:

$$\text{module chain: II.4 module} \to \text{II.5 module},$$
$$\text{Lean theorem calls: II.1 binary API} + \text{II.3} + \text{rectangle API.}$$

5.9 Relationship with Book I Infrastructure

The Book I geometry is present below the reusable rectangle layer. Rectangles, squares, right angles, parallel sides, parallelograms, and finite scissors terms all ultimately depend on the established Hilbert/Book I development. II.5 does not locally replay Euclid's diagonal-complement argument from I.43. Instead, the effect of those earlier geometric facts has already been packaged into `HilbertRectangle`, the scissors API, and the earlier Book II theorems.

This is an architectural gain: a proposition whose classical proof is diagram specific becomes a composition of stable content-level interfaces.

5.10 Formalization Notes and Hidden Geometric Data

5.10.1 Order orientation

The classical diagram commonly places the unequal cut in the right half $A - C - D - B$, whereas the Lean theorem fixes $A - D - C - B$. These are mirror cases. The source makes the chosen orientation explicit through `hADC` and then derives $D - C - B$ from the midpoint betweenness. No appeal to a picture is allowed to supply this ordering silently.

5.10.2 Midpoint as geometry, not arithmetic

`HilbertIsMidpoint` contributes both betweenness and segment congruence. The proof uses those components separately. Betweenness creates the required subsegment orders; congruence transports rectangle sides and eventually recognizes the square on the half. There is no variable standing for a numerical half-length.

5.10.3 Representative independence

The objects named “the rectangle contained by” particular segments are concrete quadrilateral representatives. Different helpers may receive different representatives of the same pair of side magnitudes. `rectangle_transport_scissorsEq`, `rectangle_contained_by_unique`, and the side-order transport theorems are therefore mathematically substantive parts of the formal statement, not mere syntactic rewrites.

5.10.4 No gnomon primitive

Euclid’s proof naturally names a gnomon. The present library does not define a `GnomonTerm` or another primitive gnomon object for II.5. No such new abstraction is needed here: the required content is expressed as a formal sum of scissors terms. The private scissors bridge then performs the exact content rearrangement that the classical gnomon packages visually.

5.10.5 No cancellation axiom

At an intermediate stage the two identities contain the common rectangle $\mathrm{Rect}(AD, DC)$. It might be tempting to introduce a cancellation principle for content. The actual source deliberately avoids this. The scissors bridge combines equalities additively and rearranges formal sums, so no new positivity or cancellation theorem is required.

5.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`. The user-verified module build is clean. The reported transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_5

'Geometry.euclid_proposition_2_5' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Accordingly II.5 should not be described as "axiom-free." The precise claim is that II.5 introduces no new proposition-specific axiom and no `sorry`; its transitive dependencies are the four declarations shown above, and the list contains no `sorryAx`.

5.12 Dependency Summary

The actual Lean dependency chain is best displayed as

$$\begin{array}{c}\text{Hilbert incidence + Euclidean plane layer}\\ \Downarrow \\ \text{Book I square, rectangle, parallelogram, and scissors infrastructure}\\ \Downarrow \\ \text{HilbertRectangle representative/transport API}\\ \Downarrow \\ \text{II.1 binary split} \quad + \quad \text{II.3 rectangle-plus-square theorem}\\ \Downarrow \\ \text{private II.5 order, transport, bridge, and assembly helpers}\\ \Downarrow \\ \text{public theorem } \texttt{euclid\_proposition\_2\_5}.\end{array}$$

The classical local chain is different:

$$\text{I.43} + \text{I.36} + \text{II.4 corollary} \longrightarrow \text{gnomon identification} \longrightarrow \text{II.5}.$$

The formal proof is therefore not a line-by-line translation of the classical one. It reconstructs the same geometric content through a different, reusable DAG.

New geometric axiom in II.5:	no
New content axiom in II.5:	no
New proposition-local helper theorem:	yes, private helpers
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.1 and II.3
Imports II.4 module:	yes
Calls `euclid_proposition_2_4`:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability:	no
Uses numerical area:	no
Uses segment multiplication:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Proposition-local `sorry/admit`:	no
Transitive `sorryAx`:	absent in reported audit

5.13 Conclusion

Proposition II.5 is an important architectural checkpoint for Book II. The classical proof organizes the equality around a gnomon inside the square on the half. The formal proof shows that a separate gnomon primitive is not needed to recover the theorem: the same content can be assembled from the binary II.1 rectangle split, II.3, midpoint congruence, exact rectangle transport, and the additive structure of `HilbertScissorsEq`.

The result also clarifies the role of the modern quadratic identity. It is a useful shadow for recognizing the proposition, but the verified argument stays synthetic throughout. Geometry supplies the order and congruences; representation supplies rectangles and squares; scissors equality supplies the content calculus. Numerical area and segment multiplication remain outside the proof.

Chapter 6

Proposition II.6

6.1 Introduction

Proposition II.6 is the external-section companion of Proposition II.5. A straight line AB is bisected at C and then produced beyond B to D. Thus the one-dimensional order used by the formal theorem is

$$A - C - B - D.$$

The proposition asserts, in geometric content language,

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) = \mathrm{Sq}(CD).$$

The familiar modern algebraic shadow is obtained by writing the half-length $AC = CB$ as a and the added segment BD as x:

$$a^2 + (2a + x)x = (a + x)^2.$$

Equivalently, if $y = CD$ and $z = CB$, one sees the difference-of-squares pattern

$$(y + z)(y - z) + z^2 = y^2.$$

Both formulas are documentary mnemonics only. The Lean proof does not define a multiplication of segment lengths and does not evaluate a numerical area. Its objects are concrete rectangle and square representatives, and the conclusion is exact scissors equality.

The production source begins with

```
import CGJteamLab.Proposition2_5
```

so Proposition II.5 is a module dependency. It is not, however, a theorem dependency of the proof term. The current II.6 proof directly calls the binary theorem from II.1, Proposition II.3, Proposition II.4, and the rectangle-transport/scissors API. This distinction between the module chain and the actual theorem-call DAG is important in the present reconstruction.

6.2 Euclid's Statement and Classical Configuration

6.2.1 The statement

In Heath's wording, the proposition begins:

> If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.

With AB bisected at C and BD added in a straight line, the concrete statement is

$$\mathrm{Rect}(AD, DB) + \mathrm{Sq}(CB) = \mathrm{Sq}(CD) \qquad [1].$$

As throughout the present Book II documentation, the notation $\mathrm{Rect}(-,-)$ and $\mathrm{Sq}(-)$ is documentary shorthand for geometric figures. It is not a definition of a numerical product or numerical square.

6.2.2 The classical diagram

Euclid constructs the square $CEFD$ on CD, joins the diagonal DE, draws through B a parallel to CE (or DF), then through the resulting intersection H a line parallel to the baseline, and finally through A a parallel to the vertical sides. In the annotated diagram we also mark N as the midpoint of BD, so that the traditional name of the gnomon NOP has a literal set of boundary labels on the figure [1]. The construction is shown in Figure 6.1.

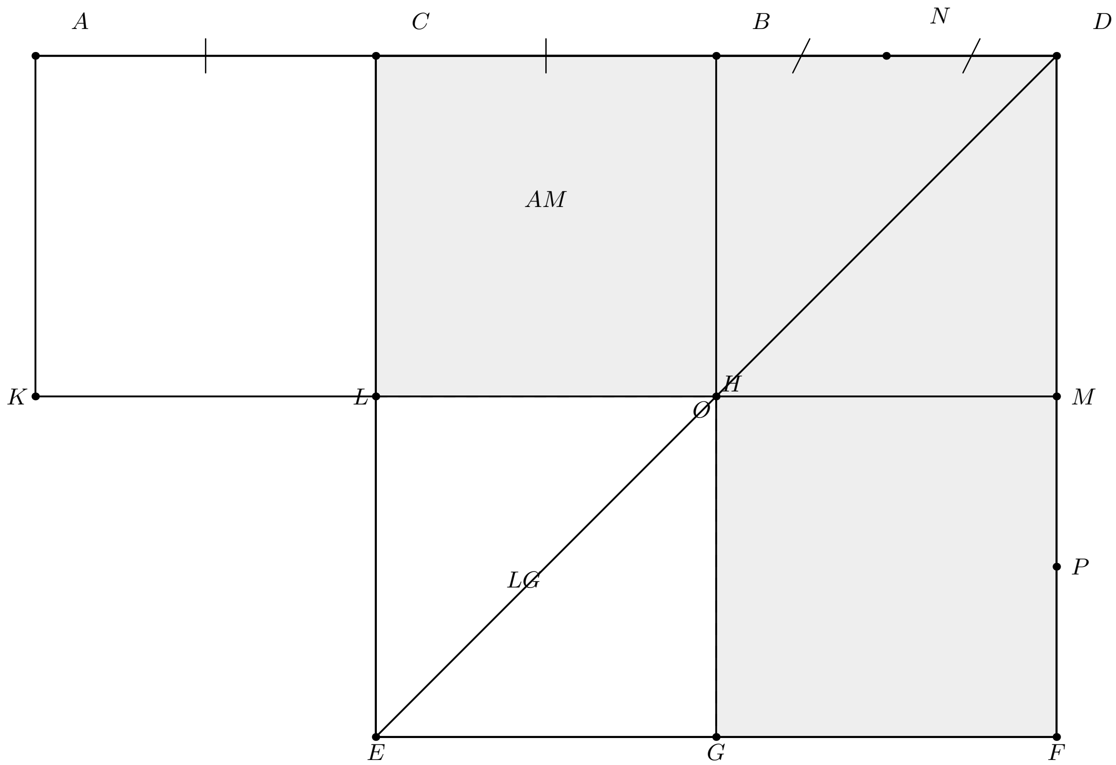


Figure 6.1: The classical mechanism of Euclid II.6. The baseline has order $A - C - B - D$ and C bisects AB. The square $CEFD$ is built on CD. The point N is the midpoint of BD. The dashed square LG is equal to the square on BC; the shaded remainder is Euclid's gnomon NOP, and the auxiliary boundary labels N, O, P are shown explicitly on the figure. The rectangle AM has adjacent side magnitudes AD and DB.

The same picture exposes the central geometry. Since $AC = CB$, the parallelograms AL and CH have equal content by I.36. The diagonal construction makes CH and HF complementary parallelograms, so I.43 gives their equality. Thus $AL = HF$. Adding the same rectangle CM identifies the whole rectangle AM with the gnomon NOP.

6.3 Classical Proof

The classical proof can be organized into four content steps.

First, the midpoint condition gives

$$AC = CB.$$

The parallelograms AL and CH stand on these equal bases and lie between the same parallels, so I.36 yields

$$AL = CH.$$

Second, CH and HF are complements about the diagonal DE of the relevant parallelogram configuration, hence I.43 gives

$$CH = HF.$$

Therefore

$$AL = HF.$$

Third, Euclid adds the same rectangle CM to both sides. The whole figure AM is therefore equal to the gnomon NOP. The construction gives $DM = DB$, so AM is the rectangle contained by AD and DB. Hence

$$\mathrm{Rect}(AD, DB) = \text{gnomon } NOP.$$

Finally, the figure LG is the square on BC. Adding it to both sides gives

$$\mathrm{Rect}(AD, DB) + \mathrm{Sq}(BC) = \text{gnomon } NOP + \mathrm{Sq}(BC).$$

The right-hand side is the whole square $CEFD$ on CD, proving II.6 [1].

The classical local DAG is therefore

$$\begin{array}{c}
\text{I.36: equal parallelograms on equal bases} \quad + \quad \text{I.43: equal complements} \\
\Downarrow \\
\mathrm{Rect}(AD, DB) = \text{gnomon } NOP \\
\Downarrow \\
\text{add the square on } BC \\
\Downarrow \\
\mathrm{Rect}(AD, DB) + \mathrm{Sq}(BC) = \mathrm{Sq}(CD).
\end{array}$$

This is not the Lean DAG. The production proof does not construct a gnomon object and does not call I.36 or I.43 locally.

6.4 The Geometric Identity in the Current Formalization

The public theorem receives

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)
```

so C is the midpoint of AB and B lies between A and D. The midpoint hypothesis gives

$$A - C - B \qquad \text{and} \qquad AC \cong CB.$$

Combining $A - C - B$ with $A - B - D$ by Hilbert betweenness transitivity yields

$$C - B - D \qquad \text{and} \qquad A - C - D.$$

Thus the full order

$$A - C - B - D$$

is derived, not read from a diagram.

At the exact-scissors level the public target is

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CD).$$

The order of the two summands on the left is the one present in the actual Lean conclusion. The theorem proves `HilbertScissorsEq`; it does not merely prove the weaker general content relation and it does not assert an equality of real numbers.

A second formal distinction is important. The square on BC in the public conclusion is the concrete square BCS_0S_1. The rectangle contained by AD, BD is represented by the supplied quadrilateral $P_0P_1P_2P_3$. The statement is therefore about concrete representatives whose side magnitudes satisfy the required contained-by predicates.

6.5 Reusable Book II Infrastructure Used

6.5.1 Proposition II.3: extend the unequal-parts rectangle

The first major exact dissection is obtained by applying `euclid_proposition_2_3` to AD cut at B, with BD as the fixed second side:

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, BD) + \mathrm{Sq}(BD).$$

This is the external-section analogue of introducing a square term in the earlier Book II identities.

6.5.2 Proposition II.1: split the old segment at its midpoint

The binary theorem `euclid_proposition_2_1_two` is then applied to AB cut at C, again with BD as the fixed side:

$$\mathrm{Rect}(AB, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD).$$

Composing with the II.3 decomposition gives

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).$$

6.5.3 Midpoint transport

The midpoint congruence $AC \cong CB$ is converted into equality of rectangle content by `rectangle_transport_scissorsEq`:

$$\mathrm{Rect}(AC, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, BD).$$

Consequently

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).$$

No numerical factor 2 is introduced. The phrase "twice the rectangle" means two scissors terms with the same geometric content.

6.5.4 Proposition II.4 on the produced segment

The decisive second decomposition calls `euclid_proposition_2_4` directly on CD cut at B. The formal substitution is

$$A_{\mathrm{II.4}} := C, \qquad B_{\mathrm{II.4}} := D, \qquad C_{\mathrm{II.4}} := B.$$

The required order hypothesis is exactly the derived betweenness $C - B - D$. The result is

$$\mathrm{Sq}(CD) \simeq_{\mathrm{sc}} \big(\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BC)\big) + \big(\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BD)\big).$$

This use of II.4 is one of the sharpest differences between II.5 and II.6 in the current formal library. II.5 imported the II.4 module but did not call `euclid_proposition_2_4`; II.6 calls it explicitly.

6.6 Proof Architecture of the Reconstruction

The production file preserves the names of the private incremental lemmas used during proof development. Their `testXX` names are implementation history only; because the declarations are `private`, they do not form part of the public API.

6.6.1 Step 1: recover the full order

`proposition2_6_test01_order` derives

$$C - B - D, \qquad A - C - D, \qquad AC \cong CB, \qquad CA \cong CB.$$

The only input is the midpoint structure together with $A - B - D$.

6.6.2 Step 2: decompose the rectangle on AD, BD

`proposition2_6_test02_ad_decomposition` calls II.3 and proves

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AB, BD) + \mathrm{Sq}(BD).$$

6.6.3 Step 3: split AB at C

`proposition2_6_test03_ab_decomposition` calls the binary II.1 theorem:

$$\mathrm{Rect}(AB, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD).$$

6.6.4 Step 4: compose the two decompositions

`proposition2_6_test04_compose_decompositions` uses the same concrete representative of $\mathrm{Rect}(AB, BD)$ in both preceding lemmas. Therefore no representative-uniqueness transport is needed at this junction. The result is

$$\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \big(\mathrm{Rect}(AC, BD) + \mathrm{Rect}(CB, BD)\big) + \mathrm{Sq}(BD).$$

6.6.5 Step 5: use the midpoint congruence

`proposition2_6_test05_midpoint_rectangle_transport` proves

$$\mathrm{Rect}(AC, BD) \simeq_{\mathrm{sc}} \mathrm{Rect}(CB, BD)$$

by transporting the first adjacent side along $AC \cong CB$ while the second side remains congruent to BD.

6.6.6 Step 6: normalize the left decomposition

`proposition2_6_test06_double_cross` inserts the midpoint transport into Step 4:

$$\boxed{\mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Rect}(CB, BD)) + \mathrm{Sq}(BD).}$$

The proof uses only addition, reflexivity, and transitivity of exact scissors equality.

6.6.7 Step 7: expand the square on CD by II.4

`proposition2_6_test07_square_cd_decomposition` applies II.4 to the segment CD cut at B:

$$\boxed{\mathrm{Sq}(CD) \simeq_{\mathrm{sc}} (\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BC)) + (\mathrm{Rect}(CB, BD) + \mathrm{Sq}(BD)).}$$

A useful implementation decision is visible here. The final assembly supplies the *same* rectangle representative $Z_0Z_1Z_2Z_3$ to both occurrences of $\mathrm{Rect}(CB, BD)$, and it reuses the same supplied square on BD as in Step 6. Hence the following algebraic bridge can match the two decompositions literally.

6.6.8 Step 8: the abstract scissors bridge

`proposition2_6_test08_scissors_bridge` is purely algebraic at the level of formal scissors sums. For terms P, Q, R, S, T, it assumes

$$P \simeq (R + R) + T, \qquad Q \simeq (R + S) + (R + T),$$

and proves

$$S + P \simeq Q.$$

The only rearrangement is associativity and commutativity of the formal sum, implemented by `ac_rfl`. The scissors equalities themselves are combined by `HilbertScissorsEq.add`, reflexivity, symmetry, and transitivity.

There is no cancellation theorem here. In particular, the proof does not infer equality by subtracting the common copies of R or T. It simply adds and rearranges exact scissors equalities.

6.6.9 Step 9: assemble II.6

`proposition2_6_test09_assembly` makes the literal substitution

$$\begin{aligned} P &:= \mathrm{Rect}(AD, BD), \\ Q &:= \mathrm{Sq}(CD), \\ R &:= \mathrm{Rect}(CB, BD), \\ S &:= \mathrm{Sq}(BC), \\ T &:= \mathrm{Sq}(BD), \end{aligned}$$

into the abstract bridge. Steps 6 and 7 are exactly its two hypotheses, so the conclusion is the proposition:

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CD).$$

6.7 Lean Formalization

6.7.1 The public theorem signature

The following is the current compiling public signature, copied from the production source. It is intentionally explicit: the theorem receives the concrete cut diagrams and rectangle/square representatives required by the reused Book II APIs.

```
theorem euclid_proposition_2_6
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hABD : Geo.Between A B D)

    ------------------------------------------------------------
    -- Data for test 06.
    ------------------------------------------------------------

    (E2 H2 L2 X2 : Geo.Point)
    (hRect2 : IsRectangle Geo A D H2 E2)
    (hDH_BD2 : Geo.Congruent D H2 B D)
    (hELE2 : Geo.Between E2 L2 H2)
    (hEXD2 : Geo.Between E2 X2 D)
    (hBXL2 : Geo.Between B X2 L2)
    (hLeftPar2 : IsParallelogram Geo L2 E2 A B)
    (hRightPar2 : IsParallelogram Geo D H2 L2 B)

    -- Square on BD, shared by tests 06 and 07.
    (F G : Geo.Point)
    (hSquareBD : IsSquare Geo B D F G)

    -- Rect(AD,BD).
    (P0 P1 P2 P3 : Geo.Point)
    (hAD_BD :
      IsRectangleContainedBy Geo
        P0 P1 P2 P3 A D B D)

    -- Rect(AB,BD).
    (V0 V1 V2 V3 : Geo.Point)
    (hAB_BD :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 A B B D)

    (E3 H3 L3 X3 : Geo.Point)
    (hRect3 : IsRectangle Geo A B H3 E3)
    (hBH_BD3 : Geo.Congruent B H3 B D)
    (hELE3 : Geo.Between E3 L3 H3)
    (hEXB3 : Geo.Between E3 X3 B)
    (hCXL3 : Geo.Between C X3 L3)
```

```
(hLeftPar3 : IsParallelogram Geo L3 E3 A C)
(hRightPar3 : IsParallelogram Geo B H3 L3 C)

-- Rect(AC,BD).
(Y0 Y1 Y2 Y3 : Geo.Point)
(hAC_BD :
  IsRectangleContainedBy Geo
    Y0 Y1 Y2 Y3 A C B D)

-- The single representative R = Rect(CB,BD).
(Z0 Z1 Z2 Z3 : Geo.Point)
(hCB_BD :
  IsRectangleContainedBy Geo
    Z0 Z1 Z2 Z3 C B B D)

------------------------------------------------------------
-- Data for test 07: II.4 on CD cut at B.
------------------------------------------------------------

(D0 E0 L0 X0 : Geo.Point)
(hSquareCD : IsSquare Geo C D E0 D0)

(hD0L0E0 : Geo.Between D0 L0 E0)
(hD0X0D : Geo.Between D0 X0 D)
(hBX0L0 : Geo.Between B X0 L0)

(hII4LeftPar :
  IsParallelogram Geo L0 D0 C B)

(hII4RightPar :
  IsParallelogram Geo D E0 L0 B)

-- Representatives required internally by II.4.
(J0 J1 J2 J3 : Geo.Point)
(K0 K1 K2 K3 : Geo.Point)

(hCD_CB :
  IsRectangleContainedBy Geo
    J0 J1 J2 J3 C D C B)

(hCD_BD :
  IsRectangleContainedBy Geo
    K0 K1 K2 K3 C D B D)

------------------------------------------------------------
-- Left II.3 decomposition inside II.4.
------------------------------------------------------------

(D1 E1 L1 X1 : Geo.Point)

(hRectLeft :
  IsRectangle Geo D C E1 D1)

(hCE1_BC :
  Geo.Congruent C E1 B C)

(hD1L1E1 : Geo.Between D1 L1 E1)
(hD1X1C : Geo.Between D1 X1 C)
```

```
    (hBX1L1 : Geo.Between B X1 L1)

    (hLeftLeftPar :
      IsParallelogram Geo L1 D1 D B)

    (hLeftRightPar :
      IsParallelogram Geo C E1 L1 B)

    -- Square on BC.
    (S0 S1 : Geo.Point)
    (hSquareBC : IsSquare Geo B C S0 S1)

    ------------------------------------------------------------
    -- Right II.3 decomposition inside II.4.
    ------------------------------------------------------------

    (D2 E4 L4 X4 : Geo.Point)

    (hRectRight :
      IsRectangle Geo C D E4 D2)

    (hDE4_BD :
      Geo.Congruent D E4 B D)

    (hD2L4E4 : Geo.Between D2 L4 E4)
    (hD2X4D : Geo.Between D2 X4 D)
    (hBX4L4 : Geo.Between B X4 L4)

    (hRightLeftPar :
      IsParallelogram Geo L4 D2 C B)

    (hRightRightPar :
      IsParallelogram Geo D E4 L4 B) :

    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B C S0 S1 +
       hilbertParallelogramTerm Geo P0 P1 P2 P3)
      (hilbertParallelogramTerm Geo C D E0 D0)
```

The large signature falls into three geometric groups.

1. The points A, B, C, D, midpoint data, and $A - B - D$ encode the baseline configuration.
2. The first block supplies the two exact decompositions used to reach Step 6: II.3 on AD cut at B, II.1 on AB cut at C, a square on BD, and representatives of the three required rectangles.
3. The second block supplies the II.4 diagram on CD cut at B, including its two internal II.3 configurations, the square on BC, and the intermediate rectangle representatives required by II.4.

The conclusion itself is much smaller than the witness data:

```
HilbertScissorsEq Geo
  (hilbertParallelogramTerm Geo B C S0 S1 +
```

```
    hilbertParallelogramTerm Geo P0 P1 P2 P3)
   (hilbertParallelogramTerm Geo C D E0 D0)
```

and is exactly the formal version of

$$\mathrm{Sq}(BC) + \mathrm{Rect}(AD, BD) = \mathrm{Sq}(CD).$$

6.7.2 The final bridge call

After the two large decompositions have been established, the conceptual end of the proof is only the following substitution into the abstract bridge:

```
exact
  proposition2_6_test08_scissors_bridge
    Geo
    (hilbertParallelogramTerm Geo P0 P1 P2 P3)
    (hilbertParallelogramTerm Geo C D E0 D0)
    (hilbertParallelogramTerm Geo Z0 Z1 Z2 Z3)
    (hilbertParallelogramTerm Geo B C S0 S1)
    (hilbertParallelogramTerm Geo B D F G)
    h06
    h07
```

This makes the proof architecture unusually transparent: geometry produces two normal forms, and the last step is an exact-scissors rearrangement of those normal forms.

6.8 Relationship with Earlier Book II Propositions

6.8.1 Proposition II.1

II.1 is called directly through `euclid_proposition_2_1_two`. Its role is to split $\mathrm{Rect}(AB, BD)$ at the midpoint C. Thus the basic rectangle-distribution theorem remains the additive engine of Book II.

6.8.2 Proposition II.3

II.3 is also called directly. Applied to AD cut at B, it introduces the square on the added segment BD:

$$\mathrm{Rect}(AD, BD) = \mathrm{Rect}(AB, BD) + \mathrm{Sq}(BD)$$

at the exact-scissors level.

6.8.3 Proposition II.4

II.4 is a direct theorem dependency and supplies the expansion of the square on CD after B has been proved to lie between C and D. Since the current II.4 theorem itself composes earlier Book II infrastructure, II.6 inherits that established internal structure rather than rebuilding it.

6.8.4 Proposition II.5

The production file imports `CGJteamLab.Proposition2_5`, so the II.5 module lies immediately below II.6 in the module graph. Nevertheless `euclid_proposition_2_5` is not called by any of the II.6 private helpers or by the public theorem.

Mathematically the two propositions are companions: II.5 concerns an unequal cut inside a bisected segment, whereas II.6 concerns an extension beyond the endpoint. Formally their proof DAGs differ. The current II.5 proof does not call II.4, while II.6 uses II.4 directly as one of its two main normal-form decompositions.

6.9 Relationship with Book I Infrastructure

The classical proof uses I.36 and I.43 explicitly. The Lean proof does not replay those steps proposition-locally. Their geometric effect has already been absorbed into the rectangle, parallelogram, square, and scissors infrastructure on which II.1–II.4 are built.

Book I geometry still remains foundational below the current theorem: right-angle propagation, squares, parallelograms, parallel lines, and the finite-scissors representation all depend on the established Hilbert/Book I layers. The gain is architectural: II.6 can be expressed as a composition of stable Book II content interfaces rather than as a second formalization of Euclid's gnomon diagram.

6.10 Formalization Notes and Hidden Geometric Data

6.10.1 The extension order is proved

The diagram suggests $A - C - B - D$, but the public theorem assumes only midpoint data for C together with $A - B - D$. The private order lemma explicitly proves $C - B - D$ and $A - C - D$. This matters because II.4 can be applied to CD cut at B only after $C - B - D$ is available as a theorem.

6.10.2 Midpoint data have two independent roles

`HilbertIsMidpoint` is not a numerical assertion that C has coordinate one half. Its betweenness component supplies $A - C - B$, while its congruence component supplies $AC \cong CB$. The former feeds the order DAG; the latter feeds rectangle transport.

6.10.3 Representative reuse is deliberate

In the final II.4 call the same Z representative is used twice for $\mathrm{Rect}(CB, BD)$, and the same square $BDFG$ is shared with the Step 6 decomposition. This is not a mathematical restriction on the content identity. It is a deliberate proof-engineering choice that lets the final abstract scissors bridge identify the repeated terms literally and avoids unnecessary representative transports.

6.10.4 The square terms remain geometric representatives

The terms called $\mathrm{Sq}(BC)$, $\mathrm{Sq}(BD)$, and $\mathrm{Sq}(CD)$ are concrete squares supplied through `IsSquare`. They are inserted into the scissors calculus as parallelogram terms. No operation sends a segment to a real number or squares a numerical length.

6.10.5 No gnomon primitive

As in II.5, the formal library does not introduce a special gnomon type. Euclid's shaded remainder is represented extensionally by a formal sum of rectangle and square terms. In II.6 the combination of II.4 with the abstract scissors bridge replaces the gnomon bookkeeping completely.

6.10.6 No cancellation or subtraction of content

The identity strongly resembles a difference-of-squares calculation, but the proof does not subtract a square or cancel equal rectangles. The abstract bridge is additive: it uses only exact scissors equality, addition, associativity/commutativity of the formal sum, symmetry, and transitivity.

6.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`. Both the module build and the full project build were reported clean.

The user-verified transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_6

'Geometry.euclid_proposition_2_6' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The precise status is therefore not "axiom-free." Rather, II.6 introduces no new proposition-specific axiom and no `sorry`; its reported transitive dependency list is exactly the four declarations above and contains no `sorryAx`.

Semantically, the theorem lies in the exact finite-scissors layer. It proves `HilbertScissorsEq` for the supplied configurations. It does not invoke a numerical area function, a segment multiplication operation, or a general cancellation principle for equal content.

6.12 Dependency Summary

The actual theorem-call DAG can be displayed as

$$\begin{gathered}
\text{Hilbert incidence + Euclidean plane layer} \\
\Downarrow \\
\text{Book I square, rectangle, parallelogram, and scissors infrastructure} \\
\Downarrow \\
\text{HilbertRectangle representative/transport API} \\
\Downarrow \\
\text{II.1 binary split} \quad + \quad \text{II.3 rectangle-plus-square} \quad + \quad \text{II.4 square expansion} \\
\Downarrow \\
\text{private II.6 order, transport, composition, and bridge helpers} \\
\Downarrow \\
\text{public theorem } \texttt{euclid_proposition_2_6}.
\end{gathered}$$

The module chain is slightly different:

$$\text{Proposition2\_5 module} \longrightarrow \text{Proposition2\_6 module,}$$

but `euclid_proposition_2_5` itself is not called.

The classical local chain is different again:

$$\text{I.36 + I.43} \longrightarrow \text{gnomon identification} \longrightarrow \text{add square on } BC \longrightarrow \text{II.6.}$$

The formal reconstruction therefore proves the same geometric-content identity through a different DAG rather than translating Euclid line by line.

New geometric axiom in II.6:	no
New content axiom in II.6:	no
New proposition-local helper theorem:	yes, nine private helpers
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.1, II.3, II.4
Imports II.5 module:	yes
Calls `euclid_proposition_2_5`:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `sorryAx`:	absent in reported audit

6.13 Conclusion

Proposition II.6 is a strong checkpoint for the reuse of the Book II rectangle calculus. Classically, Euclid proves the identity by constructing a gnomon inside the square on CD and comparing its pieces through I.36 and I.43. The current Lean proof reaches the same content equality without a gnomon primitive and without numerical algebra.

Its architecture is instead a composition of three already established Book II mechanisms: II.3 decomposes the rectangle on AD, BD, II.1 splits the old segment AB at its midpoint, and II.4 expands the square on CD at the point B. Midpoint congruence normalizes the repeated cross rectangles, and a small abstract scissors lemma performs the final associative-commutative rearrangement. The result is an exact `HilbertScissorsEq` proof with no new axiom, no `sorry`, no segment multiplication, and no numerical area.

Chapter 7

Proposition II.7

7.1 Introduction

Proposition II.7 returns from the midpoint identities II.5–II.6 to an arbitrary division point. Let C lie between A and B. Euclid's statement permits either of the two parts to be chosen as the "said segment." The current Lean theorem chooses AC. In geometric-content notation its conclusion is

$$\mathrm{Sq}(AB) + \mathrm{Sq}(AC) = 2\ \mathrm{Rect}(AB, AC) + \mathrm{Sq}(CB).$$

The modern algebraic shadow is immediate if $AC = a$ and $CB = b$:

$$(a + b)^2 + a^2 = 2(a + b)a + b^2.$$

This formula is only a mnemonic for the geometric pattern. Neither Euclid's proposition nor the current Lean theorem defines a numerical multiplication of segment lengths. The formal objects are concrete square and rectangle representatives, and the conclusion is an exact `HilbertScissorsEq`.

The production source begins with

```
import CGJteamLab.Proposition2_4
```

so Proposition II.4 is the immediate module dependency. At theorem-call level the proof uses `euclid_proposition_2_4` and `euclid_proposition_2_3`. It does not call II.1 or II.2 directly; their rectangle-dissection machinery is inherited through II.4.

7.2 Euclid's Statement and Classical Configuration

7.2.1 The statement

Euclid considers a straight line cut at an arbitrary point. The square on the whole together with the square on one chosen part equals twice the rectangle contained by the whole and that chosen part, together with the square on the remaining part [1].

A common classical presentation chooses BC as the selected part and writes

$$\mathrm{Sq}(AB) + \mathrm{Sq}(BC) = 2\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(CA).$$

The current Lean theorem instead chooses AC:

$$\mathrm{Sq}(AB) + \mathrm{Sq}(AC) = 2\ \mathrm{Rect}(AB, AC) + \mathrm{Sq}(CB).$$

These are the two symmetric readings already allowed by the wording of the proposition. The formal theorem therefore does not alter the mathematical content; it fixes one of Euclid's two admissible choices.

7.2.2 The classical diagram

In the classical proof one constructs the square on AB, draws the diagonal, and then draws parallels through the division point C. The resulting configuration contains two complementary parallelograms about the diagonal, a gnomon, and the squares on the two parts. Euclid uses I.43 to compare the complements and then adds the same figures to equal figures until the desired whole-square identity is obtained [1].

The current formal proof does not reproduce that gnomon bookkeeping line-by-line. Figure 7.1 therefore shows the normal forms actually used in Lean rather than pretending that the Lean DAG is Euclid's classical DAG.

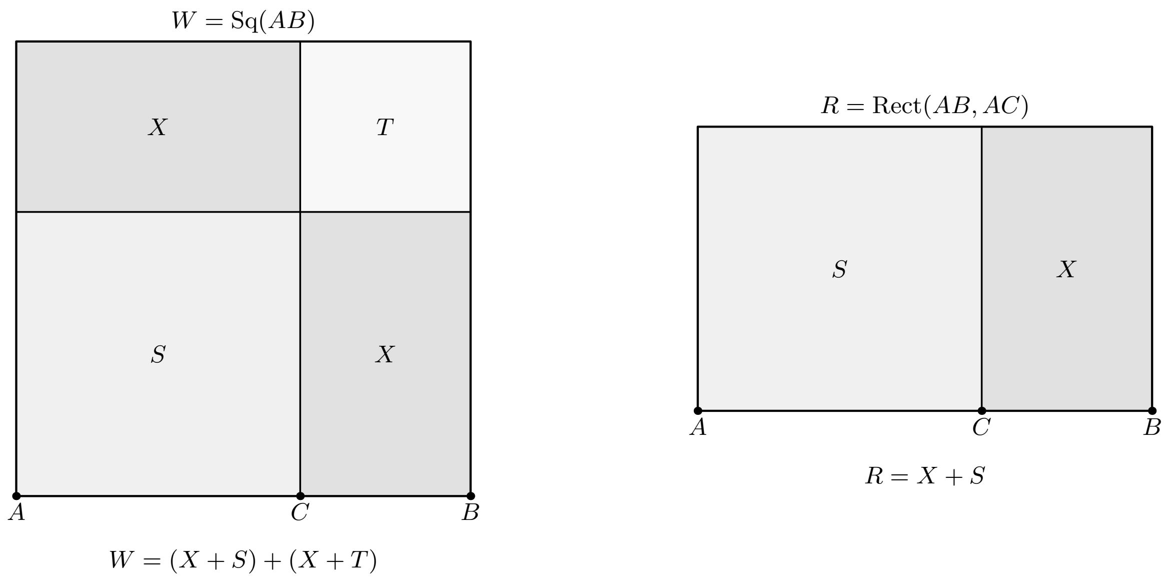


Figure 7.1: The two geometric normal forms used in the current reconstruction. On the left, II.4 decomposes the square W on AB as $(X + S) + (X + T)$, where S and T are the squares on AC and CB, and X is a rectangle contained by AC, CB. On the right, II.3 applied to the reversed division $B - C - A$ gives $R = X + S$, where R is a rectangle contained by AB, AC. Region letters are documentary labels for exact scissors terms; the Lean theorem receives concrete representatives.

7.3 Classical Proof

A convenient reading of Euclid's proof takes the selected part to be BC. Construct the square on AB and the usual parallels through C. The diagonal of the square produces complementary parallelograms; I.43 identifies the relevant pair. After adding the same adjacent rectangle to both, Euclid obtains two equal larger figures, so their sum is twice one of them.

That doubled figure is identified with twice the rectangle contained by AB and BC, because one adjacent side in the constructed rectangle is equal to BC. The remainder of the large square is organized as a gnomon together with the square on BC. Finally, adding the square on the remaining part AC converts that gnomon expression into the whole square on AB plus the square on BC. Thus

$$\mathrm{Sq}(AB) + \mathrm{Sq}(BC) = 2\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC).$$

The classical local dependency picture is therefore schematically

$$\begin{array}{c}\text{I.46 and I.31: construct square and parallels}\\ \Downarrow\\ \text{I.43: complementary parallelograms are equal}\\ \Downarrow\\ \text{gnomon and common-figure additions}\\ \Downarrow\\ \text{II.7.}\end{array}$$

The present Lean proof has a different DAG. It consumes already formalized Book II identities instead of reconstructing the gnomon from Book I each time.

7.4 The Geometric Identity in the Current Formalization

The baseline assumption is

```
(A B C : Geo.Point)
(hACB : Geo.Between A C B)
```

so the formal order is exactly

$$A - C - B.$$

No midpoint condition is present. This is the main geometric difference from II.5 and II.6.

The public conclusion is

$$\begin{aligned}&\mathrm{Sq}(AB) + \mathrm{Sq}(AC)\\ &\quad \simeq_{\mathrm{sc}} \big(\mathrm{Rect}(AB, AC) + \mathrm{Rect}(AB, AC)\big) + \mathrm{Sq}(CB).\end{aligned}$$

The notation $2\ \mathrm{Rect}(AB, AC)$ in the prose means two copies in a formal scissors sum. There is no scalar multiplication operation on content terms in the proof.

The actual Lean conclusion is even more concrete. The square on AB is the supplied square ABE_0D_0; the square on AC is supplied in reversed orientation as $CAFG$; the square on CB is $CBHK$; and the two final rectangles on AB, AC are literally the same supplied representative $U_0U_1U_2U_3$ written twice.

7.5 Reusable Book II Infrastructure Used

7.5.1 Proposition II.4: expand the square on the whole

The first normal form comes directly from `euclid_proposition_2_4`. With

$$W := \mathrm{Sq}(AB), \qquad X := \mathrm{Rect}(AC, CB), \qquad S := \mathrm{Sq}(AC), \qquad T := \mathrm{Sq}(CB),$$

the proof obtains

$$\boxed{W \simeq_{\mathrm{sc}} (X+S)+(X+T).}$$

A deliberate implementation choice makes this especially useful: the same concrete representative $T_0T_1T_2T_3$ of $\mathrm{Rect}(AC, CB)$ is supplied to both cross-rectangle positions inside II.4. Thus the two occurrences of X are literally the same scissors term, not merely representatives later proved equivalent.

7.5.2 Proposition II.3: normalize the rectangle on the whole and one part

The second normal form is

$$\boxed{R \simeq_{\mathrm{sc}} X+S,}$$

where

$$R := \mathrm{Rect}(AB, AC).$$

It comes from `euclid_proposition_2_3`, but not in the original orientation $A-C-B$. The proof first reverses the baseline to

$$B-C-A$$

and applies II.3 to that divided segment.

This reversal is geometric bookkeeping, not algebra. The contained-by data for R are reoriented from (AB, AC) to (BA, CA) by congruence symmetry. The cross rectangle is then rotated by `rectangle_contained_by_swap`, producing both the contained-by data needed by II.3 and an exact scissors equality between the rotated and original representatives.

7.5.3 The rectangle swap is geometric, not multiplicative commutativity

The call

```
have hSwapResult :=
  rectangle_contained_by_swap
    Geo
    T0 T1 T2 T3
    A C C B
    hCross
```

is the key representation step. It does not say that a numerical product $AC \cdot CB$ is commutative. It says that cyclically rotating a concrete rectangle changes the ordered side presentation while preserving its exact scissors term.

This is precisely the distinction established in the Book II rectangle API: geometric side-order transport precedes, and is independent of, any later arithmetic of segment lengths.

7.6 Proof Architecture of the Reconstruction

Unlike II.6, the production file introduces no private proposition-local helper declarations. The whole reconstruction is one public theorem containing local `let` abbreviations and `have` steps. The proof architecture is therefore visible directly in the theorem body.

7.6.1 Step 1: name the five scissors terms

The proof introduces

$$\begin{aligned} W &:= \mathrm{Sq}(AB), \\ R &:= \mathrm{Rect}(AB, AC), \\ X &:= \mathrm{Rect}(AC, CB), \\ S &:= \mathrm{Sq}(AC), \\ T &:= \mathrm{Sq}(CB). \end{aligned}$$

These are abbreviations for supplied `HilbertScissorsTerm` objects, not new geometric constructions.

7.6.2 Step 2: obtain the II.4 normal form

The local fact `hW` calls II.4 and proves

$$W \simeq_{\mathrm{sc}} (X + S) + (X + T).$$

The same X representative is passed twice to the II.4 call.

7.6.3 Step 3: reverse the betweenness orientation

From

$$A - C - B$$

the proof derives

$$B - C - A$$

using `HilbertOrder.between_incidence`. This is the exact order required by the chosen II.3 instantiation.

7.6.4 Step 4: reorient the contained-by data for R

The representative $U_0U_1U_2U_3$ initially satisfies

$$\mathrm{Rect}(AB, AC).$$

For the II.3 call its side data are re-read as

$$\mathrm{Rect}(BA, CA).$$

The underlying rectangle is unchanged; only the orientations of the two unoriented segment congruences are reversed with `CongruentSwapSecond`.

7.6.5 Step 5: rotate the cross rectangle

The call to `rectangle_contained_by_swap` produces a cyclic rotation

$$T_0T_1T_2T_3 \longmapsto T_1T_2T_3T_0.$$

The rotated rectangle can then be presented with the ordered containing sides required by II.3. The same call also supplies an exact scissors equality between the two cyclic presentations.

7.6.6 Step 6: apply II.3 on $B - C - A$

The local fact `hRawR` proves

$$R \simeq_{\mathrm{sc}} X_{\mathrm{rot}} + S.$$

The subsequent facts `hRotateCrossBack` and `hR_concrete` transport this to the original cross representative:

$$\boxed{R \simeq_{\mathrm{sc}} X + S.}$$

No representative-uniqueness theorem is needed. The proof knows exactly how the rotated rectangle is related to the original one.

7.6.7 Step 7: add the extra square S

From the II.4 normal form, `hAddS` gives

$$W + S \simeq_{\mathrm{sc}} ((X + S) + (X + T)) + S.$$

7.6.8 Step 8: regroup the formal sum

The only literal equality of terms used in the final bridge is

$$((X + S) + (X + T)) + S = ((X + S) + (X + S)) + T.$$

Lean proves it by

```
have hRegroup :
    ((X + S) + (X + T)) + S
      =
    ((X + S) + (X + S)) + T := by
  ac_rfl
```

so this step uses only associativity and commutativity of the formal sum.

7.6.9 Step 9: replace the two copies of $X + S$

Since `hR` states

$$R \simeq_{\mathrm{sc}} X + S,$$

its symmetry can replace each copy of $X + S$ by R. The proof uses `HilbertScissorsEq.add` twice and obtains

$$((X + S) + (X + S)) + T \simeq_{\mathrm{sc}} (R + R) + T.$$

7.6.10 Step 10: compose

Transitivity finally gives

$$\boxed{W + S \simeq_{\mathrm{sc}} (R + R) + T.}$$

Expanding the abbreviations is exactly the public II.7 conclusion.

This proof is additive from beginning to end. It never cancels a common content term, subtracts one figure from another, or invokes a positivity principle.

7.7 Lean Formalization

7.7.1 The public theorem signature

The following is the current compiling public signature copied from `CGJteamLab/Proposition 2_7.lean`. The amount of witness data reflects the fact that II.7 reuses the configured II.4 and II.3 interfaces rather than hiding their geometry behind an existential wrapper.

```
theorem euclid_proposition_2_7
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C : Geo.Point)

    ------------------------------------------------------------------
    -- II.2 / II.4 diagram: square on AB cut at C.
    ------------------------------------------------------------------

    (D0 E0 L0 X0 : Geo.Point)
    (hSquareAB : IsSquare Geo A B E0 D0)

    (hACB : Geo.Between A C B)
    (hD0L0E0 : Geo.Between D0 L0 E0)
    (hD0X0B : Geo.Between D0 X0 B)
    (hCX0L0 : Geo.Between C X0 L0)

    (hII2LeftPar :
      IsParallelogram Geo L0 D0 A C)
    (hII2RightPar :
      IsParallelogram Geo B E0 L0 C)

    -- Representatives of Rect(AB,AC) and Rect(AB,CB) used by II.4.
    (U0 U1 U2 U3 : Geo.Point)
    (V0 V1 V2 V3 : Geo.Point)

    (hAB_AC :
      IsRectangleContainedBy Geo
        U0 U1 U2 U3 A B A C)

    (hAB_CB :
      IsRectangleContainedBy Geo
        V0 V1 V2 V3 A B C B)

    ------------------------------------------------------------------
    -- II.3 diagram for the left II.4 term, read on B-C-A.
    ------------------------------------------------------------------

    (D1 E1 L1 X1 : Geo.Point)

    (hRectLeft :
      IsRectangle Geo B A E1 D1)

    (hAE1_CA :
      Geo.Congruent A E1 C A)

    (hD1L1E1 : Geo.Between D1 L1 E1)
```

```
  (hD1X1A : Geo.Between D1 X1 A)
  (hCX1L1 : Geo.Between C X1 L1)

  (hLeftLeftPar :
    IsParallelogram Geo L1 D1 B C)
  (hLeftRightPar :
    IsParallelogram Geo A E1 L1 C)

  -- Square on CA, i.e. the same unoriented segment as AC.
  (F G : Geo.Point)
  (hSquareCA : IsSquare Geo C A F G)

  ----------------------------------------------------------------
  -- II.3 diagram for the right II.4 term.
  ----------------------------------------------------------------

  (D2 E2 L2 X2 : Geo.Point)

  (hRectRight :
    IsRectangle Geo A B E2 D2)

  (hBE2_CB :
    Geo.Congruent B E2 C B)

  (hD2L2E2 : Geo.Between D2 L2 E2)
  (hD2X2B : Geo.Between D2 X2 B)
  (hCX2L2 : Geo.Between C X2 L2)

  (hRightLeftPar :
    IsParallelogram Geo L2 D2 A C)
  (hRightRightPar :
    IsParallelogram Geo B E2 L2 C)

  -- Square on CB.
  (H K : Geo.Point)
  (hSquareCB : IsSquare Geo C B H K)

  ----------------------------------------------------------------
  -- One representative of Rect(AC,CB), used twice in II.4.
  ----------------------------------------------------------------

  (T0 T1 T2 T3 : Geo.Point)

  (hCross :
    IsRectangleContainedBy Geo
      T0 T1 T2 T3 A C C B) :

  HilbertScissorsEq Geo
    (hilbertParallelogramTerm Geo A B E0 D0 +
     hilbertParallelogramTerm Geo C A F G)
    ((hilbertParallelogramTerm Geo U0 U1 U2 U3 +
      hilbertParallelogramTerm Geo U0 U1 U2 U3) +
     hilbertParallelogramTerm Geo C B H K)
```

The signature has four geometric blocks.

1. The points A, B, C and `hACB` specify the arbitrary internal cut.

2. The D_0, E_0, L_0, X_0 block supplies the square-on-AB diagram required by II.4, together with representatives of $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$.

3. The D_1, E_1, L_1, X_1 and D_2, E_2, L_2, X_2 blocks are the two II.3 subconfigurations consumed inside II.4, together with the squares on CA and CB.

4. The T_0, T_1, T_2, T_3 block supplies one representative of $\mathrm{Rect}(AC, CB)$, deliberately reused twice.

The conclusion contains only the square on AB, the square on AC, two copies of the same rectangle on AB, AC, and the square on CB. Much of the signature is therefore construction witness data inherited from the reusable Book II APIs rather than conceptual data in the final identity.

7.7.2 The two decisive local facts

The mathematical core of the production proof is compact:

```
have hW :
    HilbertScissorsEq Geo
      W
      ((X + S) + (X + T)) := by
  ...
  exact euclid_proposition_2_4 ...

...

have hR :
    HilbertScissorsEq Geo
      R
      (X + S) := by
  ...
```

Everything after these two normal forms is an exact-scissors rearrangement. This is the central architectural fact of II.7.

7.8 Relationship with Earlier Book II Propositions

7.8.1 Proposition II.3

II.3 is called directly on the reversed divided segment $B - C - A$. Its role is to convert the rectangle contained by the whole AB and the selected part AC into

$$\mathrm{Rect}(AC, CB) + \mathrm{Sq}(AC).$$

Because the current II.3 interface has ordered contained-by side data, the proof must explicitly reverse segment orientations and rotate the cross rectangle before the call.

7.8.2 Proposition II.4

II.4 is the main normal-form theorem:

$$\mathrm{Sq}(AB) \simeq_{\mathrm{sc}} (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(AC)) + (\mathrm{Rect}(AC, CB) + \mathrm{Sq}(CB)).$$

This is exactly the decomposition needed for II.7 once one recognizes the first parenthesis as $\mathrm{Rect}(AB, AC)$ by II.3.

7.8.3 Propositions II.1 and II.2

Neither `euclid_proposition_2_1_two` nor `euclid_proposition_2_2` is called directly by II.7. They are, however, below II.4 in the established Book II dependency chain. Thus II.7 inherits their rectangle split and square decomposition indirectly.

This distinction matters: the module/theorem DAG should not be replaced by a vague statement that II.7 "uses all previous propositions."

7.8.4 Propositions II.5 and II.6

II.5 and II.6 are not dependencies of II.7. They form the preceding midpoint block, whereas II.7 again assumes only an arbitrary cut. The production module therefore imports II.4 directly rather than continuing the linear module chain through II.6.

7.9 Relationship with Book I Infrastructure

Classically, II.7 is a gnomon argument centered on I.43 and on the square and parallel constructions supplied by I.46 and I.31. The current Lean theorem does not call those Book I propositions locally. Their geometric consequences have already been absorbed into the square, rectangle, parallelogram, and scissors infrastructure used by II.3 and II.4.

This is another clear example of the difference between the classical DAG and the formal library DAG. The reconstruction proves the same content identity without requiring every proposition to replay Euclid's local construction.

7.10 Formalization Notes and Hidden Geometric Data

7.10.1 The chosen segment differs from a common printed diagram

Many printed versions instantiate the proposition with BC as the selected part. The current Lean theorem selects AC. Because Euclid's statement says "one of the segments," this is not a mathematical strengthening or weakening. It is a concrete orientation choice that must nevertheless be recorded in technical documentation.

7.10.2 Betweenness reversal is explicit

The source assumption is `Geo.Between A C B`. II.3 is used on $B - C - A$, so the reversed betweenness statement is derived explicitly. The formal proof cannot infer the desired orientation merely from the picture.

7.10.3 Squares use unoriented side congruence

The final square on the selected part is supplied as

```
(hSquareCA : IsSquare Geo C A F G)
```

although the prose calls it the square on AC. This is legitimate because the underlying segment is unoriented at the congruence level, but the orientation is visible in the concrete Lean arguments.

7.10.4 The unused final representative V

The II.4 interface requires representatives of both $\mathrm{Rect}(AB, AC)$ and $\mathrm{Rect}(AB, CB)$. The latter, represented by $V_0V_1V_2V_3$, is internal to the II.4 call and does not appear in the final II.7 conclusion. This is an example of witness data required by a reused theorem but not by the mathematical endpoint.

7.10.5 Representative reuse avoids uniqueness transport

The same T rectangle is supplied twice to II.4. The same U rectangle is written twice in the final target. These choices make repeated terms literal. The proof therefore does not call `rectangle_contained_by_unique`.

This is not a restriction on the proposition. It is a proof-engineering normalization that removes unnecessary representative bookkeeping.

7.10.6 No cancellation principle

The algebraic shadow strongly invites subtraction. The Lean proof does not subtract X, S, or any square. It adds S, rearranges a commutative formal sum, and replaces equal-scissors subterms. No proper-part principle, positivity theorem, or content cancellation axiom enters II.7.

7.11 Axiomatic and Semantic Status

The public theorem assumes

```
[HilbertIncidence Geo]
[HilbertEuclideanPlane Geo]
```

and the production file introduces no new proposition-specific `axiom`, `sorry`, or `admit`.

The user-verified transitive axiom audit is

```
#print axioms Geometry.euclid_proposition_2_7
```

```
'Geometry.euclid_proposition_2_7' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Hence the precise status is not "axiom-free." Proposition II.7 adds no new axiom and no `sorry`; its reported transitive dependencies are the four declarations displayed above, with no `sorryAx`.

The public module and the full relevant project build were reported clean after the theorem was added to the project. Semantically, the theorem lies in the exact finite-scissors layer. It proves `HilbertScissorsEq` and does not invoke a numerical area function, segment multiplication, or general equicomplementability as its conclusion.

7.12 Dependency Summary

The theorem-call DAG is

$$\begin{array}{c}
\text{Hilbert incidence + Euclidean plane layer} \\
\Downarrow \\
\text{Book I square, rectangle, parallelogram, and scissors infrastructure} \\
\Downarrow \\
\text{HilbertRectangle contained-by and swap API} \\
\Downarrow \\
\text{II.3 rectangle-plus-square} \quad + \quad \text{II.4 square expansion} \\
\Downarrow \\
\text{betweenness/orientation transport + rectangle rotation} \\
\Downarrow \\
\text{exact-scissors addition + AC regrouping} \\
\Downarrow \\
\text{public theorem } \texttt{euclid_proposition_2_7}.
\end{array}$$

The immediate module chain is simply

$$\text{Proposition2\_4 module} \longrightarrow \text{Proposition2\_7 module.}$$

The classical local chain is different:

$$\text{I.46} + \text{I.31} + \text{I.43} \longrightarrow \text{gnomon comparison} \longrightarrow \text{common-figure additions} \longrightarrow \text{II.7.}$$

New geometric axiom in II.7:	no
New content axiom in II.7:	no
New proposition-local helper declaration:	no; local `haves` only
New reusable public helper theorem:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly reuses earlier Book II theorem:	yes, II.3 and II.4
Directly calls II.1 or II.2:	no
Imports II.5 or II.6:	no
Uses exact `HilbertScissorsEq`:	yes
Uses general equicomplementability as conclusion:	no
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation/subtraction:	no
Uses representative uniqueness:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full relevant project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `sorryAx`:	absent in reported audit

7.13 Conclusion

Proposition II.7 is an especially transparent demonstration of the reusable Book II rectangle calculus. The classical proof organizes the square on the whole through a gnomon and complementary parallelograms. The current Lean proof instead starts from two already established exact-scissors normal forms.

II.4 gives

$$W \simeq_{\mathrm{sc}} (X + S) + (X + T),$$

and II.3, after reversing the divided segment and rotating a rectangle representative, gives

$$R \simeq_{\mathrm{sc}} X + S.$$

Adding S, regrouping the formal sum by associativity and commutativity, and replacing two copies of $X + S$ by R yields

$$W + S \simeq_{\mathrm{sc}} (R + R) + T.$$

Thus the formal proof preserves the synthetic geometric-content character of Book II while avoiding numerical area, segment multiplication, subtraction, and cancellation. II.7 introduces no new axiom and no new public infrastructure; its value is architectural: it shows that II.3 and II.4 are already strong enough to recover another classical gnomon identity by exact scissors composition.

Chapter 8

Proposition II.8

8.1 Introduction

Proposition II.8 is the first proposition in the current Book II reconstruction whose classical statement explicitly asks for four copies of one rectangle. Let C lie between A and B. Euclid extends the line beyond B to a point D such that $BD = BC$. The proposition then asserts

$$4\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

Here the expression "four times" is geometric language for four equal plane figures. In the Lean theorem it becomes four literal copies of one concrete scissors term; there is no scalar multiplication operation on content terms.

A modern algebraic shadow is obtained by writing $AC = a$ and $CB = b$. Then $AB = a + b$ and the produced condition $BD = BC$ gives $AD = a + 2b$, so the mnemonic is

$$4(a + b)b + a^2 = (a + 2b)^2.$$

This identity is documentary only. Neither the classical proof nor the current formal proof introduces numerical area or multiplication of segment lengths.

The production source begins with

```
import CGJteamLab.Proposition2_7
```

so Proposition II.7 is the immediate module dependency. At theorem-call level the proof uses `euclid_proposition_2_4` and `euclid_proposition_2_7`, together with the established rectangle representative API and exact scissors addition, symmetry, and transitivity.

8.2 Euclid's Statement and Classical Configuration

8.2.1 The statement

Euclid's statement says that if a straight line is cut at random, four times the rectangle contained by the whole and one segment, together with the square on the remaining segment, is equal to the square described on the whole and the chosen segment taken together as one straight line [1].

With the current lettering, $A - C - B$ and the chosen part is BC. Euclid produces D beyond B with $BD = BC$. Thus the final square is the square on AD, and the geometric identity is

$$4\ \mathrm{Rect}(AB, BC) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

The phrase "on the whole and the chosen part as one straight line" is therefore not an abstract addition of lengths: it is represented by the actually produced collinear segment AD.

8.2.2 The classical diagram

Euclid produces BD in a straight line with AB, makes $BD = CB$, describes the square on AD, and draws the network of parallels usually called the "double" figure. The proof identifies two families of four equal regions by I.34, I.36, and I.43. Together the eight regions form a gnomon equal to four copies of the rectangle contained by AB and BD. Adding the remaining square on AC fills the square on AD [1].

Figure 8.1 deliberately separates that classical produced segment from the normal forms used by Lean. The middle panel is the II.4 normal form on $A - B - D$; the right panel records the final formal sum rather than pretending that the production proof replays Euclid's eight-region gnomon.

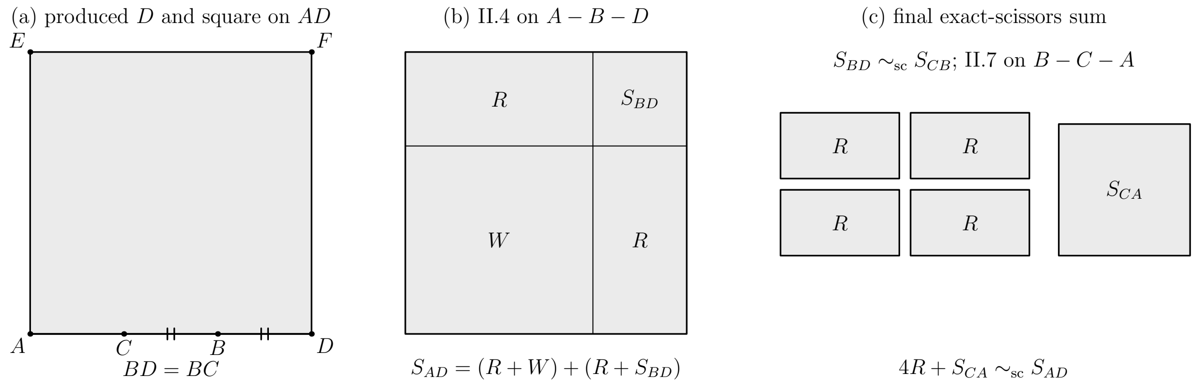


Figure 8.1: The mechanism used to read Proposition II.8. Left: the baseline order $A - C - B - D$ with $BD = BC$ and the square on AD. Middle: II.4 on $A - B - D$ gives $\mathrm{Sq}(AD) = (R + \mathrm{Sq}(BA)) + (R + \mathrm{Sq}(BD))$. Right: after transporting the square on BD to the square on CB and applying II.7 to $B - C - A$, the formal sum becomes four copies of the same rectangle term R plus the square on CA. Region labels denote concrete scissors terms, not numerical areas.

8.3 Classical Proof

Start with $A - C - B$ and produce D beyond B so that $BD = BC$. Construct the square $AEFD$ on AD and draw Euclid's double network of parallels. The proof then has two four-region stages [1].

First, the equalities $CB = BD$ are propagated through opposite sides of the constructed parallelograms. I.36 converts the equal bases into equal parallelograms, while I.43 identifies the relevant complements. Euclid obtains four equal regions DK, CK, GR, RN.

A second use of the same mechanism gives four equal regions AG, MQ, QL, RF. The eight regions together form the gnomon STU, and Euclid proves that this gnomon is four times the region AK. The latter is a rectangle contained by AB and BD, because one of its adjacent sides is equal to BD. Consequently

$$4\ \mathrm{Rect}(AB, BD) = \text{gnomon } STU.$$

The remaining region OH is the square on AC. Adding it to both sides gives

$$4\ \mathrm{Rect}(AB, BD) + \mathrm{Sq}(AC) = \mathrm{Sq}(AD).$$

Finally $BD = BC$, so the rectangle may be read as the rectangle contained by AB and BC.

The classical local dependency picture is therefore

$$\begin{gathered}
\text{I.3: lay off } BD = BC \quad + \quad \text{I.46: square on } AD \\
\Downarrow \\
\text{I.31: the parallel network} \\
\Downarrow \\
\text{I.34 and I.36: side and parallelogram equalities} \\
\Downarrow \\
\text{I.43: equal complements} \\
\Downarrow \\
\text{eight-region gnomon and common-figure addition} \\
\Downarrow \\
\text{II.8.}
\end{gathered}$$

The Lean DAG below is substantially shorter because the relevant Book II normal forms have already been proved.

8.4 The Geometric Identity in the Current Formalization

The public theorem receives the three baseline facts

```
(A B C D : Geo.Point)
(hACB : Geo.Between A C B)
(hABD : Geo.Between A B D)
(hBD_BC : Geo.Congruent B D B C)
```

so the intended order is

$$A - C - B - D, \qquad BD \cong BC.$$

Only the two displayed betweenness relations are primitive inputs; the proof reverses $A - C - B$ explicitly when II.7 needs $B - C - A$.

Let

$$R := \mathrm{Rect}(AB, BD).$$

The same concrete quadrilateral is then re-read as $\mathrm{Rect}(BA, BC)$ using endpoint reversal on the first containing side and the hypothesis $BD \cong BC$ on the second. The public conclusion is exactly

$$(R + R) + (R + R) + \mathrm{Sq}(CA) \simeq_{\mathrm{sc}} \mathrm{Sq}(AD).$$

There is no hidden coefficient object: four occurrences of R appear literally in the Lean term.

This conclusion is `HilbertScissorsEq`, not merely `HilbertScissorsEquicomplementable`. The theorem therefore certifies an exact finite-scissors relation for the configured representatives. It does not assert equality of real numbers.

8.5 Reusable Book II Infrastructure Used

8.5.1 Proposition II.4 on the produced segment

The first normal form is obtained by applying II.4 to the whole segment AD cut at B:

$$\mathrm{Sq}(AD) \simeq_{\mathrm{sc}} (R + \mathrm{Sq}(BA)) + (R + \mathrm{Sq}(BD)).$$

In this invocation the same concrete representative $T_0T_1T_2T_3$ is supplied to both cross-rectangle positions. Consequently the two occurrences of R are literally identical scissors terms.

8.5.2 Proposition II.7 on the reversed cut

From $A - C - B$ the proof derives $B - C - A$ and applies II.7 with the whole segment read as BA. The result is

$$\mathrm{Sq}(BA) + \mathrm{Sq}(CB) \simeq_{\mathrm{sc}} (R + R) + \mathrm{Sq}(CA).$$

The square on BA is deliberately the same concrete square $BAFG$ already used inside the II.4 call. This avoids a second representative-transport step.

8.5.3 Square transport through the rectangle API

II.4 supplies a square on BD, whereas II.7 uses a square on CB. The proof converts both squares to `IsRectangleContainedBy` data with `square_is_rectangle_contained_by_side`. The congruence $BD \cong BC$ is then used to read the CB square as another representative of the same rectangle contained by BD, BD. Finally `rectangle_contained_by_unique` gives

$$\mathrm{Sq}(BD) \simeq_{\mathrm{sc}} \mathrm{Sq}(CB).$$

This is representative transport in the established geometric rectangle API; it is not an appeal to numerical equality of squared lengths.

8.5.4 Exact scissors addition and regrouping

After the three normal forms are available, the remainder is formal addition, transitivity, symmetry, and associativity/commutativity of the multiset sum. No cancellation, subtraction, positivity theorem, or numerical content map is used.

8.6 Proof Architecture of the Reconstruction

The production proof can be read in seven steps.

8.6.1 Step 1: reverse the internal cut

The source assumption $A - C - B$ is converted to $B - C - A$ by

```
have hBCA : Geo.Between B C A :=
  (HilbertOrder.between_incidence A C B hACB).2.2.2.2
```

This is the orientation required by the II.7 invocation.

8.6.2 Step 2: re-read the repeated rectangle

The concrete rectangle $T_0T_1T_2T_3$ initially satisfies

$$\mathrm{Rect}(AB, BD).$$

The proof reverses AB to BA with `CongruentSwapSecond` and transports BD to BC by congruence transitivity. The same quadrilateral therefore satisfies

$$\mathrm{Rect}(BA, BC).$$

No scissors theorem is needed because the geometric representative itself is unchanged.

8.6.3 Step 3: apply II.4 to $A - B - D$

The theorem `euclid_proposition_2_4` produces

$$\mathrm{Sq}(AD) \simeq_{\mathrm{sc}} (R + W) + (R + S_{BD}),$$

where

$$W := \mathrm{Sq}(BA), \qquad S_{BD} := \mathrm{Sq}(BD).$$

8.6.4 Step 4: apply II.7 to $B - C - A$

The theorem `euclid_proposition_2_7` produces

$$W + S_{CB} \simeq_{\mathrm{sc}} (R + R) + S_{CA},$$

where

$$S_{CB} := \mathrm{Sq}(CB), \qquad S_{CA} := \mathrm{Sq}(CA).$$

8.6.5 Step 5: transport the square on the congruent produced segment

Using the square-as-rectangle conversion and `rectangle_contained_by_unique`, the proof obtains

$$S_{BD} \simeq_{\mathrm{sc}} S_{CB}.$$

This is the formal counterpart of Euclid's final use of $BD = BC$.

8.6.6 Step 6: substitute and regroup

Scissors addition first replaces S_{BD} by S_{CB} inside the II.4 normal form. Associativity and commutativity then expose the summand $W + S_{CB}$:

$$(R + W) + (R + S_{CB}) = (R + R) + (W + S_{CB}).$$

The proof uses `ac_rfl`; this is equality of the formal multiset sum, not a new geometric theorem.

8.6.7 Step 7: replace the II.7 block and close

Replacing $W + S_{CB}$ by the II.7 normal form gives

$$\begin{aligned}\mathrm{Sq}(AD) &\simeq_{\mathrm{sc}} (R+R) + ((R+R) + S_{CA}) \\ &= ((R+R) + (R+R)) + S_{CA}.\end{aligned}$$

A final symmetry puts the four rectangles plus the square on the left, exactly in the orientation of the public theorem.

8.7 Lean Formalization

8.7.1 The public theorem signature

The current compiling source declares:

```
theorem euclid_proposition_2_8
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]

    (A B C D : Geo.Point)

    (hACB : Geo.Between A C B)
    (hABD : Geo.Between A B D)
    (hBD_BC : Geo.Congruent B D B C)

    -- ==================================================================
    -- II.4 on A-B-D.
    -- ==================================================================

    (D40 E40 L40 X40 : Geo.Point)
    (hSquareAD : IsSquare Geo A D E40 D40)

    (hD40L40E40 : Geo.Between D40 L40 E40)
    (hD40X40D : Geo.Between D40 X40 D)
    (hBX40L40 : Geo.Between B X40 L40)

    (hII4LeftPar :
      IsParallelogram Geo L40 D40 A B)

    (hII4RightPar :
      IsParallelogram Geo D E40 L40 B)

    -- Rect(AD,AB), Rect(AD,BD).
    (U40 U41 U42 U43 : Geo.Point)
    (V40 V41 V42 V43 : Geo.Point)

    (hAD_AB :
      IsRectangleContainedBy Geo
        U40 U41 U42 U43 A D A B)

    (hAD_BD :
      IsRectangleContainedBy Geo
        V40 V41 V42 V43 A D B D)
```

```
-------------------------------------------------------------------
-- Left II.3 branch inside II.4.
-------------------------------------------------------------------

(D41 E41 L41 X41 : Geo.Point)

(hRect41 :
  IsRectangle Geo D A E41 D41)

(hAE41_BA :
  Geo.Congruent A E41 B A)

(hD41L41E41 : Geo.Between D41 L41 E41)
(hD41X41A : Geo.Between D41 X41 A)
(hBX41L41 : Geo.Between B X41 L41)

(hLeft41 :
  IsParallelogram Geo L41 D41 D B)

(hRight41 :
  IsParallelogram Geo A E41 L41 B)

-------------------------------------------------------------------
-- Common square on BA: used here and again as the outer square
-- in the II.7 invocation below.
-------------------------------------------------------------------

(F G : Geo.Point)
(hSquareBA : IsSquare Geo B A F G)

-------------------------------------------------------------------
-- Right II.3 branch inside II.4.
-------------------------------------------------------------------

(D42 E42 L42 X42 : Geo.Point)

(hRect42 :
  IsRectangle Geo A D E42 D42)

(hDE42_BD :
  Geo.Congruent D E42 B D)

(hD42L42E42 : Geo.Between D42 L42 E42)
(hD42X42D : Geo.Between D42 X42 D)
(hBX42L42 : Geo.Between B X42 L42)

(hLeft42 :
  IsParallelogram Geo L42 D42 A B)

(hRight42 :
  IsParallelogram Geo D E42 L42 B)

(H K : Geo.Point)
(hSquareBD : IsSquare Geo B D H K)

-------------------------------------------------------------------
-- Common repeated rectangle:
--
```

```
--   in II.4: Rect(AB,BD),
--   in II.7: Rect(BA,BC).
------------------------------------------------------------------

(T0 T1 T2 T3 : Geo.Point)

(hCross4 :
  IsRectangleContainedBy Geo
    T0 T1 T2 T3 A B B D)

-- ==================================================================
-- II.7 on the reversed division B-C-A.
--
-- The outer square is exactly the common square B-A-F-G above.
-- ==================================================================

(L70 X70 : Geo.Point)

(hGL70F : Geo.Between G L70 F)
(hGX70A : Geo.Between G X70 A)
(hCX70L70 : Geo.Between C X70 L70)

(hII7LeftPar :
  IsParallelogram Geo L70 G B C)

(hII7RightPar :
  IsParallelogram Geo A F L70 C)

-- Second outer representative Rect(BA,CA).
(V70 V71 V72 V73 : Geo.Point)

(hBA_CA :
  IsRectangleContainedBy Geo
    V70 V71 V72 V73 B A C A)

------------------------------------------------------------------
-- Left II.3 branch inside II.7.
------------------------------------------------------------------

(D71 E71 L71 X71 : Geo.Point)

(hRect71 :
  IsRectangle Geo A B E71 D71)

(hBE71_CB :
  Geo.Congruent B E71 C B)

(hD71L71E71 : Geo.Between D71 L71 E71)
(hD71X71B : Geo.Between D71 X71 B)
(hCX71L71 : Geo.Between C X71 L71)

(hLeft71 :
  IsParallelogram Geo L71 D71 A C)

(hRight71 :
  IsParallelogram Geo B E71 L71 C)

(M N : Geo.Point)
```

```
    (hSquareCB : IsSquare Geo C B M N)

    ------------------------------------------------------------------
    -- Right II.3 branch inside II.7.
    ------------------------------------------------------------------

    (D72 E72 L72 X72 : Geo.Point)

    (hRect72 :
      IsRectangle Geo B A E72 D72)

    (hAE72_CA :
      Geo.Congruent A E72 C A)

    (hD72L72E72 : Geo.Between D72 L72 E72)
    (hD72X72A : Geo.Between D72 X72 A)
    (hCX72L72 : Geo.Between C X72 L72)

    (hLeft72 :
      IsParallelogram Geo L72 D72 B C)

    (hRight72 :
      IsParallelogram Geo A E72 L72 C)

    (P Q : Geo.Point)
    (hSquareCA : IsSquare Geo C A P Q)

    ------------------------------------------------------------------
    -- Cross rectangle internal to II.7: Rect(BC,CA).
    ------------------------------------------------------------------

    (Z0 Z1 Z2 Z3 : Geo.Point)

    (hCross7 :
      IsRectangleContainedBy Geo
        Z0 Z1 Z2 Z3 B C C A) :

    HilbertScissorsEq Geo
      (((hilbertParallelogramTerm Geo T0 T1 T2 T3 +
         hilbertParallelogramTerm Geo T0 T1 T2 T3) +
        (hilbertParallelogramTerm Geo T0 T1 T2 T3 +
         hilbertParallelogramTerm Geo T0 T1 T2 T3)) +
       hilbertParallelogramTerm Geo C A P Q)
      (hilbertParallelogramTerm Geo A D E40 D40)
```

The signature is intentionally large because the theorem is configured around concrete II.4 and II.7 diagrams. The meaningful top-level geometric data are small: $A - C - B$, $A - B - D$, and $BD \cong BC$. The remaining points certify the squares, rectangles, cuts, and parallelograms consumed by the already proved Book II propositions.

8.7.2 The decisive local composition

The two principal theorem calls are

```
have hII4 :=
```

```
  euclid_proposition_2_4 Geo A D B ...

have hII7 :=
  euclid_proposition_2_7 Geo B A C ...
```

The ellipses here are documentary abbreviations only; the full public signature above records the actual configured witnesses.

The exact square transport is obtained with

```
have hSquareTransport :
    HilbertScissorsEq Geo
      (hilbertParallelogramTerm Geo B D H K)
      (hilbertParallelogramTerm Geo C B M N) :=
  rectangle_contained_by_unique
    Geo
    B D H K
    C B M N
    B D B D
    hSquareBDContained
    hSquareCBAsBD
```

so no new II.8-specific transport theorem is added to the library.

8.8 Relationship with Earlier Book II Propositions

8.8.1 Proposition II.4

II.4 is called directly on the produced whole AD cut at B. It supplies the square decomposition containing two copies of $\mathrm{Rect}(AB, BD)$, the square on BA, and the square on BD. This is the left normal form of the production proof.

8.8.2 Proposition II.7

II.7 is called directly on the same original arbitrary cut but in reversed orientation $B - C - A$. Its selected part is therefore BC, giving precisely

$$\mathrm{Sq}(BA) + \mathrm{Sq}(CB) \simeq_{\mathrm{sc}} 2\ \mathrm{Rect}(BA, BC) + \mathrm{Sq}(CA).$$

This is the block substituted into the II.4 expansion.

8.8.3 Propositions II.1–II.3

II.1–II.3 are not called directly by II.8. They lie below II.4 and II.7 in the established rectangle-calculus DAG. In particular, the direct proof does not need to reopen the binary rectangle split or the II.3 square specialization.

8.8.4 Propositions II.5 and II.6

II.5 and II.6 are not dependencies of II.8. Their midpoint hypotheses belong to a different Book II block. II.8 uses an arbitrary internal cut and a newly produced congruent external segment, not a midpoint.

8.9 Relationship with Book I Infrastructure

Classically, II.8 explicitly depends on I.3, I.31, I.34, I.36, I.43, and I.46: segment layoff, parallel construction, parallelogram side equalities, equal parallelograms on equal bases, equal complements, and square construction [1]. The present production theorem does not call those propositions locally. Their consequences have already been packaged in the square, rectangle, parallelogram, congruence, and exact-scissors infrastructure consumed by II.4 and II.7.

This is a substantial classical-DAG/Lean-DAG difference. The formal proof is not a transcription of the eight-region gnomon argument. It is a composition of previously verified geometric identities.

8.10 Formalization Notes and Hidden Geometric Data

8.10.1 The line order is split across two hypotheses

The public theorem does not take a single four-point order predicate. It takes $A - C - B$ and $A - B - D$. Together these encode the intended baseline $A - C - B - D$. The proof only derives the reversed $B - C - A$ fact it actually needs.

8.10.2 The same rectangle has two contained-by descriptions

The term $T_0T_1T_2T_3$ first represents $\mathrm{Rect}(AB, BD)$ and later represents $\mathrm{Rect}(BA, BC)$. The first change is endpoint orientation; the second is metric transport through $BD \cong BC$. This is representation data, not commutativity of a primitive segment product.

8.10.3 Representative reuse is deliberate

The same square on BA is supplied to both II.4 and II.7, and the same rectangle term R is supplied repeatedly. This design removes unnecessary applications of representative uniqueness from the final scissors bridge. Only the genuinely different squares on BD and CB require transport.

8.10.4 Four times means four summands

The public conclusion contains four literal occurrences of `hilbertParallelogramTerm` for the same rectangle. No natural-number scalar action is introduced. The proof therefore stays in the finite formal sum language already used throughout the scissors layer.

8.10.5 No cancellation or positivity

Although II.8 looks algebraically like an identity that could be proved by expanding and cancelling terms, the Lean proof performs no cancellation. It uses only forward exact-scissors equalities, addition, transitivity, symmetry, and regrouping. No positivity theorem enters.

8.11 Axiomatic and Semantic Status

The configured public theorem proves exact `HilbertScissorsEq`. It introduces no numerical area function and no multiplication of segment lengths. The formal "algebra" is therefore a calculus of geometric representatives and finite scissors sums.

The explicit axiom audit was

```
'Geometry.euclid_proposition_2_8' depends on axioms: [propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same transitive profile observed for Proposition II.7. In particular, II.8 introduces no new proposition-specific geometric axiom, no new content axiom, and no `sorryAx` dependency.

The semantic layers should therefore be kept distinct:

$$\begin{array}{c}\text{incidence/order/congruence and Euclidean rectangle geometry}\\ \Downarrow \\ \text{concrete square and rectangle representatives} \\ \Downarrow \\ \text{exact finite-scissors relation} \\ \not\Downarrow \\ \text{numerical area or segment multiplication in this proof.}\end{array}$$

8.12 Dependency Summary

The actual production chain is

$$
\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\text{Book I parallelogram/square/scissors infrastructure} \\
\Downarrow \\
\text{HilbertRectangle representative API} \\
\Downarrow \\
\text{II.1–II.3 rectangle calculus} \\
\Downarrow \\
\text{II.4 and II.7} \\
\Downarrow \\
\text{reorientation } \operatorname{Rect}(AB, BD) \to \operatorname{Rect}(BA, BC) \\
\Downarrow \\
\text{square transport } \operatorname{Sq}(BD) \simeq_{\text{sc}} \operatorname{Sq}(CB) \\
\Downarrow \\
\text{exact scissors addition and regrouping} \\
\Downarrow \\
\texttt{euclid\_proposition\_2\_8}.
\end{gathered}
$$

The implementation checkpoint is:

- new geometric axiom: no;
- new content axiom: no;
- new proposition-local theorem declaration: no;
- new reusable helper theorem: no;
- new `HilbertRectangle` theorem: no;
- new Interface / Book Zero / Scissors theorem: no;
- uses numerical area: no;
- uses segment multiplication: no;
- public theorem compiles: yes;
- `lake build CGJteamLab.Proposition2_8`: clean;
- full `lake build`: clean;
- `sorry`/`admit`: no.

8.13 Conclusion

Proposition II.8 is a useful demonstration of the point at which the established Book II API begins to replace long classical gnomon bookkeeping. Euclid's proof constructs an eight-region configuration and proves equalities among its pieces through I.34, I.36, and I.43. The Lean proof instead takes two already verified normal forms: II.4 on the produced segment AD and II.7 on the reversed original cut $B - C - A$.

The only genuinely new local bridge is representational. The segment equality $BD \cong BC$ lets the repeated rectangle and the square on the produced segment be read in the forms required by II.7. Once those transports are explicit, the fourfold rectangle identity follows from exact scissors addition and regrouping.

Thus the algebraic-looking statement

$$4(a + b)b + a^2 = (a + 2b)^2$$

is not the proof object. It is a modern mnemonic for a theorem whose formal content remains entirely synthetic: betweenness, congruence, squares, rectangles contained by segments, and exact finite scissors equality.

Chapter 9

Proposition II.9

9.1 Introduction

Proposition II.9 is a genuine architectural change in the current reconstruction of Book II. Propositions II.1–II.8 were organized around the reusable rectangle calculus and, in their configured public forms, concluded with exact `HilbertScissorsEq`. Proposition II.9 does not call any earlier Book II proposition. Instead, the current source deliberately returns to Euclid's Book I construction and angle argument and then passes through four applications of Proposition I.47 to the scissors/content layer.

Let AB be bisected at C, and let the unequal cut point D lie in the oriented branch

$$A - C - D - B.$$

The proposition asserts that the two squares on the unequal segments AD and DB have the same content as two copies of the square on the half AC plus two copies of the square on the segment CD between the two points of section. In modern notation the familiar shadow is

$$AD^2 + DB^2 = 2(AC^2 + CD^2).$$

This formula is only a mnemonic. The public Lean conclusion neither defines a numerical area nor multiplies segment lengths. It constructs four actual square representatives and relates their formal scissors sums by `HilbertScissorsEquicomplementable`.

The production source begins with the imports

```
import CGJteamLab.Proposition2_9Construction
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.HilbertSquareTransport
import CGJteamLab.Proposition06
import CGJteamLab.Proposition32
import CGJteamLab.Proposition34
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
```

The absence of a Book II import is intentional. The source comment describes the proof as the source-faithful synthetic proof of the oriented branch and records the classical route through I.5, I.29, I.32, I.34, I.6 and four uses of I.47, with explicit scissors bookkeeping.

9.2 Euclid's Statement and Classical Configuration

9.2.1 The statement

Euclid's Proposition II.9 says that if a straight line is cut into equal and unequal segments, then the sum of the squares on the two unequal segments of the whole is double the sum of the square on the half and the square on the straight line between the two points of section [1].

With the current lettering, C is the midpoint of AB and D is the second point of section. The formal theorem fixes the orientation $C-D-B$, so the unequal segments of the whole are AD and DB, the half is AC, and the segment between the points of section is CD. Thus the geometric statement is

$$\mathrm{Sq}(AD) + \mathrm{Sq}(DB) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD),$$

where the equality is an equality of geometric content, not an equality of real-valued areas.

9.2.2 The classical construction

Euclid erects CE perpendicular to AB at C and lays off

$$CE = AC = CB.$$

He joins EA and EB, draws DF through D parallel to CE, and through F draws FG parallel to the base AB. The point G lies on CE, so $DFGC$ is the auxiliary parallelogram. Finally he joins AF [1].

The current construction module reconstructs this geometry explicitly. It first produces the perpendicular and copied half, constructs the parallel through D, proves that this carrier meets EB, completes the parallelogram, and then uses order/Pasch arguments to prove the strict positions

$$E-F-B, \qquad C-G-E.$$

These order facts are not decorative: later same-ray and angle-decomposition steps depend on them.

9.3 Classical Proof

The proof has two parts: first Euclid extracts the metric equalities required by the construction; then four Pythagorean applications identify one and the same square on AF in two ways [1].

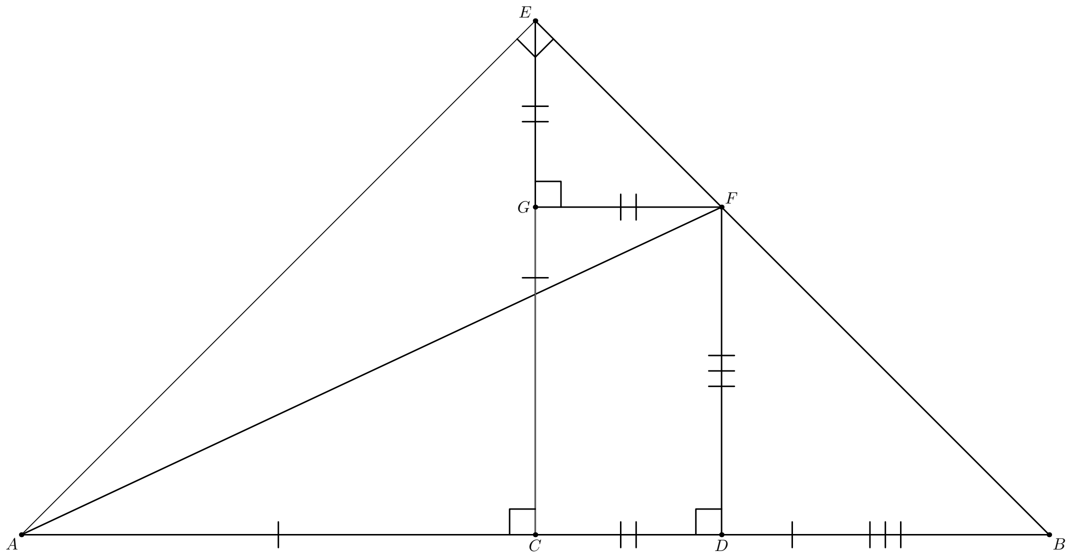
Because $AC = CE$, Proposition I.5 gives equal base angles in triangle ACE. Since the angle at C is right, Proposition I.32 shows that each of those base angles is half a right angle. The same argument in triangle CEB shows that its two base angles are also half right angles. Hence the whole angle AEB is right, and because $E-F-B$, the angle AEF is right as well.

The parallel construction gives the corresponding right-angle configurations at G and D. Using the parallel-angle theorem I.29, the triangle angle sum I.32, and the converse isosceles theorem I.6, Euclid obtains

$$GE = GF, \qquad DF = DB.$$

From the auxiliary parallelogram $DFGC$, I.34 gives

$$GF = CD.$$



$A - C - D - B$

Figure 9.1: The source-faithful configuration for Proposition II.9 in the oriented branch $A - C - D - B$. The perpendicular CE is copied from the half of AB; $DF \parallel CE$ and $FG \parallel CD$ form the parallelogram $DFGC$ with $E - F - B$ and $C - G - E$. Euclid's angle argument gives $GE = GF$ and $DF = DB$, while I.34 gives $GF = CD$. The four right triangles used by I.47 are ACE, EGF, AEF, and ADF. The auxiliary I.32 witnesses used by Lean are omitted from the drawing to keep the classical mechanism visible.

Thus GE, GF, and CD represent the same segment magnitude, while DF can be replaced by DB.

Now apply I.47 four times. In triangle ACE, right at C,

$$\mathrm{Sq}(AE) = \mathrm{Sq}(AC) + \mathrm{Sq}(CE) = 2\,\mathrm{Sq}(AC).$$

In triangle EGF, right at G,

$$\mathrm{Sq}(EF) = \mathrm{Sq}(EG) + \mathrm{Sq}(GF) = 2\,\mathrm{Sq}(CD).$$

In triangle AEF, right at E,

$$\mathrm{Sq}(AF) = \mathrm{Sq}(AE) + \mathrm{Sq}(EF) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD).$$

Finally, in triangle ADF, right at D,

$$\mathrm{Sq}(AF) = \mathrm{Sq}(AD) + \mathrm{Sq}(DF) = \mathrm{Sq}(AD) + \mathrm{Sq}(DB).$$

Comparing the two descriptions of the square on AF proves II.9.

The classical local dependency picture is therefore

$$\begin{array}{c} \text{I.11 + I.3: perpendicular and copied half} \\ \Downarrow \\ \text{I.31: parallel through } D \quad + \quad \text{join } EA, EB, AF \\ \Downarrow \\ \text{I.5 + I.29 + I.32 + I.6: } GE = GF,\ DF = DB \\ \Downarrow \\ \text{I.34: } GF = CD \\ \Downarrow \\ \text{four applications of I.47} \\ \Downarrow \\ \text{II.9.} \end{array}$$

9.4 The Geometric Identity in the Current Formalization

The public theorem assumes

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hCDB : Geo.Between C D B)
```

so the midpoint package supplies $A - C - B$ and $AC \cong CB$, while the second hypothesis selects the oriented branch $C - D - B$. The proof derives the required four-point order $A - C - D - B$ rather than taking it as a primitive datum.

The theorem then constructs concrete squares on AD, DB, AC, and CD. If their formal terms are denoted by

$$S_{AD}, \quad S_{DB}, \quad S_{AC}, \quad S_{CD},$$

the conclusion has the shape

$$S_{AD} + S_{DB} \simeq_{\text{ec}} (S_{AC} + S_{AC}) + (S_{CD} + S_{CD}).$$

Here $\simeq_{\text{ec}}$ denotes the relation formalized by `HilbertScissorsEquicomplementable`.

This distinction is essential. Unlike the configured theorems II.1–II.8, II.9 does *not* conclude `HilbertScissorsEq`. Nor does it conclude an equality of numerical areas. Its four Pythagorean blocks inherit the content strength of the current formalization of I.47 and are composed by symmetry, transitivity, additive compatibility, and transport between square representatives.

9.5 Reusable Infrastructure Used

9.5.1 The permanent II.9 construction layer

The file `Proposition2_9Construction.lean` is a separate geometric layer. Its public helpers progressively establish the source diagram rather than mixing construction with the final content proof. In particular, `proposition2_9_G_between_CE` returns points E, F, G with

$$E - F - B, \qquad C - G - E, \qquad A - C - D,$$

together with the right angle at C, $CE \cong CB$, the parallelogram $DFGC$, and its opposite-side congruences.

The construction module itself imports I.31, I.34, and I.47 infrastructure. Its initial perpendicular/copy helper packages the I.11 + I.3 construction, while its later order proofs use the established Hilbert incidence/order and Pasch machinery to locate F and G on the correct open segments.

9.5.2 Angle decomposition rather than numerical angle measure

The source proof repeatedly says that an angle is half a right angle and later subtracts equal parts from equal right angles. The formal proof does not introduce degrees or real-valued angle measures. It uses `HilbertRayMeetsSegment` as the interior-ray witness and the reusable lemmas

```
hilbert_angleDecomposition_halves_congruent_of_whole_congruent
hilbert_angleDecomposition_angle_addition_interior
hilbert_angleDecomposition_angle_subtraction
hilbert_angleDecomposition_angle_subtraction_right
```

from `HilbertAngleDecomposition.lean`.

9.5.3 Square transport

The four I.47 calls do not automatically choose the same square representative on a common side. The proof therefore uses `hilbert_square_transport` to move between squares erected on congruent segments and between different representatives of the same segment. This is most visible for the two occurrences of the square on AF and the two occurrences of the square on EF.

This is representation transport, not a hidden equation $x^2 = y^2$ in a numerical ring. The transported objects remain concrete `hilbertParallelogramTerm` values in the scissors calculus.

9.5.4 Book I square construction and Pythagoras

The public theorem constructs fresh target squares on AD, DB, AC, and CD with `euclid_proposition_46`. It then imports four already proved I.47 decompositions and transports their square terms into those target representatives. Thus I.46 controls representation while I.47 controls the content identities.

9.6 Proof Architecture of the Reconstruction

9.6.1 Step 1: recover the right-angle configuration

`proposition2_9_right_angle_configuration` starts from the permanent construction and proves the right angles needed later:

$$\angle ADF, \quad \angle EGF, \quad \angle ACE, \quad \angle BCE.$$

The proof propagates right angles through same-ray transport, opposite extensions, and adjacent angles of the parallelogram $DFGC$.

9.6.2 Step 2: reconstruct Euclid's half-right-angle block

`proposition2_9_classical_angle_block` applies I.5 in the two isosceles triangles ACE and CEB, then applies I.32 in oriented exterior-angle form. The resulting points R and S lie on EB and EA and act as explicit interior-ray witnesses for the two angle bisections at C.

Using equality of right angles and the angle-decomposition API, the proof shows that the two components AEC and BEC are congruent. Their sum is therefore a right angle, giving AEB right; the same-ray fact $E - F - B$ then gives AEF right.

9.6.3 Step 3: recover the two isosceles triangles

`proposition2_9_classical_isosceles_blocks` proves

$$GE \cong GF, \qquad DF \cong DB.$$

This is deliberately not done by segment subtraction. For each equality the proof follows the Euclidean angle route: create an I.32 decomposition witness, compare two right angles, subtract congruent components synthetically, obtain congruent base angles, and finish with `euclid_proposition_6`.

The first block introduces a witness T inside the right angle FGC; the second introduces a witness U inside the right angle CDF. These points are formal witnesses for Euclid's angle arithmetic and are not needed in the final statement.

9.6.4 Step 4: build the four Pythagorean blocks

`proposition2_9_four_pythagoras_blocks` adds the I.34 equality $GF \cong CD$ and invokes I.47 on exactly the four right triangles used in the classical proof:

$$ACE, \qquad EGF, \qquad AEF, \qquad ADF.$$

The helper returns all four existential square packages together with the segment congruences needed to identify their legs with the target sides.

9.6.5 Step 5: construct target squares

The public theorem uses I.46 four times to choose concrete squares $S_{AD}, S_{DB}, S_{AC}, S_{CD}$. This separates the statement-level representatives from whatever representatives happened to be produced inside the four I.47 applications.

9.6.6 Step 6: follow the common square on AF

The first content path begins with I.47 on ADF:

$$S_{AF} \simeq_{\text{ec}} S_{DA} + S_{DF}.$$

Square transport replaces DA by AD and DF by DB, yielding

$$S_{AF} \simeq_{\text{ec}} S_{AD} + S_{DB}.$$

The proof reverses this relation, moves to the AF representative used by the AEF block, and applies I.47 again to obtain

$$S_{AD} + S_{DB} \simeq_{\text{ec}} S_{EA} + S_{EF}.$$

9.6.7 Step 7: replace AE and EF by four leg squares

Representative transport first identifies the EA square with the AE square used by triangle ACE and the EF square with the EF square used by triangle EGF. The two remaining I.47 blocks then give

$$S_{EA} + S_{EF} \simeq_{\text{ec}} (S_{CA} + S_{CE}) + (S_{GE} + S_{GF}).$$

Finally the segment congruences

$$CA \cong AC, \quad CE \cong AC, \quad GE \cong CD, \quad GF \cong CD$$

transport these four squares to two copies of S_{AC} and two copies of S_{CD}. Additivity and transitivity close the public theorem.

9.7 Lean Formalization

9.7.1 The public theorem signature

The following is an ASCII transcription of the current public signature:

```
theorem euclid_proposition_2_9_oriented
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hCDB : Geo.Between C D B) :
    exists SAD0 SAD1 SDB0 SDB1 SAC0 SAC1 SCD0 SCD1 : Geo.Point,
      IsSquare Geo A D SAD0 SAD1 /\
      IsSquare Geo D B SDB0 SDB1 /\
      IsSquare Geo A C SAC0 SAC1 /\
      IsSquare Geo C D SCD0 SCD1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A D SAD0 SAD1 +
         hilbertParallelogramTerm Geo D B SDB0 SDB1)
        ((hilbertParallelogramTerm Geo A C SAC0 SAC1 +
          hilbertParallelogramTerm Geo A C SAC0 SAC1) +
         (hilbertParallelogramTerm Geo C D SCD0 SCD1 +
          hilbertParallelogramTerm Geo C D SCD0 SCD1)) := by
```

The suffix `_oriented` is substantive documentation. The public theorem proved in this file is the branch in which the unequal point D lies strictly between the midpoint C and endpoint B. The document must therefore not silently replace the formal statement by an un-oriented theorem over all possible positions of D.

9.7.2 The proposition-local theorem chain

The main source file is organized around four named proposition-local helpers:

```
proposition2_9_right_angle_configuration
proposition2_9_classical_angle_block
proposition2_9_classical_isosceles_blocks
proposition2_9_four_pythagoras_blocks
```

They form a genuine mathematical factorization of the proof. The public statement does not expose the auxiliary points E, F, G, R, S, T, U because they are construction and angle-decomposition witnesses rather than part of the final content identity.

9.7.3 The final content bridge

The last stage contains five conceptually different transports:

(a) I.47 on ADF converts the left-hand target sum into a square on AF;

(b) two AF representatives are identified;

(c) I.47 on AEF converts that square into squares on EA and EF;

(d) I.47 on ACE and EGF expands those two squares into four leg squares;

(e) square transport identifies the four legs with two copies of AC and two copies of CD.

No cancellation of content terms occurs. The proof is a forward chain of synthetic equicomplementability relations.

9.8 Relationship with Earlier Book II Propositions

There is no earlier Book II theorem in the production dependency chain for II.9. In particular, the theorem does not call II.1, II.4, II.7, II.8, or the `HilbertRectangle` rectangle-contained-by calculus that dominated the previous block.

This is not merely an omission at theorem-call level. The production file imports `Proposition2_9Construction`, the angle-decomposition and square transport layers, and Book I propositions I.6, I.32, I.34, I.46, and I.47. The construction file itself imports Book I I.31, I.34, and I.47 infrastructure. No `Proposition2_N` module with $N < 9$ appears in these imports.

Mathematically, II.9 therefore begins a new Book II proof mode in this project: its algebraic-looking square identity is reconstructed from the classical Euclidean configuration instead of being normalized through the previously built rectangle identities.

9.9 Relationship with Book I Infrastructure

The classical and formal DAGs are close in mathematical shape but not identical line-by-line.

Classically, the construction uses I.11 and I.3 for the perpendicular and copied half, I.31 for the parallel, I.5 for isosceles base angles, I.29 for parallel angle transport, I.32 for the triangle angle-sum/exterior-angle mechanism, I.6 for the converse isosceles conclusions, I.34 for opposite sides of the parallelogram, and I.47 four times [1].

The Lean development packages some of these into reusable lower-level infrastructure. For example, "half a right angle" becomes an explicit interior-ray decomposition, and Euclid's subtraction of equal angle parts is realized by angle-decomposition lemmas. Conversely, Lean adds an explicit use of I.46 to construct the four square representatives that occur in the public statement.

Thus the source-faithful mathematical DAG is retained, but the proof is not a literal transcription of Euclid's prose.

9.10 Formalization Notes and Hidden Geometric Data

9.10.1 The oriented branch is part of the theorem

The hypothesis `hCDB : Geo.Between C D B` records the choice of the right-hand unequal cut. Euclid's printed statement is symmetric in the two possible unequal cuts, but the current public theorem proves this one branch. Any later symmetric wrapper should be documented as an additional theorem, not read into the present signature.

9.10.2 Betweenness controls same rays

The order statements $E - F - B$, $C - G - E$, $A - C - D$, and their reversed forms are used to build `HilbertSameRay` witnesses. Those witnesses justify the replacement of an angle arm by a collinear point on the same ray. Lean cannot infer these transports from the appearance of the diagram.

9.10.3 The points R, S, T, U encode angle arithmetic

The exterior-angle theorem I.32 produces explicit points used as interior rays. The first pair R, S produces the two half-right-angle decompositions needed to prove AEB right. Later T and U support the two synthetic angle-subtraction arguments that end in $GE = GF$ and $DF = DB$.

This is the main hidden geometric data of II.9. It replaces numerical statements such as "$45° + 45° = 90°$" and "subtract equal angles from equal right angles" by incidence, betweenness, ray, and angle-congruence witnesses.

9.10.4 Noncollinearity is pervasive

Every right-angle transport, angle congruence, and application of I.6 or I.47 requires nondegeneracy. The proof repeatedly derives noncollinearity from the parallelogram, the perpendicular construction, strict betweenness, and collinearity transitivity. These obligations are mathematically meaningful but are largely suppressed by the classical diagram.

9.10.5 Square identity is weaker than exact scissors equality

The final relation is equicomplementability. It is therefore incorrect to describe II.9 as an exact cut of the left pair of target squares into the four right-hand target squares. The proof composes the equicomplementability data returned by I.47 with square-representative transport. The appropriate prose is "equal content/equicomplementable", not "the same dissection", unless a later theorem strengthens the result.

9.11 Axiomatic and Semantic Status

The current source introduces no numerical area function and no multiplication of segment lengths. Its semantic layers are

$$\begin{array}{c}\text{incidence/order/congruence and Euclidean parallel geometry}\\ \Downarrow \\ \text{right angles, angle decomposition, isosceles and parallelogram facts}\\ \Downarrow \\ \text{concrete square representatives}\\ \Downarrow \\ \text{I.47 equicomplementability and square transport}\\ \Downarrow \\ \text{II.9 equicomplementability conclusion.}\end{array}$$

A source inspection of `Proposition2_9.lean` shows no local `axiom`, `sorry`, or `admit` declaration. The present documentation pass does not contain a recorded output of

```
#print axioms Geometry.euclid_proposition_2_9_oriented
```

so no transitive axiom list is asserted here. This distinction is deliberate: "no new local axiom" is weaker than "axiom-free", and a final audit should record the actual `#print axioms` output when it is run in the project environment.

9.12 Dependency Summary

The production chain is

$$\begin{gathered}\text{Hilbert incidence, order, congruence, Euclidean parallel layer}\\ \Downarrow \\ \text{Book I construction/angle infrastructure: I.3, I.5, I.6, I.11, I.29, I.31, I.32, I.34}\\ \Downarrow \\ \text{Proposition2\_9Construction + HilbertAngleDecomposition}\\ \Downarrow \\ \text{right-angle configuration}\\ \Downarrow \\ \text{classical angle block}\\ \Downarrow \\ \text{classical isosceles blocks } GE = GF,\ DF = DB\\ \Downarrow \\ \text{I.34 + four I.47 blocks}\\ \Downarrow \\ \text{I.46 target squares + HilbertSquareTransport}\\ \Downarrow \\ \text{equicomplementability addition/transitivity}\\ \Downarrow \\ \texttt{euclid\_proposition\_2\_9\_oriented}.\end{gathered}$$

The implementation checkpoint from the inspected source is:

- new proposition-specific geometric axiom declaration: no;
- new proposition-specific content axiom declaration: no;
- proposition-local helper theorem declarations: yes;
- separate reusable infrastructure used: `HilbertAngleDecomposition` and `HilbertSquareTransport`;
- new `HilbertRectangle` theorem used or required: no;
- earlier Book II proposition used: no;
- conclusion is exact `HilbertScissorsEq`: no;
- conclusion is `HilbertScissorsEquicomplementable`: yes;
- uses numerical area: no;
- uses segment multiplication: no;
- local `sorry`/`admit` in `Proposition2_9.lean`: no;
- transitive `#print axioms` audit: not recorded in the materials available to this documentation pass.

9.13 Conclusion

Proposition II.9 is the first clear break in the current Book II documentation from the rectangle-normal-form strategy of II.1–II.8. Its formal proof is built around Euclid's own auxiliary perpendicular, parallel, and right-triangle configuration. The angle argument is reconstructed synthetically, with interior-ray witnesses replacing numerical angle arithmetic, and the two key isosceles equalities are recovered through I.32, angle subtraction, and I.6.

The four applications of I.47 then expose the real content architecture. One square on AF is simultaneously connected with the two target squares on AD, DB and with the four leg squares coming from ACE and EGF. Congruence and square transport identify those legs with two copies of AC and two copies of CD.

Accordingly, the modern identity

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

should remain exactly what it is: a useful algebraic shadow. The theorem proved by Lean is a synthetic statement about order, perpendiculars, parallels, congruent segments, square representatives, and `HilbertScissorsEquicomplementable`; it neither invokes the earlier Book II rectangle calculus nor crosses into numerical area or segment multiplication.

Chapter 10

Proposition II.10

10.1 Introduction

Proposition II.10 is the external-section companion to Proposition II.9, but the current Lean development does not prove it by invoking II.9. Instead it reconstructs Euclid's second source configuration directly. The midpoint C lies between A and B, while the new point D lies beyond B:

$$A - C - B - D.$$

Euclid then erects the same perpendicular at C, constructs a parallel parallelogram, produces two lines to a new intersection G, proves two isosceles equalities, and uses Proposition I.47 four times.

The public theorem states that the two squares on AD and DB have the same content as two copies of the square on AC together with two copies of the square on CD. A useful modern mnemonic is

$$AD^2 + DB^2 = 2(AC^2 + CD^2).$$

If $AC = CB = a$ and $BD = x$, then $AD = 2a + x$ and $CD = a + x$, so the familiar algebraic shadow is

$$(2a + x)^2 + x^2 = 2a^2 + 2(a + x)^2.$$

Neither formula is the formal object proved by Lean. The theorem constructs four actual square representatives and relates their formal sums by `HilbertScissorsEquicomplementable`. It introduces neither numerical area nor multiplication of segment lengths.

The production source begins with

```
import CGJteamLab.Proposition2_9Construction
import CGJteamLab.Proposition34
import CGJteamLab.HilbertAngleDecomposition
import CGJteamLab.Proposition06
import CGJteamLab.Proposition32
import CGJteamLab.Proposition46
import CGJteamLab.Proposition47
import CGJteamLab.HilbertSquareTransport
```

The import of `Proposition2_9Construction` is an infrastructure reuse, not a proof by Proposition II.9. II.10 does not import `Proposition2_9.lean` and does not call `euclid_proposition`

`_2_9_oriented`. It reuses the genuinely generic perpendicular-and-copy helper extracted while II.9 was being developed.

10.2 Euclid's Statement and Classical Configuration

10.2.1 The statement

Euclid's Proposition II.10 begins with a line AB bisected at C and a segment BD added in the same straight line beyond B. In the lettering used here the order is therefore

$$A - C - B - D.$$

Euclid's claim is

$$\mathrm{Sq}(AD) + \mathrm{Sq}(DB) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD),$$

where the equality is an equality of geometric content between plane figures [1].

10.2.2 The classical construction

At C erect CE perpendicular to AB and choose it so that

$$CE = AC = CB.$$

Join EA and EB. Through E draw EF parallel to AD, and through D draw DF parallel to CE. The quadrilateral $CDFE$ is therefore a parallelogram. Produce EB beyond B and FD beyond D until they meet at G, and join AG [1].

The strict positions are

$$E - B - G, \qquad F - D - G.$$

They matter formally. The first controls the extension of the right angle AEB to AEG and the vertical-angle argument at B; the second controls the right-angle and same-ray transports at D and F.

10.3 Classical Proof

The proof first turns the construction into two metric bridges and then compares two decompositions of the same square on AG.

Because $AC = CE$, Proposition I.5 gives

$$\angle EAC = \angle AEC.$$

Since $\angle ACE$ is right, Proposition I.32 makes each of these angles half a right angle. The same argument in the isosceles triangle CEB, using $CE = CB$, shows that $\angle CEB$ and $\angle EBC$ are also half right angles. Therefore

$$\angle AEB$$

is right.

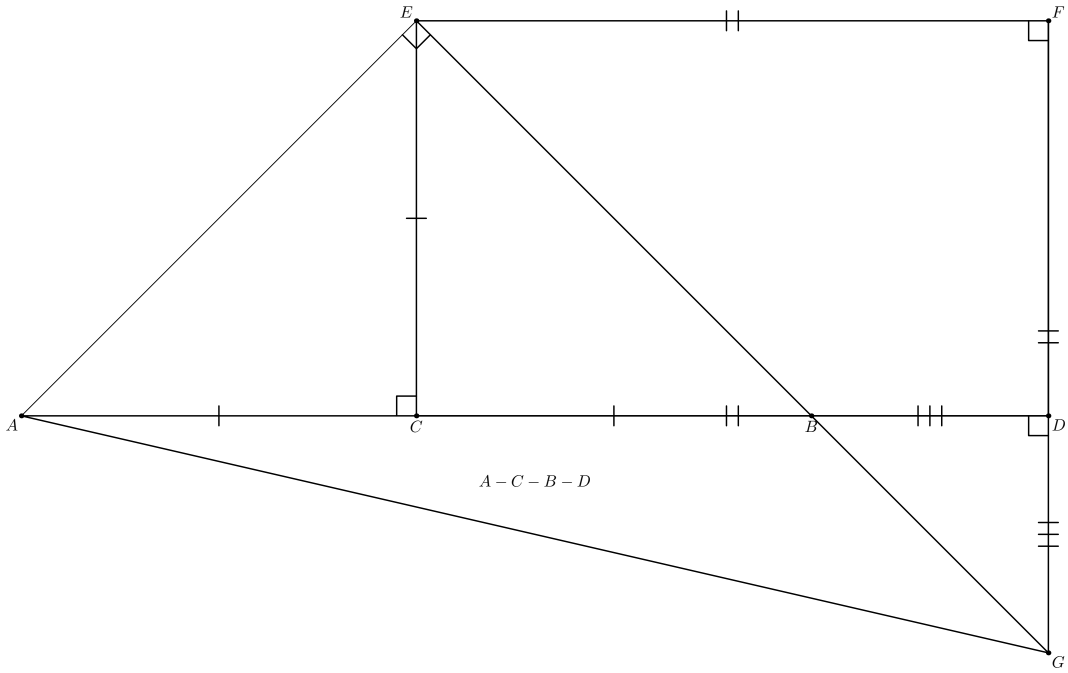


Figure 10.1: The source-faithful configuration for Proposition II.10 in the order $A - C - B - D$. The segment CE is perpendicular to the base and satisfies $CE = AC = CB$. The quadrilateral $CDFE$ is a parallelogram, while the produced lines satisfy $E - B - G$ and $F - D - G$. Euclid's angle argument gives $DB = DG$ and $GF = EF$; I.34 gives $EF = CD$. The four right triangles used by I.47 are ACE, FEG, EAG, and DAG. Auxiliary interior-ray witnesses used by Lean are omitted from the drawing.

Since EB and FD have been produced to G, the vertical-angle theorem I.15 identifies the half-right angle at B with $\angle DBG$. The angle $\angle BDG$ is right by the parallel construction and I.29. Proposition I.32 then shows that the remaining angle $\angle DGB$ is half right. Hence

$$\angle DBG = \angle DGB,$$

and Proposition I.6 gives

$$DB = DG.$$

For the second isosceles bridge, the angle $\angle EGF$ is half right. The angle at F is right because it corresponds to the right angle at C in the parallelogram/parallel configuration. Proposition I.32 therefore makes $\angle FEG$ half right as well, and I.6 gives

$$GF = EF.$$

The opposite sides of parallelogram $CDFE$ are equal by I.34, so

$$EF = CD.$$

Consequently

$$GF = EF = CD.$$

Now apply I.47 four times. In the right triangle ACE,

$$\mathrm{Sq}(AE) = \mathrm{Sq}(AC) + \mathrm{Sq}(CE) = 2\,\mathrm{Sq}(AC).$$

In the right triangle FEG,

$$\mathrm{Sq}(EG) = \mathrm{Sq}(FE) + \mathrm{Sq}(FG) = 2\,\mathrm{Sq}(CD).$$

Because $E - B - G$, the right angle AEB extends along the same ray to give $\angle AEG$ right. Thus I.47 in triangle EAG yields

$$\mathrm{Sq}(AG) = \mathrm{Sq}(AE) + \mathrm{Sq}(EG) = 2\,\mathrm{Sq}(AC) + 2\,\mathrm{Sq}(CD).$$

At D, the parallel/right-angle construction and the collinear order on the base give $\angle ADG$ right. Hence I.47 in triangle DAG gives

$$\mathrm{Sq}(AG) = \mathrm{Sq}(AD) + \mathrm{Sq}(DG) = \mathrm{Sq}(AD) + \mathrm{Sq}(DB).$$

Comparing these two descriptions of the square on AG proves II.10.

The classical local dependency picture is therefore

$$\begin{gathered}
\text{I.11 + I.3: perpendicular at } C \text{ and copied half}\\
\Downarrow\\
\text{I.31: parallels } EF \parallel AD,\ DF \parallel CE\\
\Downarrow\\
\text{Post.5: produced } EB \text{ and } FD \text{ meet at } G\\
\Downarrow\\
\text{I.5 + I.15 + I.29 + I.32 + I.6: } DB = DG,\ GF = EF\\
\Downarrow\\
\text{I.34: } EF = CD\\
\Downarrow\\
\text{four applications of I.47}\\
\Downarrow\\
\text{II.10.}
\end{gathered}$$

10.4 The Geometric Identity in the Current Formalization

The public theorem assumes

```
(A B C D : Geo.Point)
(hMidC : HilbertIsMidpoint Geo C A B)
(hABD : Geo.Between A B D)
```

The midpoint package supplies $A - C - B$ and $AC \cong CB$. The second hypothesis places D beyond B. The proof then derives, among other order facts,

$$A - C - B - D, \qquad C - B - D, \qquad A - C - D, \qquad D - B - A.$$

The theorem constructs concrete squares on AD, DB, AC, and CD. If their formal terms are denoted by

$$S_{AD}, \quad S_{DB}, \quad S_{AC}, \quad S_{CD},$$

the conclusion is

$$S_{AD} + S_{DB} \simeq_{\mathrm{ec}} (S_{AC} + S_{AC}) + (S_{CD} + S_{CD}),$$

where $\simeq_{\mathrm{ec}}$ denotes `HilbertScissorsEquicomplementable`.

This conclusion is not `HilbertScissorsEq`. The proof does not claim that one explicit finite dissection of the two left-hand target squares is the four right-hand target squares. Rather, the current I.47 theorem supplies equicomplementability, and II.10 composes four such Pythagorean blocks with square-representative transport.

10.5 Reusable Infrastructure Used

10.5.1 A generic perpendicular-and-copy construction

The helper

```
proposition2_9_erect_perpendicular
```

is defined in `Proposition2_9Construction.lean`, but its statement is not specific to II.9. It packages the source operations I.11 and I.3: given a nondegenerate segment, it produces a point off its carrier so that the new segment is perpendicular to the old one and congruent to a prescribed segment.

II.10 reuses this helper to construct E with $CE \perp AC$ and $CE \cong AC$. The midpoint congruence then gives $CE \cong CB$.

10.5.2 Parallelogram geometry

The formal proof constructs the fourth vertex F of the parallelogram $CDFE$ through the established Euclidean parallelogram interface. This packages the parallel-existence geometry corresponding to Euclid's I.31. The resulting `IsParallelogram` object supplies the two parallel pairs, while Proposition I.34 supplies opposite-side congruences such as

$$CD \cong FE.$$

Right-angle propagation through the parallelogram uses the reusable theorem `parallelogram_adjacent_right_angle`. This is the formal interface form of the I.29 parallel-angle mechanism needed in the source proof.

10.5.3 Angle decomposition without numerical measure

Euclid repeatedly reasons with "half a right angle" and subtracts equal angle components from equal right angles. The Lean proof introduces no degrees or real-valued angle measure. It uses explicit `HilbertRayMeetsSegment` witnesses together with

```
hilbert_angleDecomposition_halves_congruent_of_whole_congruent
hilbert_angleDecomposition_angle_addition_interior
hilbert_angleDecomposition_angle_subtraction
```

from `HilbertAngleDecomposition.lean`.

The exterior-angle form of I.32 supplies the points used to witness these decompositions. This is the formal replacement for Euclid's compressed angle arithmetic.

10.5.4 Directed intersection and plane separation

The source says that EB and FD, when produced in the directions B and D, meet by Postulate 5. The formal proof must recover more than the bare existence of an intersection: it needs

$$E - B - G, \qquad F - D - G.$$

The helper `proposition2_10_oriented_intersection` proves this directed form. It uses Euclidean parallel uniqueness together with same-side/opposite-side information, segment-line intersection, line uniqueness, and order trichotomy. This is a characteristic example of hidden formal data that the classical drawing suppresses.

10.5.5 Book I square construction, Pythagoras, and square transport

The proof invokes `euclid_proposition_47` four times and later uses `euclid_proposition_46` to choose the four square representatives appearing in the public theorem. Since independent I.47 calls may choose different squares on the same or congruent side, the reusable theorem `hilbert_square_transport` identifies their content.

This is representation transport. It is not an equation in a numerical ring and does not introduce a primitive square operation on segment lengths.

10.6 Proof Architecture of the Reconstruction

10.6.1 Step 1: normalize the external order

`proposition2_10_order` starts from the midpoint and extension hypotheses and establishes the order package needed later:

$$A - C - B, \qquad C - B - D, \qquad A - C - D, \qquad D - B - A.$$

The source statement only draws these positions; the formal proof records them individually because later ray transports depend on orientation.

10.6.2 Step 2: erect the perpendicular and copy the half

`proposition2_10_configuration` constructs E and proves

$$CE \cong AC, \qquad CE \cong CB,$$

together with noncollinearity and the right angles at C required for the two isosceles triangles ACE and ECB.

10.6.3 Step 3: build the auxiliary parallelogram

`proposition2_10_parallelogram` constructs F so that

$$CDFE$$

is a parallelogram. The helper returns the parallel pairs, the opposite-side congruences

$$CD \cong FE, \qquad DF \cong EC,$$

and the right angle at F that will later enter triangle FEG.

10.6.4 Step 4: reconstruct Euclid's first angle block

`proposition2_10_classical_angle_block` applies I.5 in triangles ACE and CEB and uses I.32 to create explicit angle-decomposition witnesses. It proves

$$\angle AEC \cong \angle BEC$$

and hence establishes the right angle AEB.

The formal witnesses R and S lie on the appropriate produced sides and encode the two half-right-angle decompositions. They are construction data, not part of the final theorem statement.

10.6.5 Step 5: produce the intersection in the correct direction

`proposition2_10_oriented_intersection` produces G and proves the strict source orders

$$E - B - G, \qquad F - D - G.$$

The proof does not accept an arbitrary intersection and then reason from the picture. It proves that the intersection lies on the required extensions.

10.6.6 Step 6: recover the two isosceles equalities

`proposition2_10_classical_isosceles_blocks` follows the source angle route to obtain

$$DB \cong DG, \qquad GF \cong EF.$$

At B, I.15 supplies the vertical-angle bridge; at D the parallel configuration gives a right angle; I.32 identifies the remaining half-right angle, and I.6 yields $DB \cong DG$.

The second block transports the corresponding half-right angle to triangle FEG, uses the right angle at F, and again finishes with I.6. Local interior-ray witnesses introduced by I.32 make Euclid's subtraction of angle parts explicit.

10.6.7 Step 7: package the four Pythagorean blocks

`proposition2_10_four_pythagoras_blocks` establishes the segment identifications

$$CA \cong CE, \qquad FE \cong CD, \qquad GF \cong CD, \qquad DB \cong DG$$

and applies I.47 to exactly four right triangles:

$$ACE, \qquad FEG, \qquad EAG, \qquad DAG.$$

The helper returns the four existential square packages and their `HilbertScissorsEquicomplementable` decompositions.

10.6.8 Step 8: follow the common square on AG

The final theorem uses I.46 to choose statement-level squares $S_{AD}, S_{DB}, S_{AC}, S_{CD}$. It begins from I.47 on DAG:

$$S_{AG} \simeq_{\text{ec}} S_{DA} + S_{DG}.$$

Square transport replaces DA by AD and DG by DB, yielding the left-hand target.

A second representative of the square on AG, coming from triangle EAG, is then identified with the first. I.47 on EAG gives

$$S_{AD} + S_{DB} \simeq_{\text{ec}} S_{EA} + S_{EG}.$$

The remaining I.47 blocks expand these two squares:

$$S_{EA} + S_{EG} \simeq_{\text{ec}} (S_{CA} + S_{CE}) + (S_{FE} + S_{FG}).$$

Finally

$$CA \cong AC, \quad CE \cong AC, \quad FE \cong CD, \quad FG \cong CD$$

transport the four terms to two copies of S_{AC} and two copies of S_{CD}. Additive compatibility and transitivity close the proof. No content cancellation is used.

10.7 Lean Formalization

10.7.1 The public theorem signature

The following is an ASCII transcription of the current compiling source:

```
theorem euclid_proposition_2_10_oriented
    [HilbertIncidence Geo]
    [HilbertEuclideanPlane Geo]
    (A B C D : Geo.Point)
    (hMidC : HilbertIsMidpoint Geo C A B)
    (hABD : Geo.Between A B D) :
    exists SAD0 SAD1 SDB0 SDB1 SAC0 SAC1 SCD0 SCD1 : Geo.Point,
      IsSquare Geo A D SAD0 SAD1 /\
      IsSquare Geo D B SDB0 SDB1 /\
      IsSquare Geo A C SAC0 SAC1 /\
      IsSquare Geo C D SCD0 SCD1 /\
      HilbertScissorsEquicomplementable Geo
        (hilbertParallelogramTerm Geo A D SAD0 SAD1 +
         hilbertParallelogramTerm Geo D B SDB0 SDB1)
        ((hilbertParallelogramTerm Geo A C SAC0 SAC1 +
          hilbertParallelogramTerm Geo A C SAC0 SAC1) +
         (hilbertParallelogramTerm Geo C D SCD0 SCD1 +
          hilbertParallelogramTerm Geo C D SCD0 SCD1)) := by
```

The suffix `_oriented` is substantive. The theorem fixes the external branch $A - C - B - D$ through the hypothesis `hABD : Geo.Between A B D`. A future theorem that packages both orientations would be a separate wrapper.

10.7.2 The proposition-local theorem chain

The production source is factored into

```
proposition2_10_order
proposition2_10_configuration
proposition2_10_parallelogram
proposition2_10_classical_angle_block
```

```
proposition2_10_oriented_intersection
proposition2_10_classical_isosceles_blocks
proposition2_10_four_pythagoras_blocks
euclid_proposition_2_10_oriented
```

This chain mirrors genuine mathematical stages rather than merely splitting a large tactic script. The hidden witnesses E, F, G, R, S and the local I.32 decomposition points are progressively introduced and then eliminated from the final public interface.

10.7.3 The final scissors bridge

The last stage uses only forward equicomplementability operations:

(a) I.47 on DAG connects a square on AG with the two left target squares after congruence transport;

(b) two independently constructed AG squares are identified;

(c) I.47 on EAG replaces AG by the EA and EG squares;

(d) I.47 on ACE and FEG expands those into four leg squares;

(e) square transport identifies the four legs with two copies of AC and two copies of CD.

No subtraction or cancellation in the scissors calculus is used.

10.8 Relationship with Earlier Book II Propositions

The production theorem does not call an earlier Book II proposition. In particular it does not use II.9 as an identity from which II.10 is deduced.

There is one important module-level nuance. The import `Proposition2_9Construction` supplies `proposition2_9_erect_perpendicular`, a generic construction helper encoding I.11 plus I.3. Thus II.10 reuses infrastructure that was extracted during the II.9 development, but it does not reuse the theorem `euclid_proposition_2_9_oriented` or any II.9 content identity.

This distinction is mathematically appropriate. Euclid's guide-level relationship between II.9 and II.10 is that the diagrams and arguments are companions: II.10 moves the unequal point beyond the end of the original segment. The formal library preserves that parallelism by sharing low-level construction machinery while proving the two proposition-level statements separately.

10.9 Relationship with Book I Infrastructure

The classical source uses I.11 and I.3 to erect and copy the perpendicular, I.31 for the parallels, I.29 and Postulate 5 for the produced intersection, I.5 for the isosceles base angles, I.15 for the vertical angle, I.32 for the half-right-angle arguments, I.6 for the two isosceles conclusions, I.34 for opposite sides of the parallelogram, and I.47 four times [1].

The Lean DAG packages several of these steps. The I.11+I.3 construction is the reusable perpendicular helper. I.31 is represented by the fourth-vertex/parallelogram interface. The I.29

right-angle consequences are available through parallelogram right-angle transport. Postulate 5 is replaced at the proof-script level by a Euclidean parallel-uniqueness contradiction together with plane-separation and order machinery that also determines the direction of the intersection.

Thus the formal proof is source-faithful in mathematical architecture but is not a line-by-line transcription of Euclid's prose.

10.10 Formalization Notes and Hidden Geometric Data

10.10.1 The external order is a theorem input

II.10 is not the internal-cut branch of II.9 with names changed. The hypothesis

```
hABD : Geo.Between A B D
```

places D strictly beyond B. From this the proof derives the other betweenness relations required to compare rays from C and D.

10.10.2 The intersection G needs orientation, not just existence

The classical phrase "produce the lines until they meet" hides a major formal obligation. Later angle identities require B to lie between E and G and D to lie between F and G. An arbitrary point lying on both carriers is insufficient.

The formal proof therefore combines side-of-line data, order trichotomy, and line uniqueness to establish the exact orders $E - B - G$ and $F - D - G$.

10.10.3 Same-ray witnesses control angle-arm replacement

Once the strict orders are known, `HilbertSameRay` witnesses justify replacing EB by EG, DB by DA or DC where appropriate, and FD by FG. These transports are invisible in the printed diagram but essential to every right-angle propagation.

10.10.4 The points R, S and later local witnesses encode angle arithmetic

The I.32 exterior-angle theorem produces concrete points that witness interior rays. The first pair supports the proof that AEB is right. Further local witnesses support the two synthetic angle-subtraction arguments leading to $DB = DG$ and $GF = EF$.

No statement of the form "$45° + 45° = 90°$" occurs in the Lean development. The information is carried by betweenness, ray intersection, and angle congruence.

10.10.5 Noncollinearity is structural data

Every use of right-angle transport, I.6, I.15, angle decomposition, and I.47 requires nondegeneracy. The proof repeatedly derives noncollinearity from strict betweenness, the perpendicular construction, and the parallelogram. These are genuine geometric facts, not merely type-checking noise.

10.10.6 Square representatives are not canonical

Four I.47 calls and four final I.46 constructions create several squares on the same or congruent segments. The proof must therefore identify their content explicitly with `hilbert_square_transport`. The final identity is independent of these representation choices, but the proof of that independence is part of the formal content layer.

10.11 Axiomatic and Semantic Status

The production theorem introduces no new proposition-specific geometric axiom and no new content axiom. It contains no `sorry` or `admit`. Its semantic layers are

$$\begin{gathered}\text{incidence/order/congruence and Euclidean parallel geometry}\\ \Downarrow\\ \text{perpendiculars, right angles, angle decomposition, parallelogram facts}\\ \Downarrow\\ \text{concrete square representatives}\\ \Downarrow\\ \text{I.47 equicomplementability} + \text{square transport}\\ \Downarrow\\ \text{II.10 equicomplementability conclusion.}\end{gathered}$$

The transitive axiom audit was run on the compiled production theorem:

```
#print axioms Geometry.euclid_proposition_2_10_oriented
```

and reported

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

These are inherited project dependencies. The precise conclusion is therefore: II.10 adds no new axiom and no `sorry`; it is not literally axiom-free.

The theorem uses neither numerical area nor the later arithmetic of segment multiplication. The algebraic-looking identity remains entirely in the synthetic geometry/content architecture.

10.12 Dependency Summary

The production chain is

$$\begin{gathered}
\text{Hilbert incidence, order, congruence, Euclidean parallel layer} \\
\Downarrow \\
\text{I.11 + I.3 generic perpendicular/copy infrastructure} \\
\Downarrow \\
\text{order + perpendicular configuration + parallelogram } CDFE \\
\Downarrow \\
\text{I.5 + I.15 + I.29/I.32 + angle-decomposition machinery} \\
\Downarrow \\
\text{oriented intersection } E-B-G,\ F-D-G \\
\Downarrow \\
\text{I.6: } DB = DG,\ GF = EF \quad + \quad \text{I.34: } EF = CD \\
\Downarrow \\
\text{four I.47 blocks on } ACE, FEG, EAG, DAG \\
\Downarrow \\
\text{I.46 target squares + HilbertSquareTransport} \\
\Downarrow \\
\text{equicomplementability addition/transitivity} \\
\Downarrow \\
\texttt{euclid_proposition_2_10_oriented}.
\end{gathered}$$

The implementation checkpoint is:

New geometric axiom in II.10:	no
New content axiom in II.10:	no
New proposition-local helper theorem:	yes
New reusable public helper theorem/module:	no
New `HilbertRectangle` theorem:	no
New Interface / Book Zero / Scissors theorem:	no
Directly calls an earlier Book II proposition:	no
Imports II.9 production theorem:	no
Reuses generic helper from `Proposition2_9Construction`:	yes
Conclusion is exact `HilbertScissorsEq`:	no
Conclusion is `HilbertScissorsEquicomplementable`:	yes
Uses numerical area:	no
Uses segment multiplication:	no
Uses content cancellation:	no
Public theorem compiles:	yes
Module build:	clean (user verified)
Full project build:	clean (user verified)
Proposition-local `sorry`/`admit`:	no
Transitive `#print axioms`:	`propext`, `Classical.choice`, `bookZero_nullSegment1`, `Quot.sound`

10.13 Conclusion

Proposition II.10 confirms that the architectural change first visible in II.9 is a genuine new Book II proof mode rather than a one-off exception. The rectangle-normal-form chain of II.1–II.8 is not used. Instead the formalization follows Euclid's external-section construction through perpendiculars, parallels, produced intersections, isosceles angle arguments, and four Pythagorean content identities.

The most delicate new formal feature is the direction of the intersection G. The source drawing makes $E - B - G$ and $F - D - G$ immediate; Lean must prove both. Once that order is available, the remaining geometry follows the classical metric bridges $DB = DG$ and $GF = EF = CD$.

The final common object is the square on AG. One I.47 path connects it to the two target squares on AD and DB; the other connects it to the four leg squares that become two copies of AC and two copies of CD. Consequently the mnemonic

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

is best understood as the algebraic shadow of a synthetic equicomplementability theorem. The current proof does not use II.9 as a proposition, numerical area, or segment multiplication, and the completed axiom audit shows that II.10 introduces no new axiom beyond the established transitive project dependencies.

Appendix A

Greenberg: Foundational and Semantic Control

A.1 Purpose and scope

This appendix records the Greenberg-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.10. As in Book I, the purpose is not to force a one-to-one correspondence between Euclid's proposition numbering and Greenberg's organization. The relevant question is which parts of the formal architecture Greenberg controls and where the synthetic content theory must be supplied by another source.

A.2 Greenberg's role at the beginning of Book II

Greenberg distinguishes numerical area from equal content. He describes the latter through finite dissection and reassembly of polygonal figures but does not develop the full synthetic equal-content calculus; he refers the reader to Hartshorne for that theory [4, 5].

This source boundary, already important in the area propositions of Book I, continues unchanged at the opening of Book II. Greenberg's Euclidean parallel and parallelogram theory is well suited to the geometric shell of rectangle constructions and cuts. It is not the source for the exact scissors relation used by the Lean theorems.

A.3 Comparison table

Table A.1: Euclid II.1–II.10 in Greenberg's Hilbert-style framework

Euclid	**Greenberg counterpart**	**Status in Greenberg**	**Source-audit remarks and relevance to formalization**
II.1	Rectangle dissection supported geometrically by Euclidean parallel and parallelogram theory.	No dedicated Greenberg proof of II.1 was located in the current audit. Greenberg supplies the geometric setting and the distinction between equal content and numerical area, but not the project's exact content calculus.	The formal theorem should therefore be read as Greenberg-compatible geometry plus Hilbert/Hartshorne content semantics plus Lean scissors infrastructure. The exact `HilbertScissorsEq` conclusion is not a theorem attributed to Greenberg.
II.2	Square cut into two rectangles; same Euclidean rectangle/parallelogram shell as II.1.	No separate Greenberg counterpart was located. The source-level role remains the same as for II.1.	Greenberg controls the geometry of the configured square and parallel cut, but the side-order transport and exact scissors equality belong to the formal rectangle/content layer.
II.3	Rectangle and square configuration assembled from the same parallel, parallelogram, and right-angle geometry.	No separate Greenberg counterpart was located. II.3 introduces no new Greenberg-side semantic principle beyond the boundary already identified at II.1.	The formal reduction of II.3 to the binary II.1 theorem is a library architecture fact. Greenberg remains a geometry control source rather than a proof source for the content identity.
II.4	Square on a divided line assembled from square, rectangle, parallel, and parallelogram geometry.	No dedicated Greenberg proof of II.4 was located in the current Book II audit. The source continues to supply the Euclidean geometric shell and the distinction between equal content and numerical area.	The formal theorem's exact `HilbertScissorsEq` and its composition of earlier rectangle identities are project-level content infrastructure, not a result attributed to Greenberg.
II.5	Bisected-segment configuration with an additional internal cut.	Greenberg supplies the Hilbert-style order, congruence, and Euclidean geometry needed to formulate midpoint and rectangle configurations, but no dedicated synthetic content proof of II.5 was located in the current audit.	The Lean midpoint transport and exact rectangle/scissors composition remain inside the formal project. No numerical area interpretation is imported from Greenberg.
II.6	Bisected segment extended beyond an endpoint, with the corresponding square and rectangle configuration.	The source remains useful for order, midpoint, parallel, and rectangle geometry. No dedicated Greenberg exact-content counterpart to II.6 was located in the current audit.	The formal use of II.1, II.3, II.4 and exact scissors addition is a Lean-library architecture fact. Greenberg is not cited as supplying that proof.
II.7	Arbitrary internal cut of a segment, square on the whole, squares on the parts, and a rectangle contained by the whole and one part.	No separate Greenberg proposition-level proof was located in the current audit. The same geometry/content boundary identified at II.1 remains in force.	The current Lean proof composes exact II.4 and II.3 and performs geometric side-order transport. Its exact scissors conclusion and representative reuse belong to project infrastructure rather than to a Greenberg content calculus.

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Table A.1 – continued from previous page

Euclid	Greenberg counterpart	Status in Greenberg	Source-audit remarks and relevance to formalization
II.8	Arbitrary internal cut followed by an endpoint extension $BD \cong BC$, with a square on the produced whole AD and rectangle/square representatives.	No separate Greenberg proposition-level proof was located in the current audit. The same Euclidean geometry/content boundary remains in force.	Greenberg controls the order, congruence, parallel, and rectangle geometry and the warning against numerical-area substitution. The exact II.4/II.7 composition, square representative transport, and fourfold scissors sum are project infrastructure.
II.9	Bisected segment with a second internal cut, perpendicular and parallel construction, auxiliary parallelogram, right triangles, and squares.	No separate Greenberg proposition-level proof of II.9 was located in the current audit. Greenberg remains a control source for Hilbert-style order, congruence, perpendicular/parallel geometry, and for the distinction between equal content and numerical area.	The Lean proof returns to a Book I geometric construction and ends in `HilbertScissorsEquicomplementable`, not exact `HilbertScissorsEq`. The Pythagorean/scissors composition and square representative transport are project infrastructure and are not attributed to Greenberg.
II.10	Bisected segment extended beyond an endpoint, with perpendicular/parallel construction, auxiliary parallelogram, produced intersection, right triangles, and squares.	No separate Greenberg proposition-level proof of II.10 was located in the current audit. Greenberg remains a control source for Hilbert-style order, congruence, separation, perpendicular/parallel geometry, and for the distinction between equal content and numerical area.	The formal proof is the external-section companion to II.9. Its directed intersection argument and right-triangle geometry remain on the Greenberg-side geometric layer, while the four I.47 blocks, square transport, and final `HilbertScissorsEquicomplementable` composition are project infrastructure.

A.4 Structural observations from the audit

Three observations should be retained.

- The Euclidean geometry needed to construct rectangles and parallel cuts is naturally expressible in Greenberg's Hilbert-style framework.
- Equality of content is a separate semantic layer. Greenberg's decision not to develop its full synthetic calculus is a source boundary, not a defect to be repaired by importing numerical area formulas.
- The formal project's exact scissors and representative-independence theorems should therefore be documented as project infrastructure, not as Greenberg results.

A.5 II.1–II.8: one stable geometry/content split

Nothing in II.2–II.7 changes Greenberg's role established at II.1. The block uses elementary Euclidean rectangle, square, midpoint, order, and parallel geometry, while the equality-of-content conclusions belong to the separate content calculus. The formal distinctions between binary splitting, side-order rotation, square expansion, midpoint transport, the additive II.7 bridge, and the II.8 endpoint-extension/square-transport composition occur inside the Lean API and do not create new Greenberg source layers.

This is precisely why the appendix should grow by structural blocks rather than by one narrative section per proposition.

A.6 II.9: the same semantic boundary with a different geometric engine

II.9 changes the internal Lean proof architecture without changing Greenberg's source role. The rectangle-normal-form chain of II.1–II.8 is no longer used. Instead the proof develops a perpendicular/parallel construction, isosceles and right-triangle angle geometry, a parallelogram, and four Pythagorean content blocks. These geometric ingredients remain compatible with Greenberg's Hilbert-style Euclidean framework.

The semantic boundary is, however, even more visible than before. The public II.9 theorem lands at `HilbertScissorsEquicomplementable`. Greenberg's distinction between equal content and numerical area continues to be the relevant warning, while the particular formal equicomplementability calculus is not a theorem attributed to Greenberg.

A.7 II.10: the external companion keeps the same semantic boundary

II.10 confirms that the geometric engine introduced at II.9 is not tied to an internal cut. The new point lies beyond the endpoint, so the formal proof must control the produced intersection in the strict orders $E - B - G$ and $F - D - G$. This makes plane separation, line uniqueness, order, and same-ray transport especially visible, all within the Hilbert-style Euclidean geometry for which Greenberg remains a useful control source.

The content conclusion does not move toward numerical area. As in II.9, the public type is `HilbertScissorsEquicomplementable`. The current audit therefore records II.9 and II.10 as a companion pair with different order configurations but the same geometry/content boundary.

A.8 Result of the Greenberg audit

At the II.10 checkpoint Greenberg remains a foundational and geometric control source. He supports the distinction

$$\text{synthetic content} \neq \text{numerical area}$$

and the Euclidean geometry of rectangles, parallels, and parallelograms. For II.1–II.8 the exact finite-dissection conclusions, and for II.9–II.10 the more general equicomplementability conclusions, belong to the formal development and to the Hilbert/Hartshorne content side of the source hierarchy.

A.9 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.10. No new Greenberg-specific boundary appears between these propositions. Later entries should extend the comparison table proposition by proposition, while new narrative sections should be added only when a genuinely new construction, axiomatic, or semantic block appears.

Appendix B

Hartshorne: Equal Content and the Algebra of Areas

B.1 Purpose and scope

This appendix records the Hartshorne-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.10. Its purpose is not to create a second proposition-by-proposition commentary. Instead, it records what Hartshorne contributes to the semantic interpretation of Euclid's rectangle and square equalities, which propositions he singles out, and which source-level distinctions should remain stable as Book II grows.

Hartshorne is especially important here because he separates Euclid's equality of plane figures from numerical area. In Section 22 he defines equal content synthetically and states that the propositions of Book II are valid when Euclid's equality of figures is read in that sense [5].

B.2 Hartshorne's organization of Book II

Hartshorne's treatment places Book II inside the theory of equal content rather than inside a prior arithmetic of numerical lengths. A rectangle described by two segments can therefore play a product-like geometric role without those segments first becoming numbers. This is the background of the familiar interpretation of Book II as an "algebra of areas" [5].

The phrase is structural, not a claim that Euclid has introduced a field of segment lengths. Modern formulas such as

$$a(b + c) = ab + ac$$

are useful mnemonics for the pattern of a geometric decomposition, but the objects compared by Euclid remain plane figures.

Hartshorne's brief Euclid summary explicitly lists II.1 and II.4, but does not list II.2 or II.3 among its selected Book II entries. In particular, II.4 is recorded as the statement that the square on the whole line equals the squares on its two segments plus twice the rectangle on the two segments [5]. The omission of II.2 and II.3 should not be read as a negative judgment about them: his general area discussion states that the results of Book II are valid for equal content.

B.3 Comparison table

Table B.1: Euclid II.1–II.10 in Hartshorne's equal-content framework

Euclid	**Hartshorne counterpart**	**Status in Hartshorne**	**Source-audit remarks and relevance to formalization**
II.1	The rectangle contained by two lines is the sum of the rectangles contained by one of them and the segments of the other.	Hartshorne lists II.1 explicitly in his brief Euclid summary and, more generally, places Book II inside the theory of equal content.	This is the cleanest source match for the formal rectangle-distribution API. The Lean reconstruction strengthens this direct-dissection instance to exact `HilbertScissorsEq`, but that strengthening is a project-level fact and should not be attributed to Hartshorne's general equal-content relation.
II.2	Square on the whole line decomposed into the two rectangles pairing the whole with the two parts.	II.2 is not separately listed in Hartshorne's selected brief Book II summary; it is covered by his general statement that the Book II equalities are valid for equal content.	The source supports the synthetic content reading, while the Lean proof exposes an additional representation issue: the two orders of the containing segments are reconciled geometrically by rectangle rotation rather than by numerical commutativity.
II.3	Rectangle on the whole and one part decomposed into the rectangle on the two parts together with the square on that part.	II.3 is likewise not isolated in the selected brief summary; its semantic status is supplied by Hartshorne's general Book II equal-content result.	The formal theorem is an exact-scissors specialization of the binary II.1 API. The square is used as a representative of the rectangle contained by the same segment twice. Unlike II.2, no side-order transport is required.
II.4	Square on the whole decomposed into the two squares on the parts together with twice the rectangle contained by the parts.	Hartshorne explicitly lists II.4 in his brief Euclid summary, and Section 22 supplies the equal-content semantics for the equality of the figures.	The Lean theorem strengthens the configured case to exact `HilbertScissorsEq`. It composes II.2 with two II.3 instances and treats "twice the rectangle" as two independent representatives of the same rectangle content, without numerical area or segment multiplication.
II.5	The rectangle on the unequal parts of an internally divided bisected segment, together with the square on the displacement from the midpoint, equals the square on the half.	Hartshorne explicitly singles out II.5, together with II.6 and II.11, in his discussion of the area identities used later in the construction of the regular pentagon. The proposition remains within the synthetic equal-content theory rather than numerical area.	For the formal reconstruction the significant point is that the midpoint condition and the rectangle identity still live below segment arithmetic. The current Lean proof reaches exact `HilbertScissorsEq` through earlier Book II rectangle decompositions and geometric transport; the modern difference-of-squares formula is only an interpretive shadow.

Continued on next page

Table B.1 – continued from previous page

Euclid	Hartshorne counterpart	Status in Hartshorne	Source-audit remarks and relevance to formalization
II.6	The rectangle contained by the whole extended line and the added segment, together with the square on the half, equals the square on the half plus the added segment.	Hartshorne explicitly mentions II.6 in the same geometrical-algebra block as II.5 and II.11. He also notes that Proposition III.36 is proved by several applications of II.6 together with I.47.	This gives II.6 a source role stronger than a generic Book II example: it is a reusable equal-content identity in Hartshorne's later Euclidean development. The Lean theorem again strengthens the configured statement to exact `HilbertScissorsEq`, but does so by II.1, II.3, II.4, midpoint transport, and formal scissors addition rather than by numerical algebra.
II.7	Square on the whole together with the square on one chosen part equals twice the rectangle contained by the whole and that part, together with the square on the remaining part.	II.7 is not separately singled out in Hartshorne's brief selected list in the same way as II.4–II.6; its semantic status is nevertheless covered by his general statement that the propositions of Book II are valid for equal content.	The current Lean theorem proves the configured identity by exact `HilbertScissorsEq`, composing II.4 with II.3 on the reversed divided segment and a geometric rectangle swap. The modern formula is only a documentary shadow; no numerical area or segment multiplication is used.
II.8	Four times the rectangle contained by the whole and one part, together with the square on the remaining part, equals the square on the produced segment made from the whole and a congruent copy of the chosen part.	No separate II.8 entry has been identified in the brief selected Book II list used for the present audit; its semantic status is covered by Hartshorne's general statement that Book II equalities are valid for equal content.	The Lean theorem sharpens the configured identity to exact `HilbertScissorsEq`. It composes II.4 with II.7 on the reversed cut and transports the square on BD to the square on CB through rectangle representative uniqueness, without numerical area or segment multiplication.
II.9	Sum of the squares on the unequal parts of a bisected line, expressed through Euclid's right-triangle construction and four uses of the Pythagorean theorem.	Hartshorne's equal-content framework remains the semantic control: Book II square identities are statements about content of plane figures, not numerical area formulas or a pre-existing multiplication of segment lengths.	The current Lean theorem is the first configured Book II proposition in this sequence whose final relation is `HilbertScissorsEquicomplementable` rather than exact `HilbertScissorsEq`. The modern formula $AD^2 + DB^2 = 2(AC^2 + CD^2)$ is therefore only a mnemonic for the geometric-content statement.
II.10	External-section companion to II.9: squares on the produced whole and on the added segment are compared with twice the squares on the half and on the half-plus-added segment.	No separate II.10 entry has been identified in the brief selected Book II list used for the present audit; its semantic status is covered by Hartshorne's general statement that Book II equalities are valid for equal content.	The current Lean theorem, like II.9, concludes `HilbertScissorsEquicomplementable` rather than exact `HilbertScissorsEq`. The modern identity $(2a + x)^2 + x^2 = 2a^2 + 2(a + x)^2$ is only a mnemonic; no numerical area or segment multiplication is used.

B.4 Semantic observations retained from Hartshorne

Several points should remain stable as later propositions are added.

- Euclid's equality of plane figures in the area block should not be silently replaced by equality of real-valued areas.
- The phrase "rectangle contained by" two segments names a geometric figure through its adjacent side magnitudes; it is not itself a numerical multiplication operation.
- Hartshorne's equal content is the appropriate general semantic control for Book II. Exact finite equidecomposability is a stronger local conclusion and must be justified proposition by proposition.
- Modern algebraic identities are useful documentary shadows of the synthetic configurations, not substitutes for the geometric proof.
- Hartshorne's discussion of property (Z) belongs to the converse direction in the theory of content. Nothing in II.1–II.4 requires recovering metric data from a hypothesis of equal content.

B.5 II.1–II.8: from distribution through midpoint identities to endpoint-extension composition

The first eight propositions now display three stages of Book II structure.

II.1–II.4 establish and compose the basic rectangle calculus. II.1 supplies the general additive mechanism, II.2 exposes geometric side-order symmetry, II.3 gives a direct rectangle-plus-square specialization, and II.4 composes those results into the square-of-a-sum pattern.

II.5 and II.6 then add midpoint/order information. They are companion difference-of-squares identities: II.5 treats an unequal cut inside a bisected segment, whereas II.6 treats a segment added beyond the endpoint. Hartshorne explicitly singles out both propositions in his later discussion of geometrical algebra. II.7 removes the midpoint condition again and composes the already established square and rectangle identities for an arbitrary cut. II.8 then adds a produced endpoint segment congruent to one part and combines II.4 with II.7, plus representative transport for the square on that produced segment. The formal project keeps all four results in the same synthetic content layer while allowing their Lean DAGs to differ.

For II.6 the modern mnemonic

$$a^2 + (2a + x)x = (a + x)^2$$

for II.7 the mnemonic

$$(a + b)^2 + a^2 = 2(a + b)a + b^2,$$

and for II.8

$$4(a + b)b + a^2 = (a + 2b)^2$$

should all be read exactly as mnemonics. In the current Lean theorems the repeated rectangles are concrete scissors terms, the square terms are concrete square representatives, and the final equalities are obtained inside the synthetic content layer rather than by numerical algebra.

B.6 II.9: from exact rectangle dissections to equicomplementable square identities

II.9 is the first proposition in the current Book II sequence for which the formal conclusion itself makes the distinction between exact scissors equality and the broader equal-content relation unavoidable. The configured theorems II.1–II.8 land at `HilbertScissorsEq`; II.9 lands at `HilbertScissorsEquicomplementable` because its proof is assembled from the current formal I.47 Pythagorean blocks and square-representative transports.

This does not weaken Hartshorne's semantic role. On the contrary, his interpretation of Euclid's Book II equalities as equal content is exactly the level at which II.9 should be read. The familiar algebraic shadow

$$AD^2 + DB^2 = 2(AC^2 + CD^2)$$

remains explanatory notation only; neither Euclid nor the Lean theorem has introduced numerical area or multiplication in a segment field at this stage.

B.7 II.10: the external member of the Pythagorean pair

II.10 preserves the semantic lesson of II.9 while changing the order configuration. The second point of section is moved beyond the endpoint, and the source proof again builds a common square through four applications of I.47. Hartshorne's equal-content interpretation remains exactly the right level for the result: the identity concerns contents of concrete square figures, not values of a prior numerical area function.

The current Lean implementation makes the II.9–II.10 pairing precise without making one proposition depend on the other. Each theorem reconstructs its own geometric configuration and lands at `HilbertScissorsEquicomplementable`; only lower-level geometric and representation infrastructure is shared.

B.8 Result of the Hartshorne audit

At the II.10 checkpoint Hartshorne supplies the main semantic control for the opening of Book II. His contribution is not a separate proof script for each of II.1–II.10, but a framework in which their equalities are understood as equalities of content between plane figures. II.5 and II.6 are additionally singled out by Hartshorne for their later use in geometrical constructions.

The Lean development refines this framework in two ways that are specific to the current implementation. First, the configured II.1–II.8 theorems land at exact `HilbertScissorsEq`, while II.9–II.10 land at `HilbertScissorsEquicomplementable`. Second, the formal predicate `IsRectangleContainedBy` makes representative independence and side order explicit. These refinements should remain distinguished from claims made by Hartshorne himself.

B.9 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.10. The structural picture now branches at the strength of the formal content relation:

$$\begin{gathered}\text{rectangle/square language}\\ \Downarrow\\ \text{synthetic equal content}\\ \Downarrow\\ \begin{cases}\text{exact } \texttt{HilbertScissorsEq}, & \text{II.1–II.8},\\ \texttt{HilbertScissorsEquicomplementable}, & \text{II.9–II.10}.\end{cases}\end{gathered}$$

Within the Lean rectangle calculus, II.1 is the reusable distributive core, II.2 tests geometric side-order symmetry, II.3 returns to direct specialization of the binary II.1 theorem, and II.4 demonstrates composition. II.5 and II.6 add midpoint/order geometry, II.7 returns to an arbitrary cut, and II.8 adds an endpoint extension and square representative transport. II.9 is the first architectural break in this sequence: it does not reuse a Book II proposition, but reconstructs Euclid's Book I geometric route and then composes four I.47 content blocks. Hartshorne continues to control the semantic reading as equal content; the stronger exact-scissors status of II.1–II.8 remains a proposition-specific feature of the Lean development.

Appendix C

Hilbert: Content and Segment Arithmetic

C.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.10. Hilbert is relevant in two distinct ways: Chapter IV supplies the synthetic theory of plane content, whereas Section 15 later develops an arithmetic of segments. The documentary purpose is to keep those layers separate and to identify which one is actually used by the current formal reconstruction.

C.2 Hilbert's organization of the relevant theory

Hilbert's Chapter IV distinguishes finite equidecomposability from the more general relation of equicomplementability. The theory of plane area begins on the basis of Groups I–IV and does not need to be introduced as a numerical area function [2].

For the present project the useful hierarchy is

$$\text{exact finite dissection} \Longrightarrow \text{equicomplementability} \Longrightarrow \text{later numerical area semantics},$$

with no converse inserted merely for convenience.

Section 15 is a different layer. Hilbert there defines segment addition and a geometric product, using a fixed unit segment and parallel construction, and then proves the familiar algebraic laws, including commutativity and distributivity [2]. That later arithmetic is a valuable control source precisely because it shows what would be required if Book II were to be formalized through segment multiplication. The current Lean development does not take that route.

C.3 Comparison table

Table C.1: Euclid II.1–II.10 against Hilbert's content and segment-arithmetic layers

Euclid	**Hilbert counterpart**	**Status in the Grundlagen**	**Source-audit remarks and relevance to formalization**
II.1	Chapter IV finite decomposition and content theory; later Section 15 distributivity of segment multiplication is a structural analogue.	No one-to-one numbered Hilbert theorem reproducing Euclid II.1 was located in the current audit. The relevant material is organized by content theory and, later, by segment arithmetic rather than by Euclid's Book II numbering.	The formal proof belongs to the earlier content/dissection layer. Additivity is obtained from an actual rectangle cut and representative transport, not by invoking the later distributive law for a segment product.
II.2	Chapter IV content plus the geometric symmetry of a rectangle; Section 15 commutativity is only a later arithmetic analogue.	Again the source correspondence is structural rather than proposition-level. Hilbert's later commutative segment multiplication must not be read backward into the scissors proof.	The Lean theorem makes the distinction concrete: the required exchange of the two containing sides is implemented by cyclic rotation of a rectangle representative, preserving exact scissors content.
II.3	Chapter IV content/dissection; later Section 15 distributivity gives the arithmetic pattern only after segment multiplication has been defined.	No separate proposition-level Hilbert counterpart was located. The source supports the distinction between geometric content operations and the later arithmetic of segments.	II.3 is a direct specialization of the binary II.1 cut with the chosen segment used as the fixed side. The square is packaged as a rectangle contained by the same segment twice. No segment product and no side-order transport are needed.
II.4	Chapter IV finite content composition; later Section 15 distributivity and commutativity give an arithmetic analogue of the square expansion.	No one-to-one Hilbert proposition corresponding to Euclid II.4 was located in the current audit. Hilbert's source role remains architectural.	The formal theorem composes exact II.2 with two exact II.3 instances and a geometric rectangle rotation. Despite the modern shadow $(a+b)^2 = a^2 + b^2 + 2ab$, no segment product, unit segment, or numerical area is used.
II.5	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the identity.	No one-to-one Hilbert proposition corresponding to Euclid II.5 was located in the current audit. The source role remains the separation of synthetic content from the later arithmetic of segments.	The formal theorem combines midpoint and order information with exact II.1 and II.3 rectangle decompositions, representative transport, and scissors addition. The modern shadow $(a-x)(a+x)+x^2 = a^2$ is not used as a proof object; no segment product, unit segment, numerical area, or cancellation axiom is introduced.

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Table C.1 – continued from previous page

Euclid	Hilbert counterpart	Status in the Grundlagen	Source-audit remarks and relevance to formalization
II.6	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the external-section identity.	No one-to-one Hilbert proposition corresponding to Euclid II.6 was located in the current audit. The relevant source role remains the separation between synthetic content and later segment arithmetic.	The formal theorem combines order from the midpoint/extension configuration with exact II.1, II.3, and II.4 decompositions, rectangle transport, and a purely additive scissors bridge. The modern shadow $a^2 + (2a + x)x = (a + x)^2$ is not a proof object; no segment product, unit segment, numerical area, subtraction, or cancellation axiom is introduced.
II.7	Chapter IV content/dissection; Section 15 segment multiplication supplies only a later arithmetic analogue of the arbitrary-cut identity.	No one-to-one Hilbert proposition corresponding to Euclid II.7 was located in the current audit. The relevant source role remains the separation between synthetic plane content and later segment arithmetic.	The formal theorem composes exact II.4 with II.3 on the reversed divided segment, uses a geometric rectangle rotation, and finishes by additive scissors rearrangement. No unit segment, segment product, numerical area, subtraction, cancellation, or new content principle is introduced.
II.8	Endpoint extension with $BD \cong BC$, square on AD, and four copies of a rectangle contained by the whole and the selected part.	The extension, congruence, rectangle, and square geometry belongs below the later segment-arithmetic layer; Chapter IV supplies the synthetic content setting.	The production theorem remains in exact scissors calculus: II.4 and II.7 are composed, the square on congruent sides is transported through rectangle representative uniqueness, and four copies are a finite formal sum. Hilbert's Section 15 multiplication is not used.
II.9	Right-angle, parallel, parallelogram, square, and Pythagorean configuration for a bisected segment with a second internal cut.	Hilbert's separation between elementary geometry, the Chapter IV theory of content/equicomplementability, and later segment arithmetic is directly relevant.	The current Lean proof stays below segment multiplication and numerical area. Unlike II.1–II.8, its public conclusion is `HilbertScissorsEquicomplementable`; this is a natural Chapter IV-style content statement assembled from four I.47 blocks and square transport, not an exact-dissection theorem asserted by the code.
II.10	External endpoint extension, directed intersection, right-angle and parallelogram geometry, concrete squares, and four Pythagorean content blocks.	Hilbert's separation between Groups I–IV geometry, Chapter IV content/equicomplementability, and later Section 15 segment arithmetic remains the relevant source architecture; no one-to-one numbered counterpart is required.	The Lean proof stays below segment multiplication and numerical area. As in II.9, the public conclusion is `HilbertScissorsEquicomplementable`; the directed intersection and square-representative transports are geometric/content proof obligations, not an invocation of later segment arithmetic.

C.4 Structural observations retained from Hilbert

The following distinctions should be preserved as Book II grows.

- Plane content and segment arithmetic are separate theories in the Grundlagen and should remain separate in the documentation.
- Exact finite dissection is stronger than the general content relation; when Lean proves `HilbertScissorsEq`, that fact comes from the local proof architecture rather than from a blanket identification of all equal-content figures with equidecomposable ones.
- Segment multiplication is not primitive. It requires a unit segment, proportion geometry, and parallel constructions before algebraic laws such as commutativity and distributivity can even be stated.
- Consequently, a distributive-looking Euclidean proposition can be formalized below the arithmetic layer when its geometry is literally a finite cut-and-paste argument.

C.5 II.1–II.8: exact dissection below segment arithmetic

The opening eight propositions give a useful stress test of the separation. II.1 establishes a reusable exact rectangle split. II.2 shows that the apparent exchange of factors can still be implemented geometrically by rotating a rectangle presentation. II.3 then shows that even this extra symmetry operation is unnecessary when the ordered side data already match the binary II.1 interface. II.4 is the first compositional step: it combines II.2 with two II.3 expansions and uses only exact scissors addition and representation transport. II.5 then adds midpoint and order data and composes the same rectangle/scissors mechanisms into the internal unequal-parts identity. II.6 moves the second cut outside the original segment and, in the current Lean DAG, uses II.4 directly to normalize the square on the produced segment. II.7 then returns to an arbitrary internal cut and combines II.4 with a reversed II.3 normal form. II.8 adds an external congruent segment and composes II.4 with II.7; the only new local bridge is square representative transport. All four remain entirely below segment arithmetic.

The resulting formal chain is therefore

$$\begin{array}{c}
\text{rectangle geometry}\\
\Downarrow\\
\text{exact scissors dissection}\\
\Downarrow\\
\text{II.1}\\
\Downarrow\\
\text{II.2 and II.3 specializations}\\
\Downarrow\\
\text{II.4 by composition}\\
\Downarrow\\
\text{II.5 by midpoint/order transport and composition}\\
\Downarrow\\
\text{II.6 by II.1 + II.3 + II.4 + midpoint transport}\\
\Downarrow\\
\text{II.7 by II.4 + reversed II.3 + rectangle rotation}\\
\Downarrow\\
\text{II.8 by II.4 + II.7 on reversed cut + square representative transport}
\end{array}$$

not

$$\text{segment field} \longrightarrow \text{multiplication} \longrightarrow \text{algebraic simplification.}$$

C.6 II.9: equicomplementability becomes visible in the public type

Through II.8 the configured Book II theorems could be documented at the stronger exact-dissection level `HilbertScissorsEq`. II.9 is the first current proposition for which the public type returns to the more general content relation `HilbertScissorsEquicomplementable`. This makes Hilbert's architectural separation especially useful: the theorem belongs to the plane-content layer, while the later arithmetic of segments remains uninvoked.

The formal route is also geometrically Hilbertian in spirit. Betweenness, noncollinearity, perpendiculars, parallels, right angles, congruent segments, and parallelograms are established first; concrete square representatives are then compared in the content layer. No numerical area function is used to bridge those levels.

C.7 II.10: the same Chapter IV level with an external order configuration

II.10 strengthens the evidence that the return to general equicomplementability at II.9 is an architectural feature of the current Pythagorean proof route. The new configuration adds a produced point beyond the endpoint and requires explicit plane-separation, order, and same-ray control before the four I.47 content blocks can be composed.

Nothing in this change crosses Hilbert's later arithmetic boundary. The proof still proceeds from Groups I–IV geometry to concrete square representatives and then to the Chapter IV-style content relation. No unit segment, segment product, numerical area, subtraction, or cancellation principle is introduced.

C.8 Result of the Grundlagen audit

At the II.10 checkpoint Hilbert's main contribution remains architectural. Chapter IV validates the use of a synthetic content layer independently of numerical area, while Section 15 marks a later arithmetic boundary. The current Book II formalization through II.10 remains on the earlier side of that boundary.

The rectangle API therefore has a clear foundational interpretation: congruent segments and right angles determine representative rectangles, and the scissors layer compares their content. The algebra-like behavior observed in II.1–II.3 is derived from that geometry.

C.9 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.10. At this stage the durable Hilbert-side picture is

$$\begin{array}{c}\text{Groups I–IV geometry}\\ \Downarrow\\ \text{plane-content/dissection layer}\\ \Downarrow\\ \text{exact rectangle calculus for II.1–II.8}\\ \Downarrow\\ \text{equicomplementable square identities for II.9–II.10}\end{array}$$

while Section 15 segment arithmetic remains a later comparison layer rather than a dependency of the proofs.

Appendix D

Borsuk–Szmielew: Axiomatic Layers and Representation

D.1 Purpose and scope

This appendix records the Borsuk–Szmielew-side source audit for Euclid Book II at the current checkpoint, Propositions II.1–II.10. The source is used primarily to control the axiomatic layering of incidence, order, congruence, polygons, parallels, and parallelograms. The question at the beginning of Book II is how far that geometry reaches before a separate theory of content is needed.

D.2 The relevant axiomatic boundary

Borsuk–Szmielew develop rigorous polygonal and Euclidean parallelogram infrastructure, but the current audit has not located a synthetic Euclidean theory of equal content or equicomplementability parallel to Hilbert Chapter IV or Hartshorne Section 22 [3].

That absence is structurally useful. It confirms that one can separate the representation and Euclidean geometry of rectangles from the content relation used to compare them. A rectangle is a special parallelogram, and its subdivision is ordinary polygonal geometry; the assertion that the whole and the pieces have the required common content belongs to a different layer.

D.3 Comparison table

Table D.1: Euclid II.1–II.10 against the Borsuk–Szmielew axiomatic layers

Euclid	Borsuk–Szmielew counterpart	Status / layer	Structural significance for the reconstruction
II.1	Rectangle as a special parallelogram together with polygonal subdivision.	Geometry available; no direct content counterpart located.	The geometric configuration is natural in the source architecture, but the conclusion that the whole rectangle equals the sum of the subrectangles belongs to the separate scissors/-content layer of the formal project.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Status / layer	Structural significance for the reconstruction
II.2	Square/rectangle geometry and subdivision by a parallel through the division point.	Geometry available; no direct content counterpart located.	The source can support the representation of the square and its two rectangular parts. Exact scissors equality and the formal side-order transport are not supplied by the audited Borsuk–Szmielew theory.
II.3	Rectangle plus square configuration built from the same Euclidean parallelogram/right-angle infrastructure.	Geometry available; no direct content counterpart located.	No new Borsuk–Szmielew layer appears. The reduction to binary II.1 and the identification of a square with a rectangle contained by the same segment twice belong to the formal rectangle/content API.
II.4	Square and rectangle decomposition supported by Euclidean parallelogram, parallel, right-angle, and polygonal geometry.	Geometry available; no direct content counterpart located.	The source supports the geometric representation layer. The exact scissors composition of II.2 and II.3 and representative rotation remain external formal content infrastructure.
II.5	Midpoint, betweenness, square, and rectangle configuration for an internal unequal cut.	Midpoint, order, congruence, and Euclidean geometry available; no direct content counterpart located.	Borsuk–Szmielew are a useful control source for the separation of midpoint placement from segment congruence. The exact content identity and rectangle transport are supplied by the Lean scissors layer.
II.6	Midpoint plus endpoint extension, with rectangle and square configurations.	Order/congruence and Euclidean representation geometry available; no direct content counterpart located.	The source can control the one-dimensional order and parallelogram geometry, but not the exact II.1/II.3/II.4 scissors combination used by the current Lean proof.
II.7	Arbitrary internal cut with square and rectangle representatives.	Order, congruence, polygon, parallel, and rectangle geometry available; no direct synthetic equal-content counterpart located.	The formal reversal of betweenness and cyclic rectangle representation are compatible with the source's geometric layering. Exact `HilbertScissorsEq`, representative reuse, and the final additive bridge remain separate project infrastructure.
II.8	Arbitrary cut, endpoint extension by a congruent segment, and square/rectangle representatives.	Order, congruence, polygon, parallel, and Euclidean rectangle geometry are available; no direct synthetic equal-content counterpart was located.	The source layering supports the explicit $A - C - B - D$ order and the transport $BD \cong BC$. Exact II.4/II.7 scissors composition, square representative uniqueness, and the fourfold formal sum remain separate project infrastructure.

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Table D.1 – continued from previous page

Euclid	Borsuk–Szmielew counterpart	Status / layer	Structural significance for the reconstruction
II.9	Bisected segment and unequal cut with perpendicular construction, parallel through the cut point, auxiliary parallelogram, strict order on the constructed intersections, and square representatives.	Incidence, order, congruence, perpendicular/parallel, polygonal, and parallelogram layers are relevant; no direct synthetic equal-content counterpart was located in the audited source.	II.9 makes the modular boundary especially clear: the source can control the geometric representation and construction shell, while the four Pythagorean `HilbertScissorsEquicomplementable` blocks and their additive composition remain external formal content infrastructure.
II.10	Bisected segment with an endpoint extension, perpendicular and parallel construction, auxiliary parallelogram, produced intersection with strict order, and square representatives.	Incidence, order, congruence, perpendicular/parallel, polygonal, and parallelogram layers are again the relevant source-side controls; no direct synthetic equal-content counterpart was located in the audited source.	The external order $A - C - B - D$ and the directed positions $E - B - G$, $F - D - G$ belong naturally to the geometry/representation layer. The four I.47 `HilbertScissorsEquicomplementable` blocks, square transport, and their additive composition remain separate formal content infrastructure.

D.4 Structural observations retained from the audit

The beginning of Book II reinforces the following separation:

$$\text{polygon/rectangle geometry} \quad + \quad \text{scissors/content relation} \quad + \quad \text{representative independence.}$$

Borsuk–Szmielew are strongest on the first term. The second and third are additional layers of the present formal development. This prevents the source audit from silently attributing to the book a theory of equal content that the current audit has not located there.

D.5 II.1–II.8: representation is stable, content is external

Across II.1–II.8 the source role is stable. Rectangle, square, midpoint, order, parallel, and parallelogram geometry remain within the axiomatic geometric layers. What changes from proposition to proposition is the way the formal scissors API packages and reuses that geometry: general split in II.1, side-order transport in II.2, direct specialization in II.3, composition in II.4, midpoint configurations in II.5–II.6, and the arbitrary-cut II.4/II.3 composition in II.7, and the endpoint-extension II.4/II.7 composition with representative transport in II.8.

These are important formal distinctions, but they are not reasons to invent different Borsuk–Szmielew content counterparts. The proper source conclusion is that the same geometry/content boundary persists throughout the opening eight propositions.

D.6 II.9: a richer construction while the content layer remains external

II.9 uses substantially more of the geometric side of the architecture than the preceding rectangle identities. The formal proof tracks midpoint and strict betweenness data, erects a perpendicular, constructs parallels and an auxiliary parallelogram, proves the order of two constructed intersection points, and transports segment congruences through that configuration. These are precisely the kinds of incidence/order/congruence and Euclidean representation questions for which Borsuk–Szmielew remain a useful control source.

What does not change is the content boundary. The present audit has not located in Borsuk–Szmielew a synthetic equal-content calculus matching the project's `HilbertScissorsEquicomplementable` layer. The four Pythagorean content blocks and their final composition must therefore continue to be documented as formal project infrastructure rather than as a Borsuk–Szmielew theorem.

D.7 II.10: external order adds geometry, not a new content theory

II.10 is a particularly clear example of the source role assigned to Borsuk–Szmielew. Compared with II.9, the second point lies beyond the endpoint, so the formal proof must establish a richer one-dimensional order package and prove that the constructed intersection lies on the intended extensions. Incidence, order, congruence, parallelism, and parallelogram representation are therefore central.

The source boundary nevertheless stays fixed. The current audit has not located a Borsuk–Szmielew theory matching the project's `HilbertScissorsEquicomplementable` calculus. II.10's four Pythagorean content blocks and square-representative transports are consequently documented as project infrastructure, exactly as for II.9.

D.8 Consequences for the formal library

The audit supports a modular architecture in which

- Euclidean incidence, parallelism, right-angle, and parallelogram facts live below the Book II content layer;
- rectangle representatives are built from that geometry;
- exact scissors comparison and representative independence are supplied separately;
- later Book II propositions may reuse the rectangle/content API without rebuilding the underlying Euclidean geometry.

D.9 Current checkpoint

This appendix now concerns Euclid Book II, Propositions II.1–II.10. No direct Borsuk–Szmielew content counterpart has been located for the block. The source continues to function as a geometry-and-representation control source, while the equality-of-content semantics is supplied elsewhere.

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